

Mitochondrial Genomes

Numerous mitochondrial DNAs have been sequenced, including the human mitochondrial DNA, which is 16,569 base pairs long. It is a model of economy, with very few noncoding regions and no introns (fig. 17.4). Each strand of the duplex is transcribed into a single RNA product that is then cut into smaller pieces, primarily by freeing the twenty-two transfer RNAs interspersed throughout the genome. Also formed are a 16S and a 12S ribosomal RNA. Although proteins and small molecules such as ATP and tRNAs can move in and out of the mitochondrion, large RNAs cannot. Thus, the mitochondrion must be relatively self-sufficient in terms of the RNAs needed for protein synthesis. We previously discussed mitochondrial protein synthesis when we looked at unique attributes of the mitochondrial genetic code in chapter 11.

Oxidative phosphorylation, the process that occurs within the mitochondrion, requires at least sixty-nine polypeptides. The human mitochondrion has the genes for thirteen of these: cytochrome b, two subunits of ATPase, three subunits of cytochrome-c oxidase, and seven subunits of NADH dehydrogenase. The remaining polypeptides needed for oxidative phosphorylation are transported into the mitochondrion; they are synthesized in the cytoplasm under the control of nuclear genes. Proteins targeted for entry into the mitochondrion have special signal sequences (see chapter 11).

The signal sequences range up to eighty-five amino acids long. Signal sequences examined so far do not have consensus amino acids but do have certain attributes (fig. 17.5), including a somewhat regular alternation of basic (positively charged) and hydrophobic (negatively charged) residues. In addition, they form α helices with opposite hydrophobic and hydrophilic faces that must somehow be important in the protein's ability to enter the mitochondrion. When a signal sequence (such as that in fig. 17.5) is attached to nonmitochondrial proteins by DNA manipulations, those proteins are transported into the mitochondrion.

The mitochondrial ribosomal RNA is more similar to prokaryotic ribosomal RNA than to eukaryotic ribosomal RNA. The mitochondrial ribosome, although constructed of imported cellular proteins, is sensitive to prokaryotic antibiotics; for example, streptomycin and chloramphenicol inhibit their function. This affinity (close resemblance) between mitochondria and prokaryotes is strong support for the symbiotic origin of mitochondria. That is, we now accept the model advocated by L. Margulis that organelles such as mitochondria and chloroplasts were originally free-living bacteria and cyanobacteria, respectively. These prokaryotes invaded or were eaten by early cells and, over evolutionary time, became the organelles we see today. Since they arose as prokaryotes, these organelles retain certain evolutionary similarities to other prokaryotes.

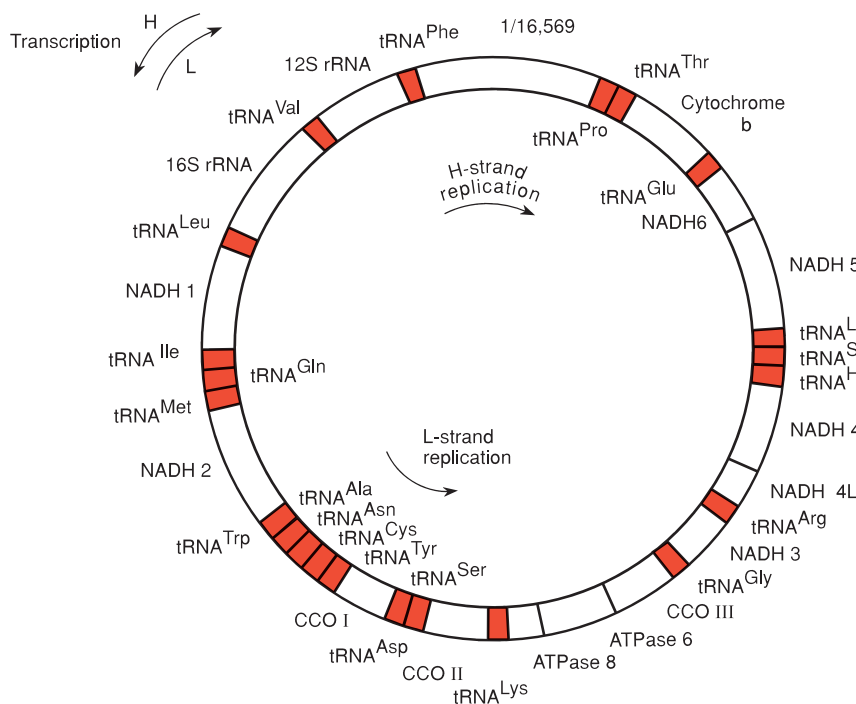


Figure 17.4 Gene map of the human mitochondrial chromosome. All but nine loci are on the heavy (H) strand. The light-strand (L) loci are labeled inside the circle; the H-strand loci are labeled on the outside. Also shown are the origins of H- and L-strand replication and the directions of transcription. The twenty-two tRNA genes are colored red. NADH refers to NADH dehydrogenase (subunits 1–4, 4L, 5, and 6); CCO refers to cytochrome-c oxidase (subunits I–III). (Source: Data from V. McKusick, *Mendelian Inheritance in Man*, 7th edition, 1986.)



Lynn Margulis (1938–).
(Courtesy of Lynn Margulis, Boston
University Photo Services.)

Among the mitochondrial DNAs that have been sequenced from different organisms, we see great variation in content and organization. Yeast mitochondrial DNA, for example, is not as economical as human mitochondrial DNA. Yeast mitochondrial DNA, about five times larger than human mitochondrial DNA, has non-coding regions as well as introns. Because mitochondria are similar in structure and biochemistry to prokaryotic cells, given the general lack of introns in

prokaryotic genes, it was surprising to find introns in yeast mitochondrial DNA. These genes most probably arose later as nuclear genes that were then “captured” by the mitochondria, possibly by recombination with nuclear DNA.

Of the many mitochondria sequenced to date (about 175 at the beginning of 2001), the sizes range from less than 6 to more than 200 kilobases and from 3 to 97 genes. With this wide range of genes present, the only generality we can make about mitochondrial DNA is that the large and small segments of the mitochondrial ribosomal RNA, as well as most of the mitochondria’s transfer RNAs, are usually coded by the mitochondria’s own genome, as are several proteins in respiratory complexes III and IV (cytochrome c oxidase and cytochrome c oxidoreductase). Once the interaction within the mitochondrial-nuclear genetic system is clearly understood, we might expect to see several different inheritance patterns—following either cytoplasmic or nuclear lines—for the genetic defects that lead to interruption of cellular respiration. Among the best-studied phenotypes with such inheritance patterns are the *petite* mutations of yeast.

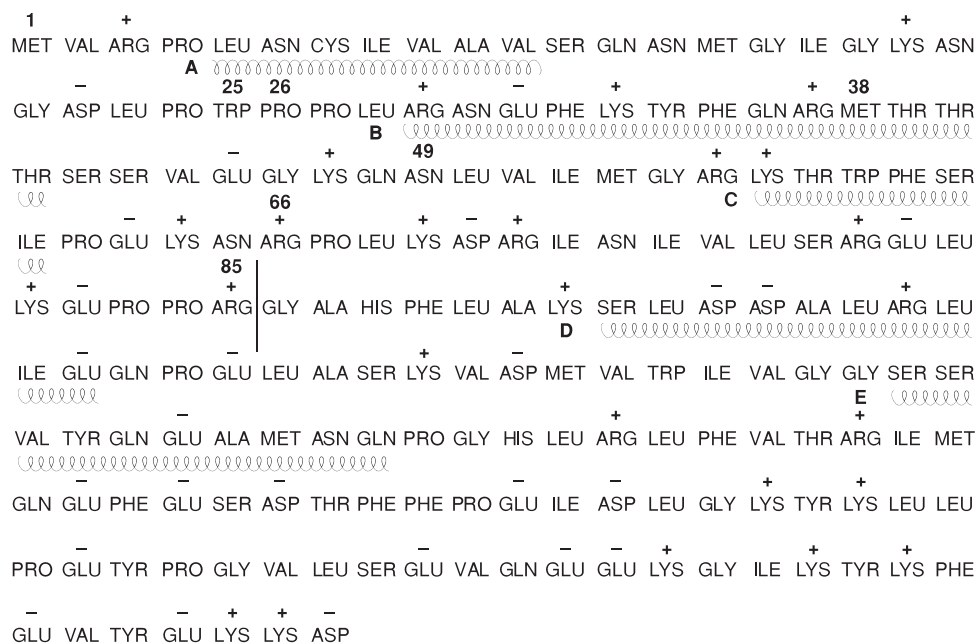


Figure 17.5 The amino acid sequence of mouse dihydrofolate reductase. Numbers refer to sequential amino acids. The first eighty-five amino acids serve as the signal sequence for transport into mitochondria. Five α -helical regions exist in the protein (A–E). Positively and negatively charged amino acids are marked with (+) and (–) signs. (Reprinted by permission from E. C. Hurt and G. Schatz, “A cytosolic protein contains a cryptic mitochondrial targeting signal,” *Nature*, Volume 325, p. 499, 1987. Copyright © 1987 Macmillan Magazines, Ltd.)

Petites

Under aerobic conditions, yeast grows with a distinctive colony morphology. Under anaerobic conditions, the colonies are smaller, and the structures of the mitochondria are reduced. Occasionally, when growing aerobically, small, anaerobiclike colonies appear; but in these colonies, the mitochondria appear perfectly normal. These colonies are caused by **petite mutations**. When petites are crossed with the wild-type, three modes of inheritance emerge (fig. 17.6). The *segregational petite*, caused by mutation of a chromosomal gene, exhibits Mendelian inheritance. The *neutral petite* is lost immediately upon crossing to the wild-type. The *suppressive petite* shows variability in expression from one strain to the next but is able to convert the wild-type mitochondria to the petite form. All petites represent failures of mitochondrial function, whether the function is controlled by the mitochondria themselves or by the cell's nucleus; they usually lack one or another cytochrome.

Although the mechanisms that produce neutral and suppressive petites are not known with certainty, their DNA has supplied some interesting information. In some petites, no change in the buoyant density of the DNA is found. (*Buoyant density*, a term that describes the position at which the DNA equilibrates during density-gradient centrifugation, is a measure of the composition of the molecule; see chapter 15.) In other petites, changes

in buoyant density range from very small to the complete absence of DNA.

Petites, therefore, can be the result of an approximation to a point mutation (with no measurable change in the buoyant density of the DNA), marked changes in the DNA, or the total absence of DNA. In most petites, protein synthesis within the mitochondrion is lacking. Any and all of these changes produce the petite (anaerobic-like) phenotype.

Neutral petites seem to have mitochondria that entirely lack DNA. When neutral petites are crossed with the wild-type to form diploid cells, the normal mitochondria dominate. During meiosis, virtually every spore receives large numbers of normal mitochondria; the progeny are, therefore, all normal.

Suppressive petites could exert their influence over normal mitochondria in one of two ways. The suppressive mitochondria might simply out-compete the normal mitochondria and take over; they might simply reproduce faster within a cell. Alternatively, crossing over between the DNA of the suppressive petite and the wild-type might affect the normal DNA if the suppressive petite's DNA were severely damaged. Presumably, recombination in mitochondrial DNAs occurs when two or more mitochondria fuse, bringing the two different sets of DNA in contact within the same organelle. Recombination would presumably take place by normal crossover mechanisms.

If large portions of the DNA from the suppressive mitochondria were missing or altered, recombination with the normal mitochondria's DNA might exchange some of this damaged DNA. Several experiments have crossed a suppressive petite and a wild-type, each with mitochondrial DNA of known buoyant density. The DNAs of the offspring colonies, which were petites, were of various buoyant densities. For example, when a normal strain with mitochondrial DNA with a buoyant density of 1.684 g/cm³ was crossed with a suppressive petite with a buoyant density of 1.677 g/cm³, the offspring colonies' mitochondrial DNA had buoyant densities of 1.671, 1.674, and 1.683 g/cm³. Such information supports the notion that the suppressive character takes over a colony by way of recombination.

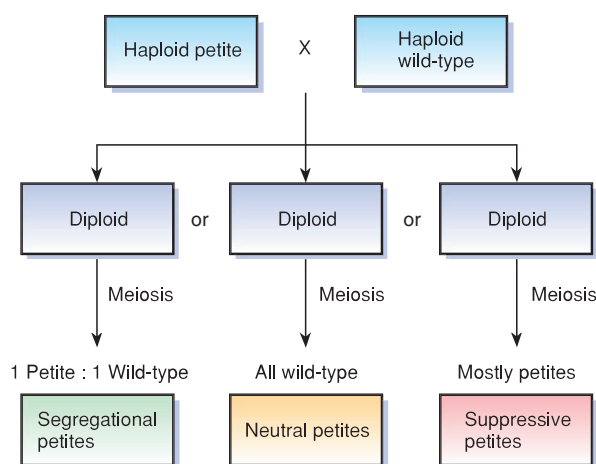


Figure 17.6 Petite yeasts categorized on the basis of segregation patterns. Three types of petites are recognized (segregational, neutral, and suppressive), depending on the meiotic segregation pattern of petite × wild-type diploids. Segregational petite heterozygotes segregate a 1:1 ratio of spores; neutral petites are lost when heterozygous; and suppressive petites act in a dominant fashion under the same circumstances.

Human Mitochondrial Inheritance

In human beings, certain diseases trace their dysfunction to mitochondrial pathologies. The first such disease, Luft disease, characterized by excessive sweating and general weakness, was reported in 1962. In 1988, Douglas Wallace and his colleagues showed that *Leber optic atrophy* is a cytoplasmically inherited disease. This disease causes blindness, with a median age of onset of twenty to twenty-four years. The onset age and phenotype are variable, depending on the degree of heteroplasmy in the in-

dividual. Apparently, defects in mitochondria are not tolerable in the optic nerve, which demands a great deal of energy. The disease also does some damage to the heart. Pedigrees showed that Leber optic atrophy is transmitted only maternally. Sequencing of mitochondrial DNAs in affected families pinned down the disease to a point mutation, a change in nucleotide 11,778, which is in the gene for NADH dehydrogenase subunit 4 (see fig. 17.4). A guanine is changed to an adenine at codon 340, which converts an arginine to a histidine. This is the first human disease traced to a specific mitochondrial DNA mutation. Since 1962, over one hundred diseases, including some of the general symptoms of aging and cancer, have been attributed to mitochondrial pathology.

Antibiotic Influences

Since the machinery of mitochondrial protein synthesis is prokaryotic in nature, antibiotics such as chloramphenicol and erythromycin can inhibit it. These antibiotics elicit a petite-type growth response in yeast. Antibiotic-resistant strains can be obtained by growing yeast on the antibiotic; only resistant mutants will grow. The resistance appears to be inherited in the mitochondrial, not the cellular, DNA. A mitochondrial inheritance pattern results, with crosses between a resistant and a sensitive (wild-type) yeast, as shown in figure 17.7. The resulting diploid colonies segregate both resistant and sensitive cells. Although not expected on the basis of a chromosomal gene, the random sorting of mitochondria through cell division could result in a wild-type cell containing only sensitive mitochondria. Since some yeast have only one to ten mitochondria per cell, this random assortment of sensitive mitochondria can be expected to occur at a relatively high rate.

Chloroplasts

The **chloroplast** is the chlorophyll-containing organelle that carries out photosynthesis and starch-grain formation in plants (fig. 17.8). Chloroplasts are referred to as **plastids** before chlorophyll develops. However, when grown in the dark (and under some other circumstances), plastids do not develop into chloroplasts, but remain reduced in size and complexity. These undeveloped plastids, referred to as **proplastids**, are each about the size and shape of a mitochondrion.

Like mitochondria, chloroplasts contain DNA and ribosomes, both with prokaryotic affinities. The DNA of chloroplasts (cpDNA) is a circle that ranges in size from 85 kilobases (kb) in the green alga *Codium* to as large as 2,000 kilobases in the green alga *Acetabularia*. Thus, chloroplast DNA is minimally about five times the size of an animal mitochondrial DNA. The chloroplast DNA, like mitochondrial DNA, controls the production of transfer

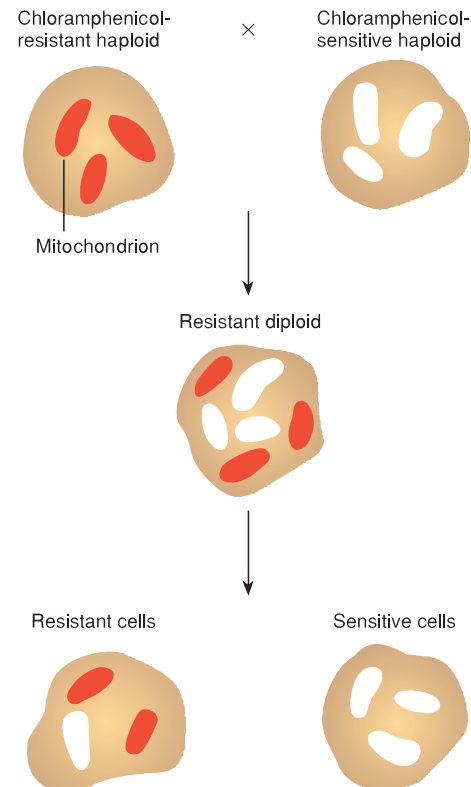


Figure 17.7 Inheritance of antibiotic (chloramphenicol) resistance in yeast. Resistant and sensitive cells are produced by a diploid cell that resulted from a cross of resistant and sensitive haploids. The segregation is not in a simple Mendelian ratio, but depends on the random assortment of mitochondria. Sensitive cells have no resistant mitochondria. Resistant cells have resistant mitochondria.

RNAs, ribosomal RNAs, and some of the proteins found within the organelle. From the more than nineteen chloroplast DNAs that have been sequenced, there seem to be about one hundred genes in the chloroplast genome. About thirty code for the subunits of the five photosynthetic protein complexes: photosystem I, photosystem II, ribulose biphosphate carboxylase-oxygenase, cytochrome *b6-f* complex, and ATP synthase. About sixty genes code for the protein synthesis apparatus of the chloroplast. Scientists believe that the chloroplast evolved from symbiotic cyanobacteria (blue-green algae), which have many affinities with the chloroplast: The ribosomal RNA of cyanobacteria will hybridize with the DNA of chloroplasts.

The similarities between mitochondria and chloroplasts make it possible to predict the inheritance patterns of chloroplast mutations on the basis of existing

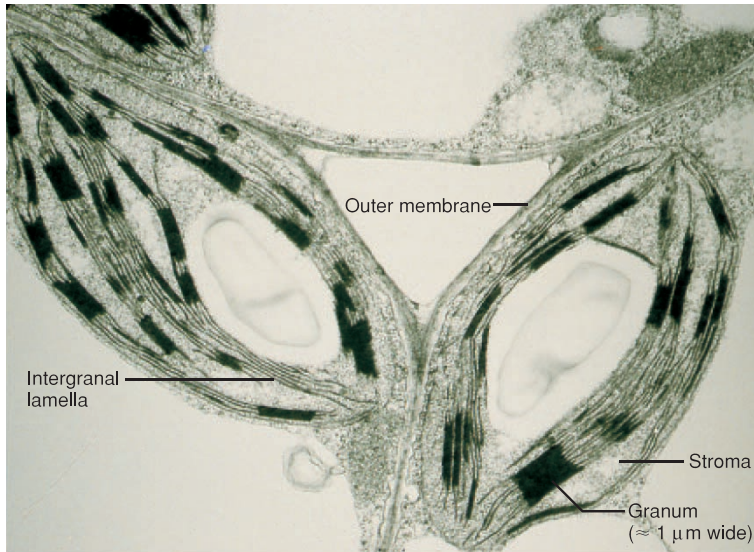


Figure 17.8 Electron micrograph of lettuce chloroplasts. The chloroplast consists of an outer membrane, stacks of grana, lamellae, and stroma. Magnification 3,570 $\times$. (© Dr. J. Burgess/Science Photo Library/Photo Researchers, Inc.)

knowledge of mitochondrial genetics: We should find both chromosomal and plastid mutants of chloroplast functions. Simple segregation should occur in the chromosomal mutations, and cytoplasmic patterns of inheritance should occur with the chloroplast DNA mutations. Investigation of these inheritance patterns is complicated by the fact that plant cells have both mitochondria and chloroplasts. Since both have prokaryotic affinities, it is sometimes difficult to determine whether a genetic trait is due to a defect in the genetic system of the chloroplast or the mitochondrion. Like mitochondria, chloroplasts generally show homoplasmy and maternal inheritance, although, as in mitochondria, there are exceptions. For example, gymnosperms usually have paternal inheritance of chloroplasts.

Lesions in the photosystems of the chloroplast result in proplastid formation, with a loss of green color. When proplastid formation occurs in a particular tissue of a plant, variegation results. That is, there are both green and white parts, often as stripes. Some interesting genetic studies have focused on the inheritance of variegation, especially in the interaction of chloroplast and chromosomal genes.

Zea mays

M. Rhoades worked on the variegation in corn (*Zea mays*) controlled by the *iojap* chromosomal locus, which, when homozygous, prevents proplastids from developing into chloroplasts and thus results in variegation. The *iojap*-affected plastids do not contain ribosomes or ribosomal RNA; they therefore lack protein synthesis.



Marcus M. Rhoades
(1903–1991). (Courtesy of
Indiana University Office of
Communications and Marketing.)

The interaction of chromosomal and extrachromosomal inheritance is shown in the reciprocal crosses depicted in figure 17.9. One cross produces results exactly as would be predicted on the basis of simple Mendelian inheritance, with the homozygous recessive genotype (*ijij*) inducing variegation. When the reciprocal cross is carried out, blotch variegation is seen in both the F_1 and F_2 that carry the dominant *Ij* allele.

This inheritance pattern is caused by the fact that the pollen grain in corn does not carry any chloroplasts, whereas the ovule does. Thus, the first cross in figure 17.9 deals with the passage of normal chloroplasts only into the F_2 generation. In the F_2 , the *ijij* genotype then induces variegation. The chloroplasts of the pollen parent are unimportant because they do not enter the F_1 . In the reciprocal cross, however, because the stigma parent is variegated, the F_1 is heterozygous but carries proplastids from the ovule that remain proplastids even under the dominant normal (*Ij*) allele. Therefore, regions of colorless cells produce white spots (blotchy variegation). Once the *ij* allele induces chloroplasts to become pro-

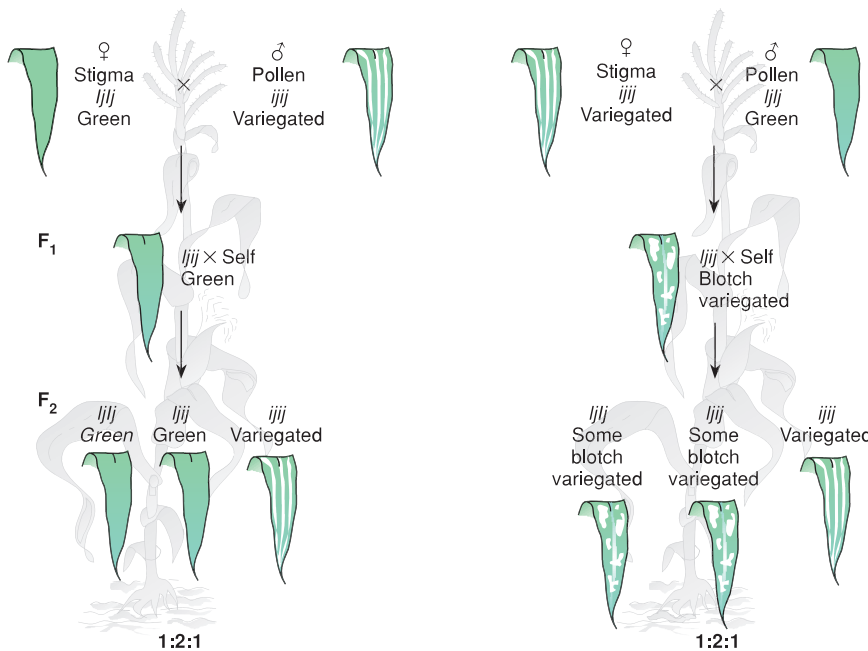


Figure 17.9 Reciprocal crosses involving the chromosomal gene *iojap* in corn. The homozygous recessive condition (*ijij*) induces variegation (representative corn leaves are shown). (Blotch variegation consists of irregularly shaped white areas rather than striping.) However, plants with the dominant allele (*ljlj*, *ljlj*) can still be variegated if their mothers were variegated, since mothers pass on their chloroplasts to their offspring; males (pollen parents) do not pass on their chloroplasts. *iojap* homozygotes induce variegation. The defective chloroplasts are then inherited in a cytoplasmic fashion.

plastids, they do not revert to normal type even under the *Ij* allele. Thus, we see the interaction of a chromosomal gene and the chloroplast itself, which “inherits” a changed condition.

There is some evidence that *iojap* may suppress the chloroplast rather than cause a mutation of some function. There are loci in corn and in other species that can induce back mutation in the chloroplasts. Removal of suppression rather than an actual reversion is more likely to occur because the reversion rate is too high to be due to simple back mutation.

Four-O’clocks

The first work with corn variegation was done by Carl Correns, one of Mendel’s rediscoverers. Correns also found maternal inheritance of variegation in the four-o’clock plant, *Mirabilis jalapa*. He could predict color and variegation of offspring solely on the basis of the region of the plant on which the stigma parent was located. A flower from a white sector, when pollinated by any pollen, would produce white plants; a flower on a green sector or a variegated sector produced green or variegated plants, respectively, when pollinated by pollen from any region of a plant. We thus see the simple maternal nature of the inheritance of the variegation. A chromosomal gene, like *iojap*, induces variegation. Inheritance of this induced variegation follows the “maternal” pattern of chloroplast inheritance.

Chlamydomonas

The single-celled green alga, *Chlamydomonas reinhardtii*, has been used in the study of extrachromosomal inheritance for several reasons. First, it has a single, large chloroplast; it can survive by culture technique even when the chloroplast is not functioning; and finally, it shows some interesting non-Mendelian inheritance patterns related to mating type. R. Sager has done extensive work on the inheritance of streptomycin resistance in *Chlamydomonas*.

Streptomycin resistance can be selected for in *Chlamydomonas* in several ways. Normal cells, sensitive to the antibiotic, are killed in its presence. If cells are grown in low levels of the antibiotic (100 g/ml), some cells show resistance to it. When these cells are crossed



Ruth Sager (1918–1997).
(Courtesy of Dr. Ruth Sager.)

with the wild-type, the resistance segregates in a 1:1 ratio, indicating that streptomycin resistance is controlled by a chromosomal locus. The same experiment can be repeated using high levels of the antibiotic in the medium (500–1,600 g/ml). Again, resistant colonies grow. If they are crossed with the wild-type, a 1:1 ratio does not ensue.

Chlamydomonas does not have sexes but does have mating types mt^+ and mt^- . Only individuals of opposite type can mate. Mating type is inherited as a single locus with two alleles. When two haploid cells of opposite mating type fuse, they form a diploid zygote, which then undergoes meiosis to produce four haploid cells, two of mt^+ and two of mt^- . The high-level resistance always segregates with the mt^+ parent (fig. 17.10). It is as if the mt^+ parent were contributing the cytoplasm to the zygote in a manner similar to maternal plastid inheritance in plants. The mt^- parent acts like a pollen parent by making a chromosomal contribution but not a cytoplasmic one.

The mechanism of the extrachromosomal inheritance pattern of *Chlamydomonas* is the preferential di-

gestion of the DNA of the chloroplast from the mt^- parent. Currently, we believe that streptomycin's target is the chloroplast.

More recent work has shown that the mt^+ inheritance is only 99.98% effective—that is, 0.02% of the offspring in crosses of the type shown in figure 17.10 have the streptomycin phenotype of the mt^- parent. Thus, we have the possibility of studying recombination in chloroplast genes. Although most of the evidence is only indirect and plagued by the previously mentioned problems of separating chloroplast and mitochondrial effects, some initial mapping studies have been done.

Infective Particles

Paramecium

Tracy Sonneborn discovered the killer trait in *Paramecium*. Before analyzing this trait, we must digress a moment to look at the life cycle of *Paramecium*, a ciliated

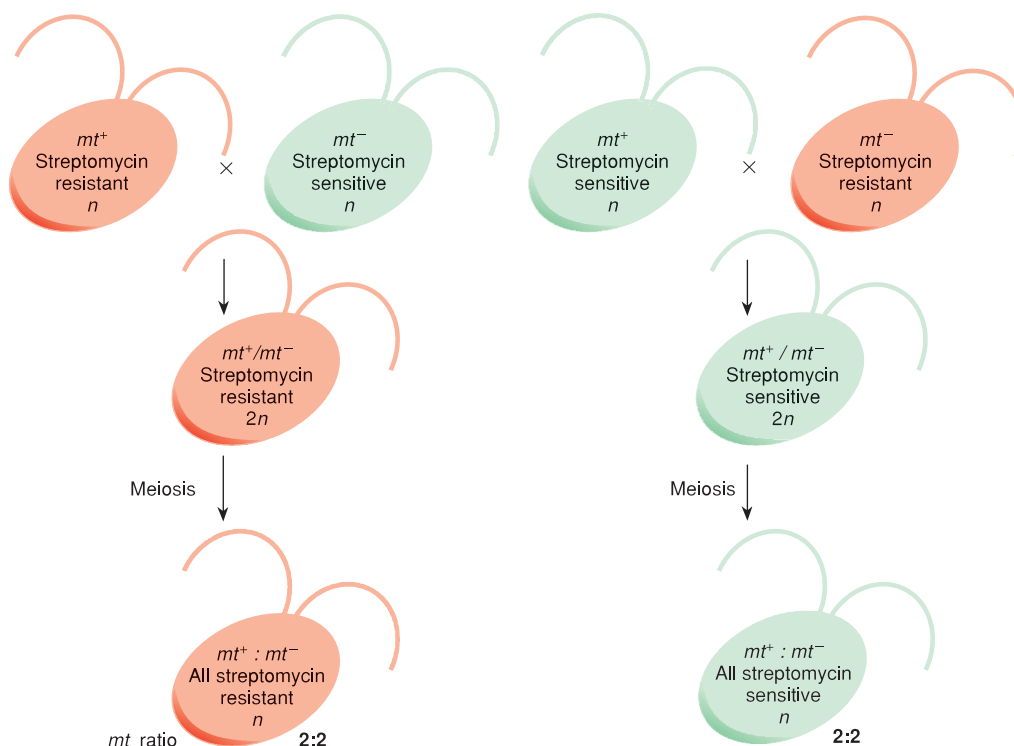


Figure 17.10 Inheritance pattern of streptomycin resistance in *Chlamydomonas* is dependent on the genotype of the mt^+ parent. (The n and $2n$ refer to the ploidy of the cells.) If the mt^+ parent is streptomycin resistant (red), then the diploid heterozygote, as well as the meiotic products, will be streptomycin resistant. If, however, the mt^+ parent is streptomycin sensitive (green), the diploid heterozygote, as well as the meiotic products, will be streptomycin sensitive.



Tracy M. Sonneborn
(1905–1981). (Courtesy: Indiana
University Archives.)

protozoan familiar to most biologists. Ciliates have two types of nuclei: macronuclei and micronuclei. In *Paramecium*, there are two micronuclei, which are primarily reproductive nuclei, and one macronucleus, which is a polyploid nucleus concerned with the vegetative functions of the cell. During cell division, termed **binary fission**, the micronuclei divide by mitosis and the macronucleus constricts and is pulled in half.

Paramecium undergoes two types of nuclear arrangements, during conjugation and **autogamy**. In conjugation, individuals of two mating types come to-

gether and form a connecting bridge. The nuclear events are shown in figure 17.11. Briefly, the macronucleus of each cell disintegrates while the micronuclei undergo meiosis. Of the resulting eight micronuclei per cell, seven disintegrate and one remains; this one undergoes mitosis to form two haploid nuclei per cell. A reciprocal exchange of nuclei across the bridge then occurs. Each cell now has two haploid nuclei, one original and one migrant. The two nuclei fuse to form a diploid nucleus. The diploid nuclei in the two conjugating cells are genetically identical because of the reciprocity of the process. These nuclei then undergo two mitoses each to form four diploid nuclei per cell. Two nuclei become macronuclei, which separate at the next cell division; two remain as micronuclei that divide by mitosis at the next cell division. The two cells that separate are known as **exconjugants**. Depending primarily on the amount of time conjugating cells remain united, an exchange of cytoplasm may occur along with the exchange of nuclei.

In the second type of process, autogamy, only one *Paramecium* is involved (fig. 17.12). The nuclear events are the same as in conjugation except that, at the point where a reciprocal exchange of nuclei would take place, the two haploid nuclei within the cell fuse. All cells after autogamy are homozygous.

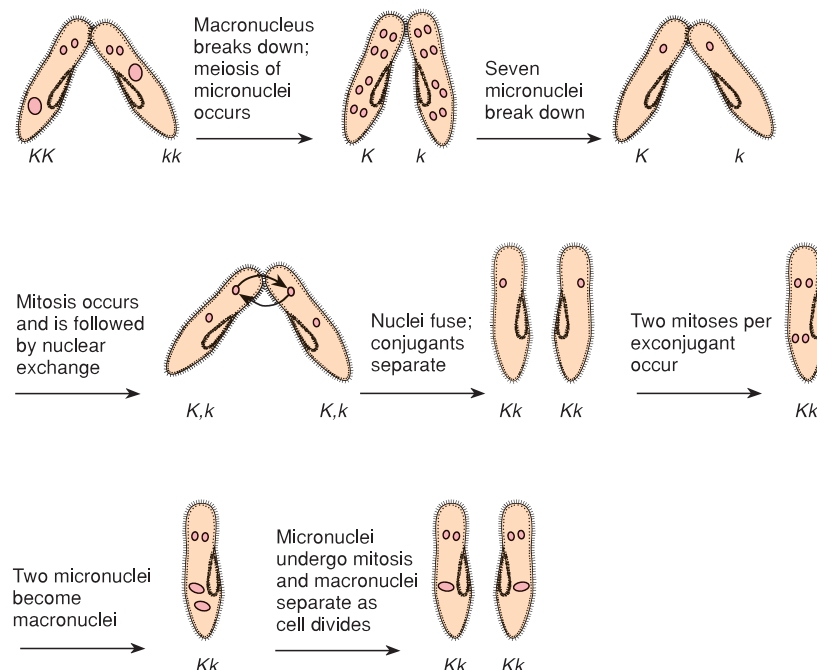


Figure 17.11 Conjugation in *Paramecium*. The letters K and k represent alleles of a gene in each micronucleus. When a KK and a kk individual conjugate, the exconjugants have the identical Kk genotype.

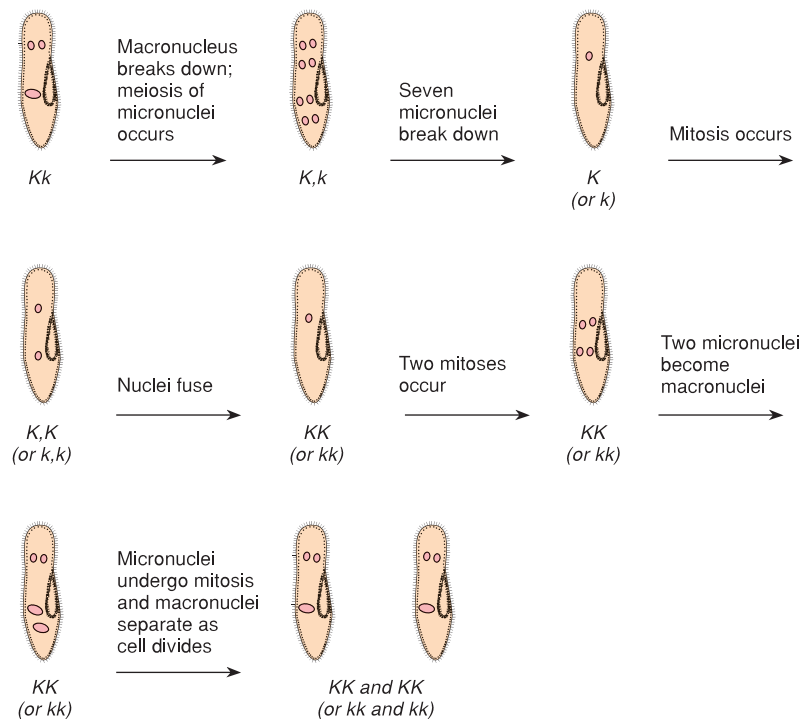


Figure 17.12 Autogamy in *Paramecium*. The letters K and k represent alleles of a gene in each micronucleus. If a heterozygote undergoes autogamy, it becomes homozygous for one of the alleles (KK or kk).

Killer *Paramecium* and Kappa Particles

Sonneborn and his colleagues found that when certain stocks of *Paramecium* were mixed together, one stock had the ability to cause the individuals of the other stock to die. Those individuals causing death were called “killers” and those dying were referred to as “sensitives.” During conjugation, the sensitives are temporarily resistant to the killers. If cytoplasm is not exchanged during the conjugation, the exconjugants retain their original phenotypes so that killers stay killers and sensitives stay sensitives. When an exchange of cytoplasm occurs between sensitive and killer cells, both exconjugants are killers. The transfer of some cytoplasmic particle seems to be implied. Indeed, Sonneborn observed such particles in the cytoplasm of killers and called them **kappa particles** (fig. 17.13).

Although the occurrence of killer *Paramecium* does not appear to involve chromosomal genes, Sonneborn reported one case in which exconjugant killer parametia of hybrid origin underwent autogamy. He found that half of the resulting cells had no kappa particles and had become sensitives. He concluded that a gene is required for the presence of kappa particles, which has subsequently

been verified by numerous crosses. Figure 17.14 illustrates the sequence of genetic events that would produce a heterozygous killer *Paramecium* that, upon autogamy, would have a 50% chance of becoming sensitive.

Although not yet cultured outside of a *Paramecium*, kappa is presumably a bacterium because it has many bacterial attributes including size, cell wall, presence of DNA, and presence of certain prokaryotic reactions (fig. 17.15). J. Preer and his colleagues, who studied kappa itself, named it *Caedobacter taeniospiralis*. Kappa occurs in at least two forms. The N form, the infective form that passes from one *Paramecium* to another, does not confer killer specificity on the host cell. The N form is attacked by bacteriophages that induce formation of inclusions, called R bodies, inside the kappa particle and thus convert it to the B form. These R bodies are visible under the light microscope as refractile bodies (fig. 17.15).

In the B form, kappa can no longer replicate; it is often lysed within the cell. It confers killer specificity on the host cell, however. The sensitives are killed by the toxin **paramycin**, which is released by the killer *Paramecium* into the environment. Precisely what steps are involved in its formation are not known, although it is

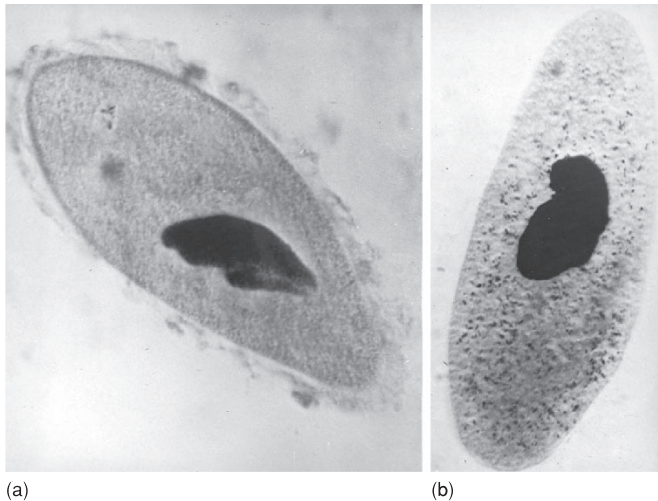


Figure 17.13 (a) Normal (sensitive) *Paramecium*. (b) Kappa-containing (killer) *Paramecium*. A *Paramecium* is about 200 μm long. (Source: T. M. Sonneborn, figure 29.3, p. 373 in I. H. Herskowitz, *Genetics*, 2nd ed. [Boston: Little, Brown, 1965]. Reproduced by permission.)

plain that the virus plays an integral role. Whether the viral DNA or the kappa DNA codes the toxin is also not known at present.

Mate-Killer Infection and Mu Particles

Kappa is not the only infective agent known in *Paramecium*. Another agent is the **mate-killer** infection. Here again, killer cells have visible, bacteriallike particles, called **mu particles**, in the cytoplasm. Preer and his colleagues have named them *Caedobacter conjugatus*. Mate-killers do not release a toxin into the environment, but instead kill their mates during conjugation. One of two unlinked dominant genes, M_1 and M_2 , is required for the presence of mu particles. An interesting phenomenon occurs when a mate-killer becomes homozygous $m_1m_1 m_2m_2$ by autogamy. Although the offspring eventually lose their mu particles, virtually no loss of particles occurs until about the eighth generation, when some offspring lose all their mu. Up to this generation, all the cells maintain a full complement of mu. In the fifteenth generation, only about 7% of the cells still have mu particles.

This phenomenon is explained as the diluting out not of the mu themselves, but of a factor called **metagon**, which is necessary for the maintenance of mu in the cell. Once the cell becomes homozygous recessive, no further metagon production occurs. The verification that metagon is subsequently diluted out is evident in fifteenth-generation cells that still have their mu. We would expect that after fission, one daughter cell would have a metagon and the other would not. What we expect, in fact, happens. The rate of dilution is consistent with an original number of about one thousand metagons per cell. The metagon appears to be

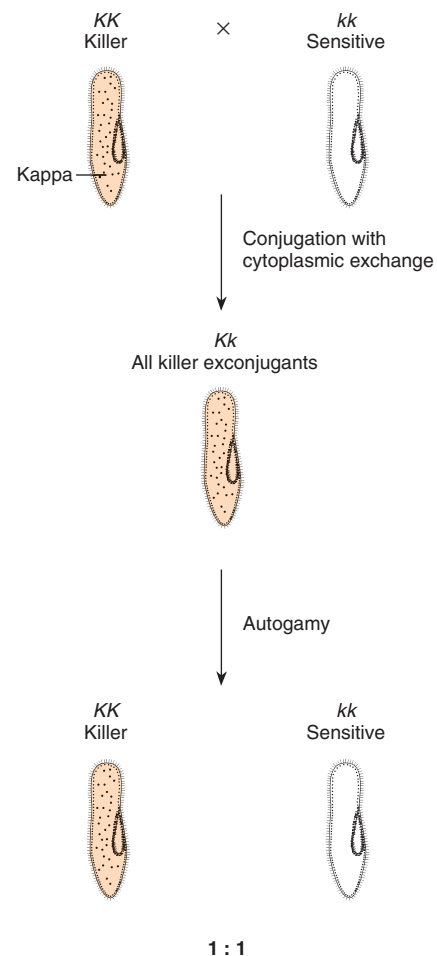


Figure 17.14 Autogamy in a heterozygous (Kk) killer *Paramecium* (formed by conjugation, with cytoplasmic exchange, of a KK killer and a kk sensitive cell). Upon autogamy, the heterozygote has a 50% chance of becoming a homozygous (KK) killer or a homozygous (kk) sensitive cell that loses its kappa particles.



Figure 17.15 Electron micrograph of a sectioned kappa particle (*Caedobacter taeniospiralis*). Phage particles appear as dark inclusions. The plane of the section cuts through a rolled-up *R* body. Magnification 61,200 $\times$. (Reproduced by permission of J. R. Preer, Jr.)

messenger RNA because it is destroyed by RNase. Its protein product is presently unknown.

We thus see several instances of infective particles that interact with the *Paramecium* genome to produce interesting phenotypic results. Similar interactions are known in other organisms—for example, the killer trait in yeast.

Drosophila

Several infective particles mimic patterns of inheritance in insects. In *Drosophila*, we find forms of the **sex-ratio phenotype** in which females produce mostly, if not exclusively, daughters. One form is inherited as a chromosomal gene; another form, however, is not chromosomal. In the nonchromosomal form, females usually produce a few sons. These sons do not pass on the sex-ratio trait, but the daughters of sex-ratio females do. Because the trait persisted even after all the chromosomes had been substituted out of the stock by appropriate crosses, it was proven to be extrachromosomal.

In addition, about half the eggs of a sex-ratio female fail to develop. Cytoplasm can be withdrawn from the undeveloped eggs and used to infect other females. The trait, then, is caused by some cytoplasmic factor that could infect other females and is not passed on by sperm. Detailed cytological examination of the cytoplasm of sex-ratio females has revealed a spirochete (fig. 17.16) that has been isolated and used to infect other female *Drosophila* with the sex-ratio trait; it is, therefore, the causal agent of this phenotype.

Prokaryotic Plasmids

In chapters 7 and 13, we discussed the role of plasmids in the study of prokaryotic genetics and in recombinant DNA work. They are mentioned again here because they represent extrachromosomal genetic systems, primarily



Figure 17.16 Electron micrograph of the spirochete associated with the extrachromosomal sex-ratio trait in *Drosophila*. Magnification 22,700 $\times$. (K. Oishi and D. F. Poulson, "A virus associated with SR-spirochetes of *Drosophila nebulosa*," *Proceedings of The National Academy of Sciences, USA*, 67 [1970]:1565–72. Reproduced by permission of the authors.)

in prokaryotes. The autonomous segments of DNA known as plasmids are, for the most part, known from bacteria, in which they occur as circles of DNA within the host cell (noncircular DNA is soon degraded). When plasmids become integrated into the chromosomes, they become indistinguishable from chromosomal material.

R and Col Plasmids

In addition to the F factor found frequently in bacteria, a variety of other plasmids occur, including the R and Col plasmids. The **R plasmids** carry genes for resistance to various antibiotics, and the **Col plasmids** have genes that are responsible for producing proteins called *colicins*, which are toxic to strains of *E. coli* (fig. 17.17). Plasmids containing genes for Col-like toxins specific for other bacterial species are also known. Col and R plasmids can exist in two states. In one state, the plasmid has

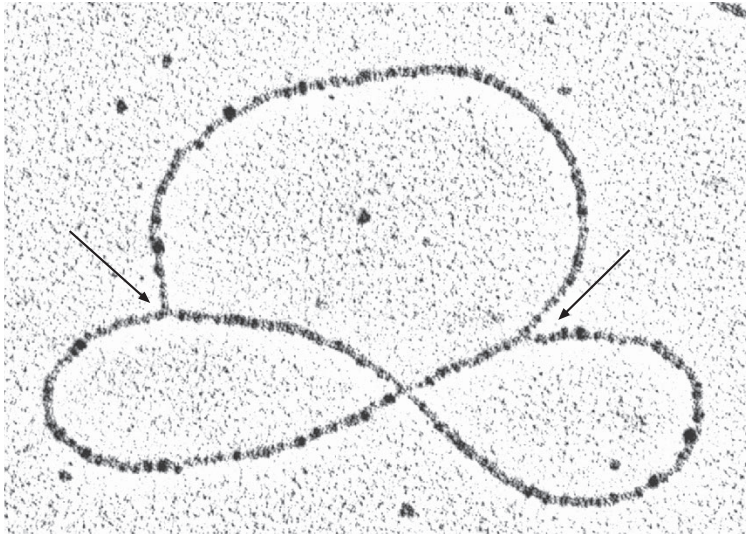


Figure 17.17 Electron micrograph of replication of Col E1 circular plasmid. The arrows mark the branch points of the theta structure. Magnification 90,000 $\times$. (Source: J. I. Tomizawa, Y. Sakakibara, and T. Kakefuda, "Replication of Colicin E1 plasmid DNA in cell extracts: Origin and direction of replication," *Proceedings of The National Academy of Sciences, USA*, 71 [1974]:2260–64.)

a sequence of genes called the **transfer operon (*tra*)**, which makes the plasmids similar to F factors in that they can transfer their genes from one bacterium to the next. In the other state, the plasmids lack this operon and cannot transfer their loci to another cell. Thus, Col and R plasmids are actually made of two parts: the loci for antibiotic resistance or colicin production and the part responsible for infectious transfer. In R plasmids, the infectious transfer part is called the **resistance transfer factor (RTF)**.

The occurrence of resistance plasmids was observed in Japan in the late 1950s, when it was discovered that bacteria were simultaneously acquiring resistance to several antibacterial agents. When cultures of *Shigella*, a dysentery-causing bacterium, were exposed to streptomycin, sulfonamide, chloramphenicol, or tetracycline, the bacteria exhibited resistance not only to the one particular agent they were exposed to but to one or more of the others as well. The plasmid responsible for this multiple resistance was named R222.

The Col plasmids contain loci that produce proteins that are toxic, for various reasons, to strains of bacteria not carrying the plasmids. Colicins attack sensitive bacterial cells at bacterial surface receptors. They have been classified into twenty or more categories according to the types of receptors they attack. Some colicins may enter the cell directly, but others do not. For example, colicin K appears to kill sensitive cells by inhibiting DNA, RNA, and protein synthesis, although not directly entering the cell. Colicin E3, however, acts as an intracellular ribonuclease that cleaves off about fifty nucleotides from the 3' end of the 16S ribosomal RNA within the ribosome. The cleavage inactivates the sensitive cell's ribosomes and is, of course, lethal.

Since many R plasmids, Col plasmids, and F factors, as well as host chromosomes, have insertion sequences (chapter 14), a good deal of exchange occurs among the plasmids, and many are able to integrate into the host chromosome. Although their mobility makes it easier to map and study plasmids, it also poses a human health problem. Resistance to various antibacterial agents is easily transferred among enterobacteria worldwide. This can even occur outside of host organisms (people) where pollution or sewage is found. In addition, resistance found in relatively harmless enterobacteria, such as *E. coli*, can easily pass to more pathogenic bacteria, such as *Shigella* and *Salmonella*. Since we are selecting for resistance every time we use antibacterial drugs, we should not use these drugs indiscriminately. For some time, health workers have been concerned about excessive medical use of antibacterial drugs as well as about the large quantities of antibiotics used in animal feed.

Uncovering Plasmids

How do we know when the phenotype is controlled by a plasmid rather than by the chromosomal genes of a bacterium? Plasmids can be seen with an electron microscope or by density-gradient centrifugation of the cell's DNA. But several less direct lines of evidence also supply the answer. To begin with, multiple aspects of the phenotype (e.g., resistance to several antibacterial agents) change simultaneously, as with plasmid R222. Another clue is that the phenotypic change is infectious: Japanese workers found that with R222, resistant cells converted nonresistant cells. As B. Lewin stated, "Resistance is infectious."

Several other clues point to the presence of a plasmid. In linkage studies, using transduction for example,

plasmid loci show no linkage to host loci; plasmids themselves can be mapped because their loci are linked to each other. Since the plasmid DNA replicates at its own speed, it can miss being incorporated into a daughter cell. Thus, many spontaneous losses of the plasmid occur. And finally, certain treatments—with acridine dyes, for example—have little effect on the replication of the host chromosome, but selectively prevent the plasmid from replicating; thus the plasmid can be eliminated from the cell population. The existence of plasmids in a bacterial population can, therefore, be verified with morphological, physiological, and analytical evidence.

IMPRINTING

Although sex linkage alters inheritance patterns, we do not expect different inheritance patterns, dependent on the parent of origin, from genes located on autosomal chromosomes. That is, the genotype of an offspring should be predicted by its alleles regardless of which parent donated which allele. That understanding has now been shown to be incorrect for a group of genes whose phenotypic effects are determined by the parent that donated a particular allele. This phenomenon is called *imprinting* (or *molecular imprinting* or *parental imprinting*). It falls under the general classification of an **epigenetic effect**, a term that has come to mean an effect due to an environmentally induced change in the genetic material but not causing a change in base pairs. It is a phenomenon of differential expression of the alleles at a locus depending on which parent the gene originated with.

A striking example of imprinting in human beings involves two medical syndromes, both resulting in mental retardation. In Prader-Willi syndrome, affected persons are extremely obese; in Angelman syndrome, those affected are thin and sometimes referred to as “happy puppets,” because they exhibit a happy facial expression and erratic, jerky movements. It turns out that both syndromes are associated with deletions in the long arm of

chromosome 15, in bands 15q11–q13. The effect is seen in an individual arising from a gamete missing a 15q11–q13 region. If the remaining region is of paternal origin, due to a deletion of the maternal gene, the offspring will have Angelman syndrome; if the remaining region is of maternal origin, the offspring will have Prader-Willi syndrome. This unusual situation indicates that the phenotype is dependent on the parent from which the region comes. Recently, this region of chromosome 15 has come under intense scrutiny. In males, from five to seven genes are expressed from this area, and in females, one gene has been identified as the cause of Angelman syndrome, *UBE3A* (E3 ubiquitin protein ligase). It has been hypothesized that there is an **imprinting center (IC)**, a region responsible for the control of imprinting. The imprinting mark is almost certainly DNA methylation, which has the property of turning off gene transcription. Stretches of CG repeats (called **CpG islands**, in which CpG indicates sequential bases on the same strand of DNA rather than a C-G base pair) have been found in these imprinting centers. The imprinting center would be the site of the erasure of past imprinting and the initiation of new imprinting during gametogenesis. Over twenty genes exhibit imprinting, and the epigenetic phenomenon also appears in proteins, with differential acetylation of proteins as the imprinting mark.

The question arises as to how imprinting evolved; that is, what evolutionary advantages come from silencing an allele from one of the parents? Although we don't really know at this point, several hypotheses have been suggested, including competition among maternal and paternal alleles for expression (see chapter 21). For example, the *Igf-2* gene (insulin-like growth factor) places demands on pregnant females to produce larger fetuses. This is advantageous to the father (assuming that the female will have offspring from several fathers), but not the mother. So, the mother's gene is usually methylated and therefore inactive. It is as if the genes are in competition with each other, with the father's gene promoting the formation of a large fetus and the mother's gene promoting the formation of a smaller fetus. Currently, the phenomenon of imprinting is under active study.

S U M M A R Y

STUDY OBJECTIVE 1: To analyze the inheritance patterns of maternal effects 509–510

Patterns of non-Mendelian inheritance fall into two categories: maternal effects and cytoplasmic inheritance. Maternal effects are illustrated by snail-shell coiling. The direction of coiling is determined by the genotype of the

maternal parent, with dextral coiling dominant to sinistral coiling.

STUDY OBJECTIVE 2: To analyze the patterns of cytoplasmic inheritance 511–524

Cytoplasmic inheritance is usually seen in organelles, sym-

bionts, or parasites that have their own genetic material. Chloroplasts and mitochondria have relatively small, circular chromosomes with prokaryotic affinities. An interaction exists between organelles and nuclei; the organelles do not encode all their own proteins and enzymes. Mitochondrial defects can be inherited through nuclear genes or through the mitochondrion itself. A similar pattern is seen in chloroplasts. The processes of cytoplasmic inheritance are exemplified by symbiotic bacteria in *Paramecium*.

Plasmids are autonomous segments of DNA. In prokaryotes, R and Col plasmids, as well as the F factor, have been well studied. Plasmids usually carry an operon for transfer and insertion sequences for attachment to cell chromo-

somes and to each other. Hence, they represent highly mobile segments of genetic material.

STUDY OBJECTIVE 3: To analyze the patterns of imprinting 524

Imprinting is a phenomenon of gene activity affected by the parent of origin. Due to a pattern of gene methylation that differs in male and female parents, a gene may show differential activity depending on the parent from which it came. More than twenty genes exhibiting this epigenetic phenomenon are known.

SOLVED PROBLEMS

PROBLEM 1: What possible phenotypes and genotypes could the female parent of a sinistrally coiled snail have?

Answer: If a snail is sinistrally coiled, its mother must have had the *dd* genotype, since sinistrality is recessive. If the female parent is a recessive homozygote, its mother must have contributed a recessive *d* allele. Therefore its mother (the grandmother) could have had either a *Dd* or *dd* genotype. Its daughter could therefore be either dextrally or sinistrally coiled (respectively). Thus, to answer the question, a sinistrally coiled snail could have had a mother that was either dextrally or sinistrally coiled, but only of the *dd* genotype.

PROBLEM 2: You have just noticed a petite yeast colony growing in a petri plate under aerobic conditions. What type of petite is it?

Answer: The simplest way to determine the nature of the lesion resulting in the petite phenotype is to make a cross of the petite strain with a wild-type strain. After

meiosis, isolate the four products (spores) and allow them to grow separately under normal, aerobic conditions. If the ratio of petite to wild-type is 1:1, the mutation is of a nuclear gene. If progeny are wild-type, the mutation is in the mitochondrial genome and is of the neutral type. If progeny are mostly petites, the mutation is also in the mitochondrial genome, but it is of the suppressive type.

PROBLEM 3: Killer *Paramecium* with the genotype *KK* are mated with *kk* cells under a situation that allows cytoplasmic exchange. If the exconjugants undergo autogamy, what types of progeny would you expect?

Answer: Both exconjugants will be *Kk*, and since cytoplasmic exchange occurred, both cytoplasms will contain kappa. Autogamy will produce either *KK* or *kk* cells. Since at least one *K* gene is needed for the maintenance of kappa, the *kk* cells eventually lose the kappas and become sensitive. Thus, we expect 1/2 sensitive:1/2 killers.

EXERCISES AND PROBLEMS*

DETERMINING NON-MENDELIAN INHERITANCE

1. J. Christian and C. Lemunyan have shown that mice raised under crowded conditions produce two generations with reduced growth rates. What sort of genetic control might exist, and how could this control be demonstrated?
2. Describe the types of evidence that could be gathered to determine whether a trait in *E. coli* is controlled by chromosomal or plasmid genes. (See also CYTOPLASMIC INHERITANCE)

3. The maroon-like (*ma-l*) locus in *Drosophila* is inherited in an X-linked recessive fashion. If you cross a heterozygous female with a maroon-like male, all the progeny are wild-type. If the female progeny from this cross are mated again with maroon-like males, half of the females produce all maroon-like progeny, and the other half produce all wild-type progeny. Explain these results. (See also MATERNAL EFFECTS)

*Answers to selected exercises and problems are on page A-20.

MATERNAL EFFECTS

4. Snail coiling is called a maternal trait. Is it possible that it is caused by an allele at a sex-linked locus?
5. How would you rule out a viral origin for snail-shell coiling?
6. Give the genotypes involved when a sinistral female snail produces dextral offspring. What genotypes could the male parent of the sinistral female have?
7. A dextral snail is self-fertilized and produces only sinistral progeny. What is the probable genotype of this snail and its parents?
8. In corn, male sterility is controlled by a maternal cytoplasmic element. A dominant nuclear gene, Restorer (*Rf*), restores fertility to male sterile lines. If pollen from a homozygous *RfRf* plant is used to pollinate a male sterile plant, what genotypes and phenotypes would you expect in the progeny?

CYTOPLASMIC INHERITANCE

9. What evidence indicates that it is not absolutely essential, in an evolutionary sense, for mitochondria to have genes for specific components of oxidative phosphorylation?
10. How would you determine that a segregative petite mutant in yeast is controlled by a chromosomal gene?
11. What results would you obtain by making all possible pairwise crosses of the three types of yeast petites?
12. An ornamental spider plant has green and white striped leaves. How can you determine whether cytoplasmic inheritance is responsible for the striping and whether there is interaction with an *iojap*-type chromosomal gene?
13. In *Cblamydomonas*, 0.02% of the meiotic products are of the *mt*[−] parental type. How can you use this information in mapping? (Use streptomycin sensitivity, *str*^s, and resistance, *str*^r, as an example.)
14. What similarities do mitochondria and plastids share?
15. What evidence is there that mitochondria and chloroplasts originated from prokaryotes?
16. Individuals from killer and nonkiller strains of *Paramecium* are mixed together. Cytoplasmic exchange occurs during conjugation. Approximately 25% of the exconjugants are sensitive, and the remaining 75% are killers. What are the genotypes of the individuals of the two strains, and what ratios of sensitives and killers would result if the various exconjugants underwent autogamy?
17. What genetic tests could you conduct to show that the mate-killer phenotype in *Paramecium* requires a dominant allele at any one of two loci?
18. Resistant and sensitive strains of *Drosophila melanogaster* differ in their ability to tolerate CO₂—anesthetization with it kills sensitive flies. What genetic experiments would you perform to determine whether the trait is caused by a virus? How would you rule out chromosomal genes?
19. Suppose you have identified a person who has introns in his or her mitochondrial DNA. What would you deduce about the origin of this DNA?
20. A mutation in the mitochondrial genome in people causes blindness. If reciprocal matings between affected and normal individuals occur in a family pedigree, what types of children would you expect from each cross?
21. When chloroplast DNA from *Cblamydomonas* is digested with a particular restriction enzyme and then hybridized with a particular probe, two bands are detected. Some strains (type 1) yield bands of 1.5 and 3.7 kilobases; other strains (type 2) yield bands of 2.5 and 6.0 kilobases. For the following crosses, predict the progeny:
 - a. *mt*⁺, strain 1 × *mt*[−], strain 2
 - b. *mt*⁺, strain 2 × *mt*[−], strain 1
22. What type of asci do you expect if you cross a yeast strain carrying an antibiotic resistance gene in its mitochondria with a strain that has normal (sensitive) mitochondria?
23. In *Paramecium*, the maintenance of kappa particles requires the dominant nuclear gene *K*. A *Kk* killer cell conjugates with a sensitive cell of the same genotype without cytoplasmic exchange. Predict the genotypes and phenotypes that result if each exconjugant then undergoes autogamy.
24. In *Neurospora*, the slow-growing trait *poky* is inherited maternally and is due to an abnormal respiratory protein. A nuclear gene *F* makes *poky* individuals grow faster, even though the protein is still defective. Such strains are called *fast-poky* (*F* is normal *poky*). *Poky* cytoplasm is not altered by *F* in a zygote, and *F* has no effect on normal cytoplasm. What genotypes and phenotypes do you expect if the maternal parent is *fast-poky* and the paternal parent is normal?
25. In corn, two independent, recessive nuclear genes, *japonica* (*j*) and *iojap* (*ij*), produce variegation (green and white striped leaves). Matings between individuals heterozygous for *japonica* always produce 3 green:1 striped individuals regardless of how the cross is performed. The behavior of *iojap* was described in figure 17.9. You have a variegated plant that could be either *jj* or *ijij*. What cross can you make to determine the genotype of this plant, and what results do you expect in the F₁ generation in each case?

26. If *Paramecium* cells heterozygous for both genes involved in the maintenance of the mate-killer trait are forced to undergo autogamy, what phenotypic ratios do you expect?
27. A petite yeast strain is crossed with a wild-type strain. What phenotypic ratio do you expect after meiosis if the petite is
- nuclear?
 - suppressive?
 - neutral?

CRITICAL THINKING QUESTIONS

- When a eukaryotic cell divides, cell organelles such as mitochondria and chloroplasts are distributed to the daughter cells. What mechanisms might exist to ensure an even distribution of these organelles?
- Lamarckian inheritance, the inheritance of acquired characteristics, is generally discounted as a major evolutionary mechanism (chapter 21). (For example,

Lamarck suggested that the long neck of the giraffe came about by as giraffes stretched for food, followed by the inheritance of this longer, stretched neck.) Is the progression of the lysogenic state of *E. coli* from one generation to the next an example of Lamarckian inheritance? Why or why not?

Suggested Readings for chapter 17 are on page B-18.

18

QUANTITATIVE INHERITANCE

STUDY OBJECTIVES

1. To understand the patterns of inheritance of phenotypic traits controlled by many loci 531
2. To investigate the way that geneticists and statisticians describe and analyze normal distributions of phenotypes 535
3. To define and measure heritability, the unit of inheritance of variation in traits controlled by many loci 542

STUDY OUTLINE

Traits Controlled by Many Loci 531

- Two-Locus Control 531
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- Location of Polygenes 533
- Significance of Polygenic Inheritance 534

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Box 18.1 Mapping Quantitative Trait Loci 537

Box 18.2 Human Behavioral Genetics 547



Much human variation is quantitative.

(© Jim Cummins/FPG International.)

When we talked previously of genetic traits, we were usually discussing traits in which variation is controlled by single genes whose inheritance patterns led to simple ratios. However, many traits, including some of economic importance—such as yields of milk, corn, and beef—exhibit what is called *continuous variation*.

Although some variation occurred in height in Mendel's pea plants, all of them could be scored as either tall or dwarf; there was no overlap. Using the same methods that Mendel used, we can look at ear length in corn (fig. 18.1). With Mendel's peas, all of the F_1 were tall. In a cross between corn plants with long and short ears, all of the F_1 plants have ears intermediate in length between the parents. When both pea and corn F_1 plants are self-fertilized, the results are again different. In the F_2 generation, Mendel obtained exactly the same height categories (tall and dwarf) as in the parental generation. Only the ratio was different—3:1.

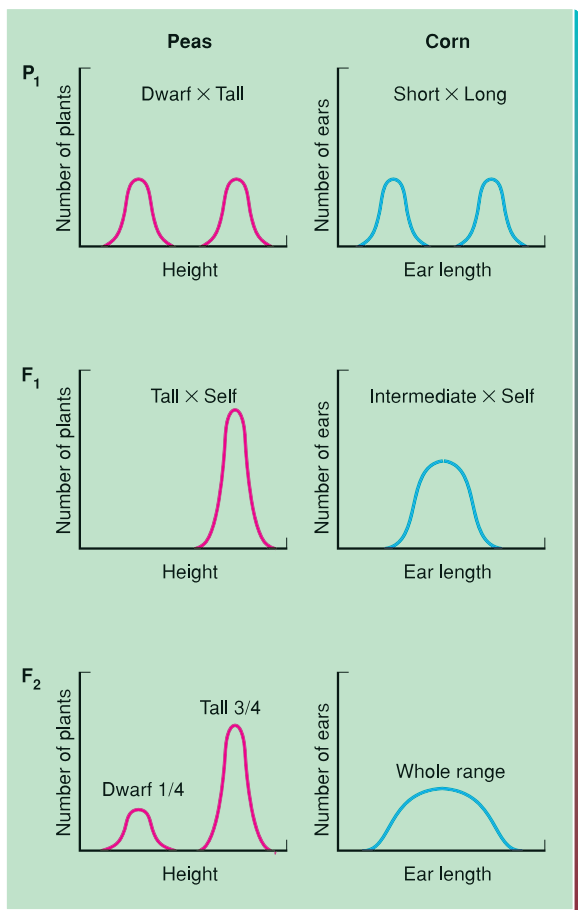


Figure 18.1 Comparison of continuous variation (ear length in corn) with discontinuous variation (height in peas).

In corn, however, ears of every length, from the shortest to the longest, are found in the F_2 ; there are no discrete categories. A genetically controlled trait exhibiting this type of variation is usually controlled by many loci. In this chapter, we study this type of variation by looking at traits controlled by progressively more loci. We then turn to the concept of heritability, which is used as a statistical tool to evaluate the genetic control of traits determined by many loci.

TRAITS CONTROLLED BY MANY LOCI

Let us begin by considering grain color in wheat. When a particular strain of wheat having red grain is crossed with another strain having white grain, all the F_1 plants have kernels intermediate in color. When these plants are self-fertilized, the ratio of kernels in the F_2 is 1 red:2 intermediate:1 white (fig. 18.2). This is inheritance involving one locus with two alleles. The white allele, a , produces no pigment (which results in the background color, white); the red allele, A , produces red pigment. The F_1 heterozygote, Aa , is intermediate (incomplete dominance). When this monohybrid is self-fertilized, the typical 1:2:1 ratio results. (For simplicity, we use dominant-recessive allele designations, A and a . Keep in mind, however, that the heterozygote is intermediate in color.)

Two-Locus Control

Now let us examine the same kind of cross using two other stocks of wheat with red and white kernels. Here, when the resulting intermediate (medium-red) F_1 are self-fertilized, five color classes of kernels emerge in a ratio of 1 dark red:4 medium dark red:6 medium red:4 light red:1 white (fig. 18.3). The offspring ratio, in sixteenths, comes from the self-fertilization of a dihybrid in which the two loci are unlinked. In this case, both loci affect the same trait in the same way. In figure 18.3, each capital



Figure 18.2 Cross involving the grain color of wheat in which one locus is segregating.

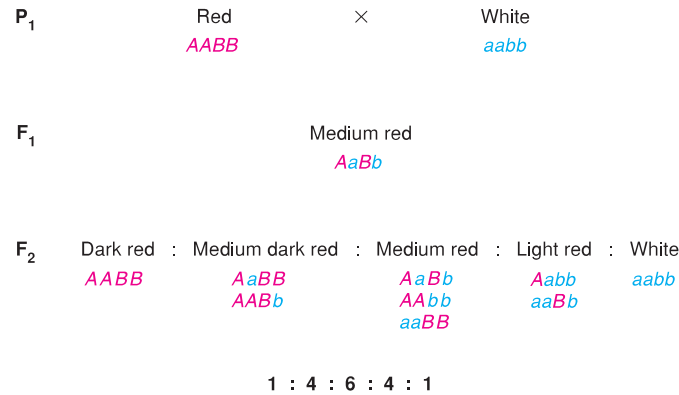


Figure 18.3 Another cross involving wheat grain color in which two loci are segregating.

letter represents an allele that produces one unit of color, and each lowercase letter represents an allele that produces no color. Thus, the genotype $AaBb$ has two units of color, as do the genotypes $AAbb$ and $aaBB$. All produce the same intermediate grain color. Recall from chapter 2 that a cross such as this produces nine genotypes in a ratio of 1:2:1:2:4:2:1:1. If these classes are grouped according to numbers of color-producing alleles, as shown in figure 18.3, the 1:4:6:4:1 ratio appears. This ratio is a product of a binomial expansion.

Three-Locus Control

In yet another cross of this nature, H. Nilsson-Ehle in 1909 crossed two wheat strains, one with red and the other with white grain, that yielded plants in the F_1 generation with grain of intermediate color. When these plants were self-fertilized, at least seven color classes, from red to white, were distinguishable in a ratio of 1:6:15:20:15:6:1 (fig. 18.4). This result is explained by assuming that three loci are assorting independently, each with two alleles, so that one allele produces a unit of red color and the other allele does not. We then see seven color classes, from red to white, in the 1:6:15:20:15:6:1 ratio. This ratio is in sixty-fourths, directly from the 8×8 (trihybrid) Punnett square, and comes from grouping genotypes in accordance with the number of color-producing alleles they contain. Again, the ratio is one that is generated in a binomial distribution.

Multilocus Control

From here, we need not go on to an example with four loci, then five, and so on. We have enough information to draw generalities. It should not be hard to see how discrete loci can generate a continuous distribution (fig. 18.5). Theoretically, it should be possible to distinguish

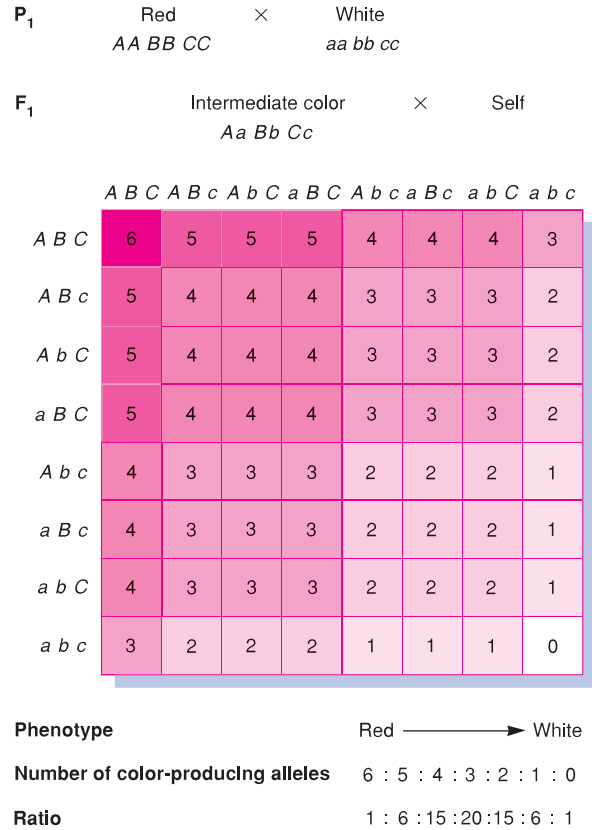


Figure 18.4 One of Nilsson-Ehle's crosses involving three loci controlling wheat grain color. Within the Punnett square, only the number of color-producing alleles is shown in each box to emphasize color production.

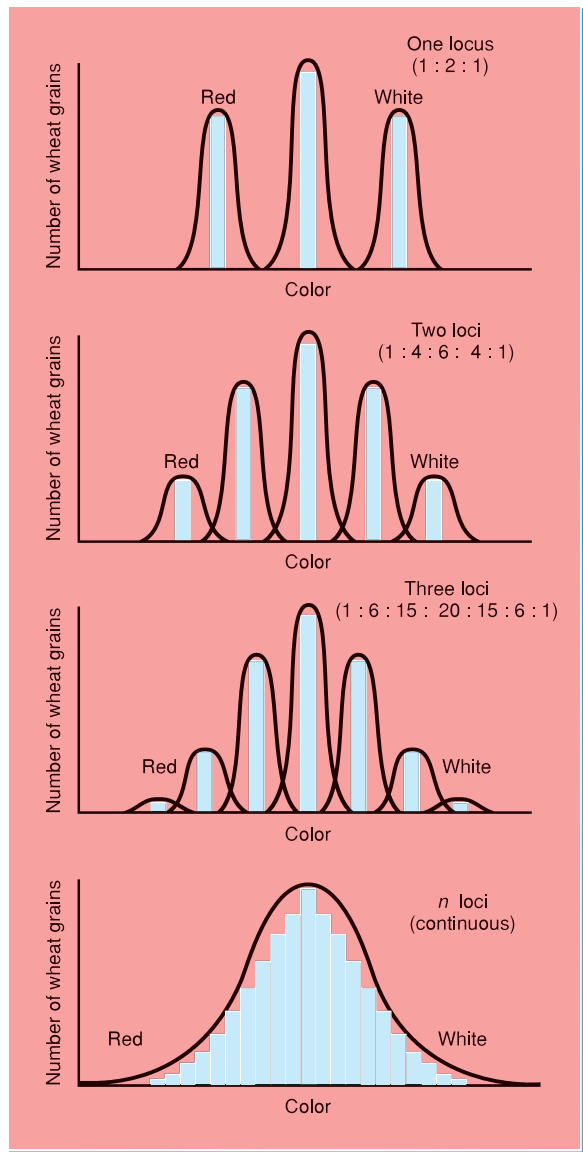


Figure 18.5 The change in shape of the distribution as increasing numbers of independent loci control grain color in wheat. If each locus is segregating two alleles, with each allele affecting the same trait, eventually a continuous distribution will be generated in the F_2 generation.

different color classes down to the level of the eye's ability to perceive differences in wavelengths of light. In fact, we rapidly lose the ability to assign unique color classes to genotypes because the variation within each genotype soon causes the phenotypes to overlap. For example, with three loci, a color somewhat lighter than medium dark red may belong to the medium-dark-red class with

three color alleles, or it may belong to the medium-red class with only two color alleles (fig. 18.5).

The variation within each genotype is due to the environment—that is, two organisms with the same genotype may not necessarily be identical in color because nutrition, physiological state, and many other variables influence the phenotype. Figure 18.6 shows that it is possible for the environment to obscure genotypes even in a one-locus, two-allele system. That is, a height of 17 cm could result in the F_2 from either the aa or Aa genotype in the figure when there is excessive variation (fig. 18.6, column 3). In the other two cases in figure 18.6, there would be virtually no organisms 17 cm tall. Systems such as those we are considering, in which each allele contributes a small unit to the phenotype, are easily influenced by the environment, with the result that the distribution of phenotypes approaches the bell-shaped curve seen at the bottom of figure 18.5.

Thus, phenotypes determined by multiple loci with alleles that contribute dosages to the phenotype will approach a continuous distribution. This type of trait is said to exhibit **continuous, quantitative, or metrical variation**. The inheritance pattern is **polygenic** or **quantitative**. The system is termed an **additive model** because each allele adds a certain amount to the phenotype.

From the three wheat examples just discussed, we can generalize to systems with more than three polygenic loci, each segregating two alleles. From table 18.1, we can predict the distribution of genotypes and phenotypes expected from an additive model with any number of unlinked loci segregating two alleles each. This table is useful when we seek to estimate how many loci are producing a quantitative trait, assuming it is possible to distinguish the various phenotypic classes. For example, when a strain of heavy mice was crossed with a lighter strain, the F_1 were of intermediate weight. When these F_1 were interbred, a continuous distribution of adult weights appeared in the F_2 generation. Since only about one mouse in 250 was as heavy as the heavy parent stock, we could guess that if an additive model holds, then four loci are segregating. This is because we expect $1/(4)^n$ to be as extreme as either parent; one in 250 is roughly $1/(4)^4 = 1/256$.

Location of Polygenes

The fact that traits with continuous variation can be controlled by genes dispersed over the whole genome was shown by James Crow, who studied DDT resistance in *Drosophila*. A DDT-resistant strain of flies was created by growing them on increasing concentrations of the insecticide. Crow then systematically tested each chromosome for the amount of resistance it conferred. Susceptible flies were mated with resistant flies, and the sons from this cross were backcrossed. Offspring were

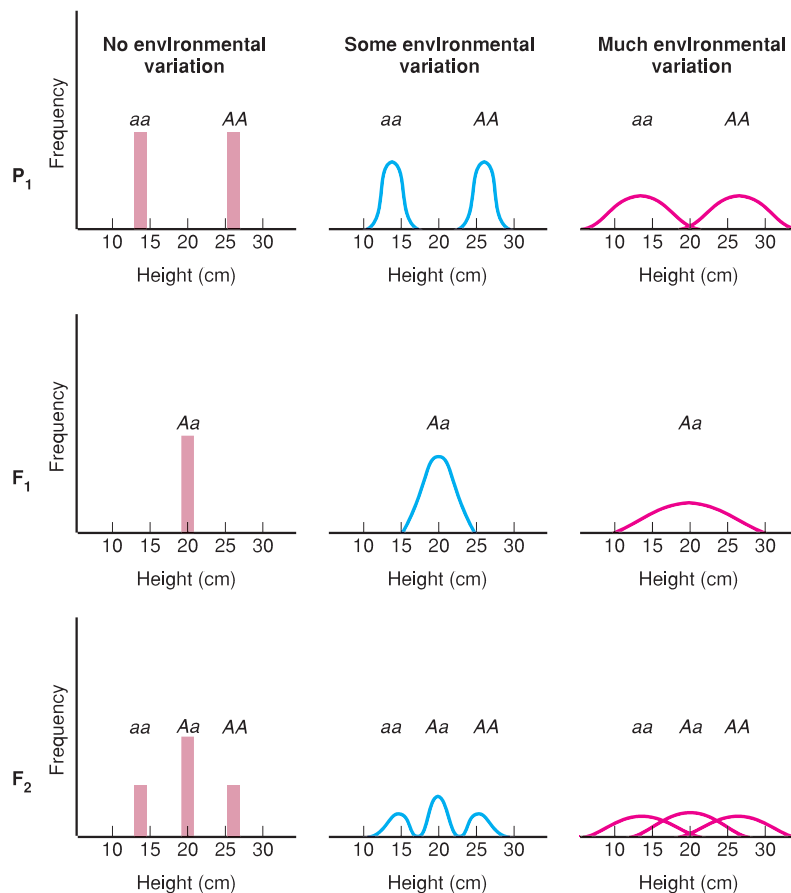


Figure 18.6 Influence of environment on phenotypic distributions.



James F. Crow (1916–).
(Courtesy of Dr. James F. Crow.)

then scored for the particular resistant chromosomes they contained (each chromosome had a visible marker) and were tested for their resistance to DDT. Sons were used in the backcross because there is no crossing over in males. Therefore, the sons would pass resistant and susceptible chromosomes on intact. Crow's results are shown in figure 18.7. As you can see, each chromosome has the potential to increase the fly's resistance to DDT. Thus, each chromosome contains

loci (polygenes) that contribute to the phenotype of this additive trait (box 18.1).

Significance of Polygenic Inheritance

The concept of additive traits is of great importance to genetic theory because it demonstrates that Mendelian rules of inheritance can explain traits that have a continuous distribution—that is, Mendel's rules for discrete characteristics also hold for quantitative traits. Additive traits are also of practical interest. Many agricultural products, both plant and animal, exhibit polygenic inheritance, including milk production and fruit and vegetable yield. In addition, many human traits, such as height and IQ, appear to be polygenic, although with substantial environmental components.

Historically, the study of quantitative traits began before the rediscovery of Mendel's work at the turn of the century. In fact, biologists in the early part of this century debated as to whether the "Mendelians" were correct or whether the "biometricians" were correct in regard to

Table 18.1 Generalities from an Additive Model of Polygenic Inheritance

	One Locus	Two Loci	Three Loci	n Loci
Number of gamete types produced by an F_1 multihybrid	2 (A, a)	4 (AB, Ab, aB, ab)	8 ($ABC, ABc, AbC, Abc, aBC, aBc, abC, abc$)	2^n
Number of different F_2 genotypes	3 (AA, Aa, aa)	9 ($AABB, AABb, AAbb, AaBB, AaBb, Aabb, aaBB, aaBb, aabb$)	27 ($AABBCC, AABBCc, AABbCC, AABbcc, AaBBCC, AaBBCc, AaBbCC, AaBbcc, AabbCC, AabbCc, Aabbcc, aaBBCC, aaBBcC, aaBbCC, aaBbcc, aaBbCC, aaBbCc, aaBbcc, aabbCC, aabbCc, aabbcc$)	3^n
Number of different F_2 phenotypes	3	5	7	$2n + 1$
Number of F_2 as extreme as one parent or the other	$1/4$ (AA or aa)	$1/16$ ($AABB$ or $aabb$)	$1/64$ ($AABBCC$ or $aabbcc$)	$1/4^n$
Distribution pattern of F_2 phenotypes	1:2:1	1:4:6:4:1	1:6:15:20:15:6:1	$(A + a)^{2n}$

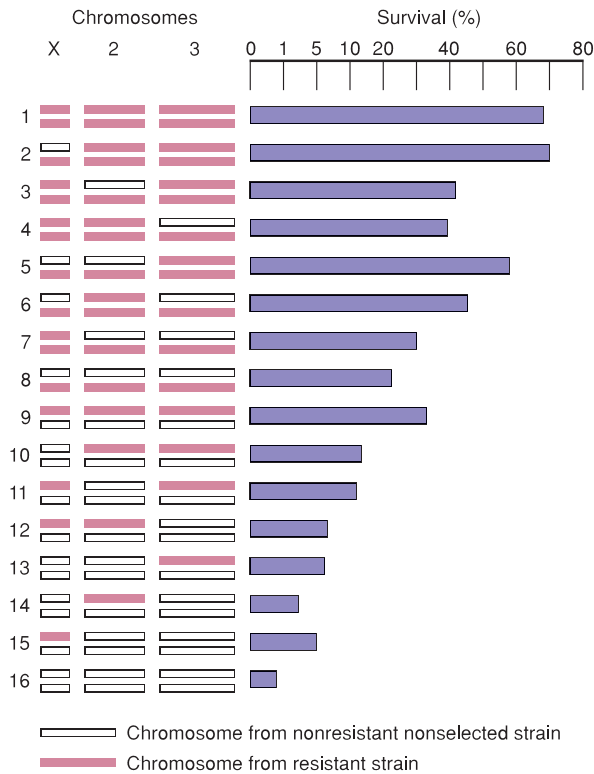


Figure 18.7 Survival of *Drosophila* in the presence of DDT. Numbers and arrangements of DDT-resistant and susceptible chromosomes vary. (Reproduced with permission from the *Annual Review of Entomology*, Volume 2, © 1957 by Annual Reviews, Inc.)

the rules of inheritance. Biometricians used statistical techniques to study traits characterized by continuous variation and claimed that single discrete genes were not responsible for the observed inheritance patterns. They were interested in evolutionarily important facets of the phenotype—traits that can change slowly over time. Mendelians claimed that the phenotype was controlled by discrete “genes.” Eventually the Mendelians were proven correct, but the biometricians’ tools were the only ones suitable for studying quantitative traits.

The biometric school was founded by F. Galton and K. Pearson, who showed that many quantitative traits, such as height, were inherited. They invented the statistical tools of correlation and regression analysis in order to study the inheritance of traits that fall into smooth distributions.

POPULATION STATISTICS

A distribution (see fig. 18.5, *bottom*) can be described in several ways. One is the formula for the shape of the curve formed by the frequencies within the distribution. A more functional description of a distribution starts by defining its center, or **mean** (fig. 18.8). As we can see from the figure, the mean is not itself enough to describe the distribution. Variation about this mean determines the actual shape of the curve. (We confine our discussion to symmetrical, bell-shaped curves called **normal distributions**. Many distributions approach a normal distribution.)

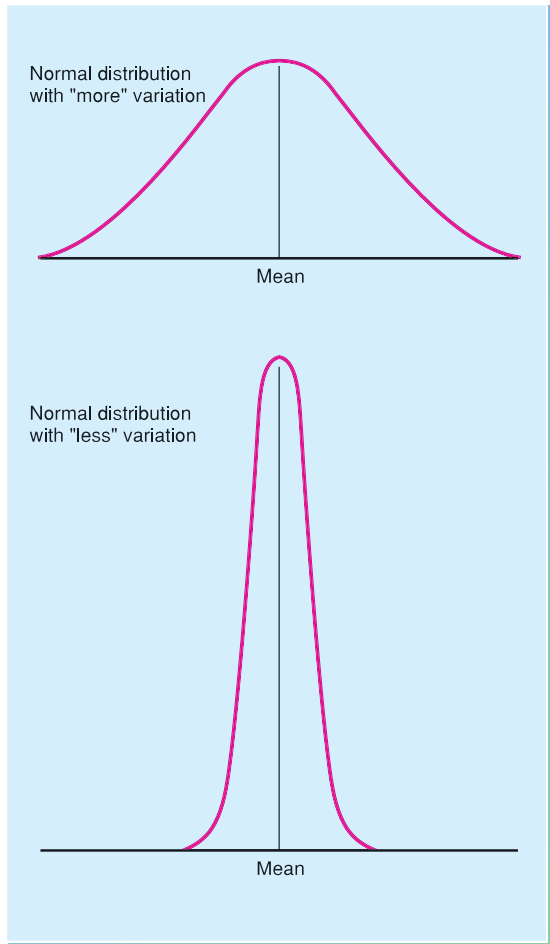


Figure 18.8 Two normal distributions (bell-shaped curves) with the same mean.

Mean, Variance, and Standard Deviation

The mean of a set of numbers is the arithmetic average of the numbers and is defined as

$$\bar{x} = \sum x / n \quad (18.1)$$

in which

$\bar{x}$ = the mean

$\sum x$ = the summation of all values

n = the number of values summed

In table 18.2, the mean is calculated for the distribution shown in figure 18.9. The variation about the mean is calculated as the average squared deviation from the mean:

$$s^2 = V = \frac{\sum (x - \bar{x})^2}{n - 1} \quad (18.2)$$

Table 18.2 Hypothetical Data Set of Ear Lengths (x) Obtained When Corn Is Grown from an Ear of Length 11 cm

x	$(x - \bar{x})$	$(x - \bar{x})^2$
7	-4.12	16.97
8	-3.12	9.73
9	-2.12	4.49
9	-2.12	4.49
10	-1.12	1.25
10	-1.12	1.25
10	-1.12	1.25
10	-1.12	1.25
10	-1.12	1.25
10	-1.12	1.25
11	-0.12	0.01
11	-0.12	0.01
11	-0.12	0.01
11	-0.12	0.01
11	-0.12	0.01
11	-0.12	0.01
11	-0.12	0.01
12	0.88	0.77
12	0.88	0.77
12	0.88	0.77
13	1.88	3.53
13	1.88	3.53
13	1.88	3.53
14	2.88	8.29
14	2.88	8.29
16	4.88	23.81
$\sum x = 278$		$\sum (x - \bar{x})^2 = 96.53$
$n = 25$		
$\bar{x} = \frac{\sum x}{n} = \frac{278}{25} = 11.12$		
$s^2 = V = \frac{\sum (x - \bar{x})^2}{n - 1} = \frac{96.53}{24} = 4.02$		
$s = \sqrt{s^2} = \sqrt{4.02} = 2.0$		

This value (V or s^2) is called the **variance**. Observe that the flatter the distribution is, the greater the variance will be.

The variance is one of the simplest measures we can calculate of variation about the mean. You might wonder why we simply don't calculate an average deviation from the mean rather than an average squared deviation. For example, we could calculate a measure of variation as

$$\frac{\sum (x - \bar{x})}{n - 1}$$

BOX 18.1

Mapping the location of a standard locus is conceptually relatively easy, as we saw in the mapping of the fruit fly genome. We look for associations of phenotypes that don't segregate with simple Mendelian ratios and then map the distance between loci by the proportion of recombinant offspring. However, with quantitative loci we have a problem: We can't do simple mapping because genes contributing to the phenotype are often located across the genome. Thus, a particular continuous phenotype will be controlled by loci linked to numerous other loci, many unlinked to each other. However, with the advent of molecular techniques, it has become feasible to map polygenes.

Experimental
MethodsMapping Quantitative
Trait Loci

In chapter 13, we showed how a locus can be discovered and mapped in the human genome (and other genomes) by association with molecular markers. That is, as the Human Genome Project has progressed, we have discovered restriction fragment length polymorphisms (RFLPs) that mark every region of all the chromosomes. Conceptually, there is not

much difference between finding the gene for cystic fibrosis and finding the gene that contributes to a quantitative trait.

In theory, we look at a population of organisms and note various RFLPs or other molecular markers. We then look for the association of a marker and a quantitative trait. If an association exists, we can gain confidence that one or more of the polygenes controlling the trait is located in the chromosomal region near the marker. The closer the polygenes are to the markers, the more reliable our estimates are, because they depend on few crossovers taking place in that population. With many crossovers, the association between a particular marker and a particular effect diminishes. Since we don't know immediately from this method whether the region of interest has one or more polygenic loci, a new term has been coined to indicate that ambiguity. Instead of talking about polygenic loci directly, we talk of **quantitative trait loci**.

For example, consider the search for polygenes associated with geotactic behavior in fruit flies (see fig. 18.13). As selection proceeds, flies in the high and low lines diverge in their geotactic scores. The lines are also becoming homozygous for many loci since only a few parents are chosen to begin each new generation (see chapter 19). Thus, quantitative trait loci can become associated with different molecular markers in each line (fig. 1). If flies from each line are crossed, heterozygotes will be produced of both the markers and the quantitative trait loci. If there is very little crossing over between the two, three classes of F_2 offspring will be produced. These offspring can be grouped according to their RFLPs and then tested for their geotactic scores. If, as figure 1 suggests, a relationship exists between a locus influencing geotactic score and an RFLP,

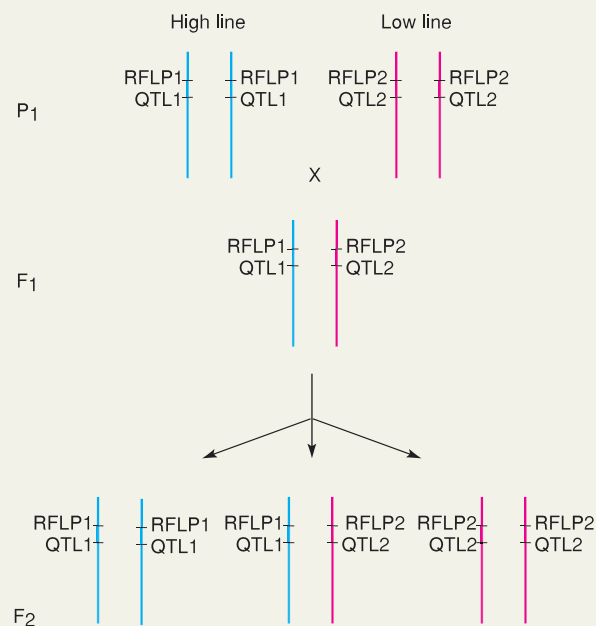


Figure 1 Mapping a quantitative trait locus (QTL) to a particular chromosomal region using a restriction fragment length polymorphism (RFLP) marker. A hypothetical chromosome pair in the fruit fly is shown. The flies have been selected for a geotactic score; QTL1 is the locus in the high line, and QTL2 is the locus in the low line. RFLP1 is homozygous in the high line and RFLP2 is homozygous in the low line.

continued

BOX 18.1 CONTINUED

then the three groups will have different geotactic scores. We can then conclude that the region of the chromosome that contains the RFLP also contains a quantitative trait locus. Finding the right RFLP is, of course, a tedious and time-consuming task.

In a recent summary of the literature, Steven Tanksley reported that numerous quantitative trait loci have been mapped in tomatoes, corn, and other organisms. For example, five quantitative trait loci have been mapped in tomatoes for fruit growth, and eleven quantitative trait loci have been mapped in corn for plant height. Enough data seem to be present to recognize an interesting gener-

ality. That is, our definition of an additive model may need to be rethought because it appears that in almost every case studied so far, one or more of the quantitative trait loci account for a major portion of the phenotype, whereas most of the loci had very small effects. Thus, the additive model that assumes that all polygenes contribute equally to the phenotype may be wrong. However, additive models that allow different loci to contribute different degrees to the phenotype are still supported.

Also of value from locating quantitative trait loci is a new ability to estimate the number of loci affecting a quantitative trait. In this chapter, we

use an estimate of extreme F_2 offspring to estimate the number of polygenes. There are other methods, including sophisticated statistical methods, that we will not develop here. Mapping quantitative trait loci gives us a third method, that is, simply counting the number of quantitative trait loci mapped.

As the methods of mapping quantitative trait loci have been developed, they have also been refined. High-resolution techniques under development will help us determine whether quantitative trait loci are, in fact, individual polygenes or clusters of polygenes.

(We will get to why we use $n - 1$ rather than n in the denominator in a moment.) Note, however, that the above measure is zero. By the definition of the mean, the absolute value of the sum of deviations above it is equal to the absolute value of the sum of deviations below it—one is negative and the other is positive. However, by squaring each deviation, as in equation 18.2, we create a relatively simple index—the variance—which is not zero and has useful properties related to the normal distribution.

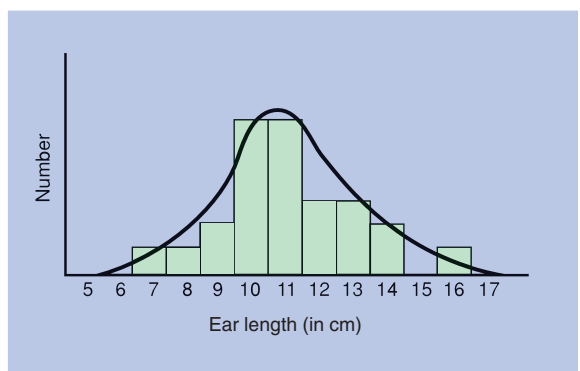


Figure 18.9 Normal distribution of ear lengths in corn. Data are given in table 18.2.

The ear lengths measured in table 18.2 are a sample of all ear lengths in the theoretically infinite population of ears in that variety of corn. Statisticians call sample values **statistics** (and use letters from the Roman alphabet to represent them), whereas they call population values **parameters** (and use Greek letters for them). The sample value is an estimate of the true value for the population. Thus, in the variance formula (equation 18.2), the sample value, V or s^2 , is an estimate of the population variance, σ^2 . When sample values are used to estimate parameters, one degree of freedom is lost for each parameter estimated. To determine the sample variance, we divide not by the sample size, but by the degrees of freedom ($n - 1$ in this case, as defined in chapter 4). The variance for the entire population (assuming we know the population mean, μ , and all the data values) would be calculated by dividing by n . The sample variance is calculated in table 18.2.

The variance has several interesting properties, not the least of which is the fact that it is additive. That is, if we can determine how much a given variable contributes to the total variance, we can subtract that amount of variance from the total, and the remainder is caused by whatever other variables (and their interactions) affect the trait. This property makes the variance extremely important in quantitative genetic theory.

The **standard deviation** is also a measure of variation of a distribution. It is the square root of the variance:

$$s = \sqrt{V} \quad (18.3)$$

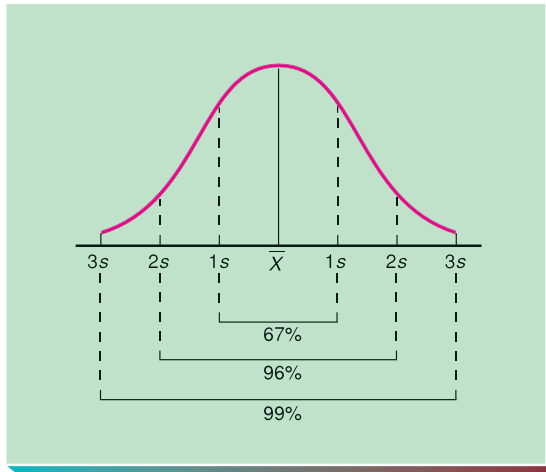


Figure 18.10 Area under the bell-shaped curve. The abscissa is in units of standard deviation (s) around the mean ($\bar{x}$).

In a normal distribution, approximately 67% of the area of the curve lies within one standard deviation on either side of the mean, 96% lies within two standard deviations, and 99% lies within three standard deviations (fig. 18.10). Thus, for the data in table 18.2, about two-thirds of the population would have ear lengths between 9.12 and 13.12 cm (mean $\pm$ standard deviation).

One final measure of variation about the mean is the **standard error of the mean (SE)**:

$$SE = s/\sqrt{n}$$

The standard error (of the mean) is the standard deviation about the mean of a distribution of sample means. In other words, if we repeated the experiment many times, each time we would generate a mean value. We could then use these mean values as our data points. We would expect the variation among a population of means to be less than among individual values, and it is. Data are often summarized as “the mean $\pm$ SE.” In our example of table 18.2, $SE = 2.0/\sqrt{25} = 2.0/5.0 = 0.4$. We can summarize the data set of table 18.2 as 11.1 ± 0.4 (mean $\pm$ SE).

Covariance, Correlation, and Regression

It is often desirable in genetic studies to know whether a relationship exists between two given characteristics in a series of individuals. For example, is there a relationship between height of a plant and its weight, or between scholastic aptitude and grades, or between a phenotypic measure in parents and their offspring? If one increases, does the other also? An example appears in table 18.3; the same data set is graphed in figure 18.11, in what is referred to as a scatter plot. A relation does appear between

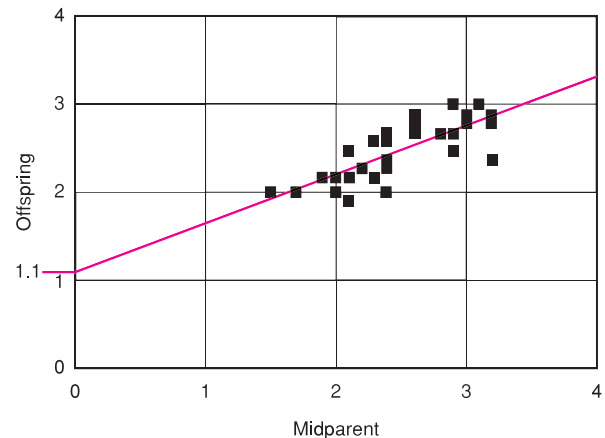


Figure 18.11 The relationship between two variables, parental and offspring wing length in fruit flies, measured in millimeters. *Midparent* refers to the average wing length of the two parents. The line is the statistical regression line. (Source: Data from D. S. Falconer, *Introduction to Quantitative Genetics*, 2d ed. [London: Longman, 1981].)

the two variables. With increasing wing length in midparent (the average of the two parents: x-axis), there is an increase in offspring wing length (y-axis). We can determine how closely the two variables are related by calculating a **correlation coefficient**—an index that goes from -1.0 to $+1.0$, depending on the degree of relationship between the variables. If there is no relation (if the variables are independent), then the correlation coefficient will be zero. If there is perfect correlation, where an increase in one variable is associated with a proportional increase in the other, the coefficient will be $+1.0$. If an increase in one is associated with a proportional decrease in the other, the coefficient will be -1.0 (fig. 18.12). The formula for the correlation coefficient (r) is

$$r = \frac{\text{covariance of } x \text{ and } y}{s_x \cdot s_y} \quad (18.4)$$

where s_x and s_y are the standard deviations of x and y , respectively.

To calculate the correlation coefficient, we need to define and calculate the **covariance** of the two variables, $\text{cov}(x, y)$. The covariance is analogous to the variance, but it involves the simultaneous deviations from the means of both the x and y variables:

$$\text{cov}(x, y) = \frac{\sum(x - \bar{x})(y - \bar{y})}{n - 1} \quad (18.5)$$

The analogy between variance and covariance can be seen by comparing equations 18.5 and 18.2. The variances, standard deviations, and covariance are calculated

in table 18.3, in which the correlation coefficient, r , is 0.78. (There are computational formulas available that substantially cut down on the difficulty of calculating these statistics. If a computer or calculator is used, only the individual data points need to be entered—most computers and many calculators can be programmed to do all the computations.)

Many experiments deal with a situation in which we assume that one variable is dependent on the other (in a cause-and-effect relationship). For example, we may ask, what is the relationship of DDT resistance in *Drosophila* to an increased number of DDT-resistant alleles? With more of these alleles (see fig. 18.7), the DDT resistance of the flies should increase. Number of DDT-resistant alleles is the independent variable, and resistance of the flies is the dependent variable. That is, a fly's resistance is dependent on the number of DDT-resistant alleles it has,

Table 18.3 The Relationship Between Two Variables, x and y (x = the midparent—average of the two parents—in wing length in fruit flies in millimeters; y = the offspring measurement)

x	y	x	y	x	y	x	y
1.5	2	2.2	2.3	2.4	2.7	2.9	2.7
1.7	2	2.3	2.2	2.4	2.7	2.9	2.7
1.9	2.2	2.3	2.6	2.6	2.7	2.9	3
2	2	2.4	2	2.6	2.7	3	2.8
2	2.2	2.4	2.3	2.6	2.8	3	2.8
2	2.2	2.4	2.4	2.6	2.9	3	2.9
2.1	1.9	2.4	2.6	2.8	2.7	3.1	3
2.1	2.2	2.4	2.6	2.8	2.7	3.2	2.4
2.1	2.5	2.4	2.6	2.9	2.5	3.2	2.8
						3.2	2.9
$\Sigma x = 92.7$		$n = 37$		$\Sigma y = 93.2$			
$\bar{x} = \frac{\Sigma x}{n} = 2.51$				$\bar{y} = \frac{\Sigma y}{n} = 2.52$			
$s_x^2 = \frac{\Sigma(x - \bar{x})^2}{n - 1} = 0.19$				$s_y^2 = \frac{\Sigma(y - \bar{y})^2}{n - 1} = 0.10$			
$s_x = \sqrt{s_x^2} = 0.44$				$s_y = \sqrt{s_y^2} = 0.32$			
$\text{cov}(x, y) = \frac{\Sigma(x - \bar{x})(y - \bar{y})}{n - 1} = 0.11$							
$r = \frac{\text{cov}(x, y)}{s_x s_y} = \frac{0.11}{(0.44)(0.32)} = 0.78$							

Source: Data from D. S. Falconer, *Introduction to Quantitative Genetics*, 2d ed. (London: Longman, 1981).

Note: Data are graphed in figure 18.11.

not the other way around. Going back to figure 18.11, we could make the assumption that offspring wing length is dependent on parental wing length. If this were so, a technique called *regression analysis* could be used. This analysis allows us to predict an offspring's wing length (y variable) given a particular midparental wing length (x variable). (It is important to note that regression analysis assumes a cause-and-effect relationship, whereas correlation analysis does not.)

The formula for the straight-line relationship (regression line) between the two variables is $y = a + bx$, where b is the slope of the line (change in y divided by change in x , or $\Delta y / \Delta x$) and a is the y -intercept of the line (see fig. 18.11). To define any line, we need only to calculate the slope, b , and the y intercept, a :

$$b = \text{cov}(x, y) / s_x^2 \quad (18.6)$$

$$a = \bar{y} - b\bar{x} \quad (18.7)$$

Thus equipped, if a cause-and-effect relationship does exist between the two variables, we can predict a y value given any x value. We can either use the formula $y = a + bx$ or graph the regression line and directly determine the y value for any x value. We now continue our examination of the genetics of quantitative traits.

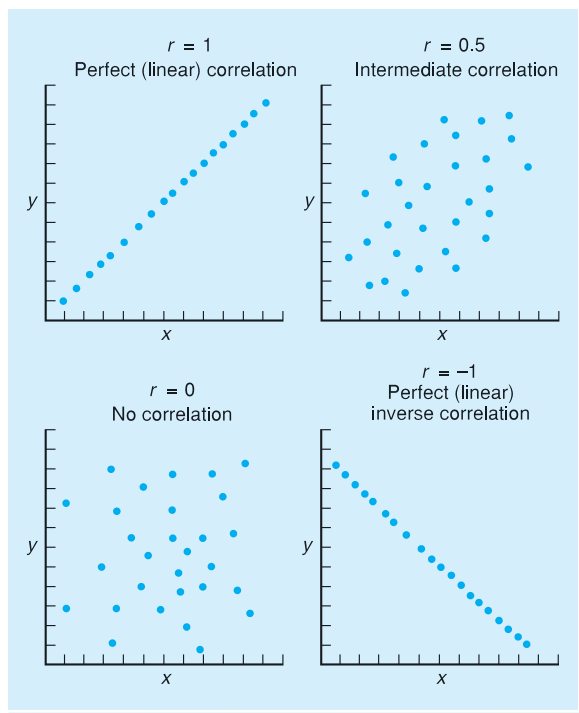


Figure 18.12 Plots showing varying degrees of correlation within data sets.

Table 18.4 Johannsen's Findings of Relationship Between Bean Weights of Parents and Their Progeny

Weight of Parent Beans	Weight of Progeny Beans (centigrams)																n	Mean $\pm$ SE
	15	20	25	30	35	40	45	50	55	60	65	70	75	80	85	90		
65-75				2	3	16	37	71	104	105	75	45	19	12	3	2	494	58.47 $\pm$ 0.43
55-65			1	9	14	51	79	103	127	102	66	34	12	6	5		609	54.37 $\pm$ 0.41
45-55			4	20	37	101	204	287	234	120	76	34	17	3	1		1,138	51.45 $\pm$ 0.27
35-45	5	6	11	36	139	278	498	584	372	213	69	20	4	3			2,238	48.62 $\pm$ 0.18
25-35		2	13	37	58	133	189	195	115	71	20	2					835	46.83 $\pm$ 0.30
15-25			1	3	12	29	61	38	25	11							180	46.53 $\pm$ 0.52
Totals	5	8	30	107	263	608	1,068	1,278	977	622	306	135	52	24	9	2	5,491	50.39 $\pm$ 0.13

POLYGENIC INHERITANCE IN BEANS

In 1909, W. Johannsen, who studied seed weight in the dwarf bean plant (*Phaseolus vulgaris*), demonstrated that polygenic traits are controlled by many genes. The parent population was made up of seeds (beans) with a continuous distribution of weights. Johannsen divided this parental group into classes according to weight, planted them, self-fertilized the plants that grew, and weighed the F_1 beans. He found that the parents with the heaviest beans produced the progeny with the heaviest beans, and the parents with the lightest beans produced the progeny with the lightest beans (table 18.4). There was a significant correlation coefficient between parent and progeny bean weight ($r = 0.34 \pm 0.01$). He continued this work by beginning nineteen lines (populations) with beans from various points on the original distribution and selfing each successive generation for the next several years. After a few generations, the means and variances stabilized within each line. That is, when Johannsen chose, within each line, parent plants with heavier-than-average or lighter-than-average seeds, the offspring had the parental mean with the parental variance for seed size. For example, in one line, plants with both the lightest average bean weights (24 centigrams) and plants with the heaviest average bean weights (47 cg) produced offspring with average bean weights of 37 cg. By selfing the plants each generation, Johannsen had made them more and more homozygous, thus lowering the number of segregating polygenes. Therefore, the lines became homozygous for certain of the polygenes (different in each line), and any variation in bean weight was then caused only by the environment. Johannsen thus showed that quantitative traits were under the control of many segregating loci.

SELECTION EXPERIMENTS

Selection experiments are done for several reasons. Plant and animal breeders select the most desirable individuals as parents in order to improve their stock. Population geneticists select specific characteristics for study in order to understand the nature of quantitative genetic control.

For example, *Drosophila* were tested in a fifteen-choice maze for geotactic response (fig. 18.13). The maze was on its side, so at every intersection, a fly had to make a choice between going up or going down. The flies with the highest scores were chosen as parents for the "high" line (positive geotaxis; favored downward direction), and the flies with the lowest score were chosen as parents for the "low" line (negative geotaxis; favored upward direction). The same selection was made for each generation. As time progressed, the two lines diverged quite significantly. This tells us that there is a large genetic component to the response; the experimenters are successfully amassing more of the "downward" alleles in the high line and more of the "upward" alleles in the low line. Several other points emerge from this graph. First, the high and low responses are slightly different, or asymmetrical. The high line responded more quickly, leveled out more quickly, and tended toward the original state more slowly after selection was relaxed. (The relaxation of selection occurred when the parents were a random sample of the adults rather than the extremes for geotactic scores.) The low line responded more slowly and erratically. In addition, the low line returned toward the original state more quickly when selection was relaxed.

The nature of these responses (fig. 18.13) indicates that the high line became more homozygous than the low line. This is shown by the former's response when selection is relaxed: It has exhausted a good deal of its variability for the polygenes responsible for geotaxis. The low line,

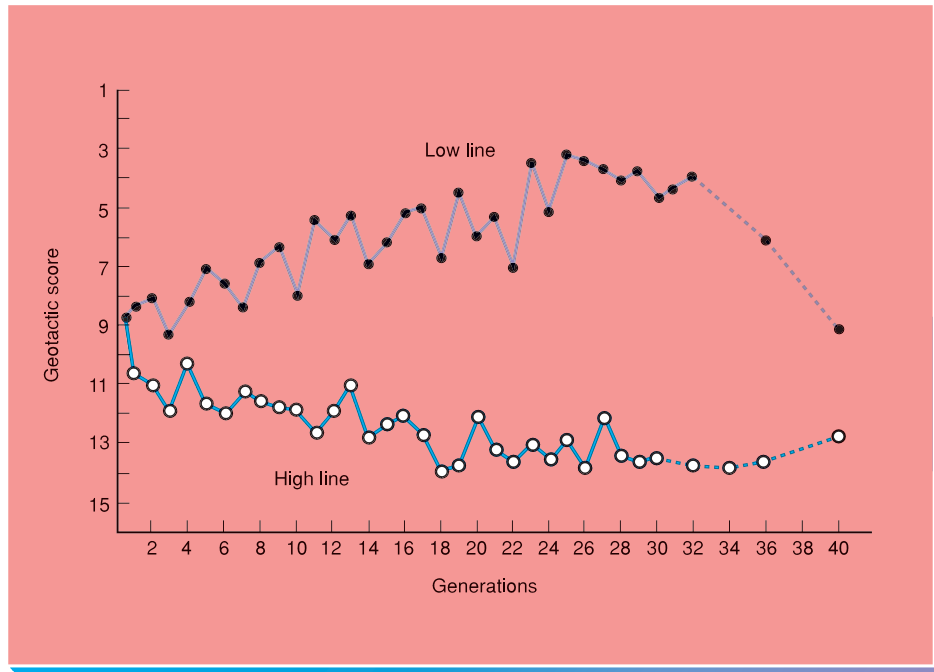


Figure 18.13 Selection for geotaxis. The dotted lines represent relaxed selection. (Source: Data from T. Dobzhansky and B. Spassky, "Artificial and natural selection for two behavioral traits in *Drosophila pseudoobscura*," *Proceedings of the National Academy of Sciences, USA*, 62:75–80, 1969.)

however, seems to have much of its original genetic variability, because the relaxation of selection caused the mean score of this line to increase rapidly. It still had enough genetic variability to head back to the original population mean. The response to a selection experiment is one way that plant and animal breeders can predict future response.

HERITABILITY

Plant and animal breeders want to improve the yields of their crops to the greatest degree they can. They must choose the parents of the next generation on the basis of this generation's yields; thus, they are continually performing selection experiments. Breeders run into two economic problems. They cannot pick only the very best to be the next generation's parents because (1) they cannot afford to decrease the size of a crop by using only a very few select parents and (2) they must avoid **inbreeding depression**, which occurs when plants are self-fertilized or animals are bred with close relatives for many generations. After frequent inbreeding, too much homozygosity occurs, and many genes that are slightly or partially deleterious begin to show themselves, depressing vigor and yield. (Chapter 19 presents more on in-

breeding.) Thus, breeders need some index of the potential response to selection so that they can then get the greatest amount of selection with the lowest risk of inbreeding depression.

Realized Heritability

Breeders often calculate a **heritability** estimate, a value that predicts to what extent their selection will be successful. Heritability is defined in the following equation:

$$H = \frac{Y_O - \bar{Y}}{Y_P - \bar{Y}} = \frac{\text{gain}}{\text{selection differential}} \quad (18.8)$$

in which

H = heritability

Y_O = offspring yield

$\bar{Y}$ = mean yield of the population

Y_P = parental yield

From this equation, we can see that heritability is the gain in yield divided by the amount of selection practiced (fig. 18.14). $Y_O - \bar{Y}$ is the improvement over the population average due to $Y_P - \bar{Y}$, which is the amount of difference between the parents and the population average. If there is no gain ($Y_O = \bar{Y}$), then the heritability

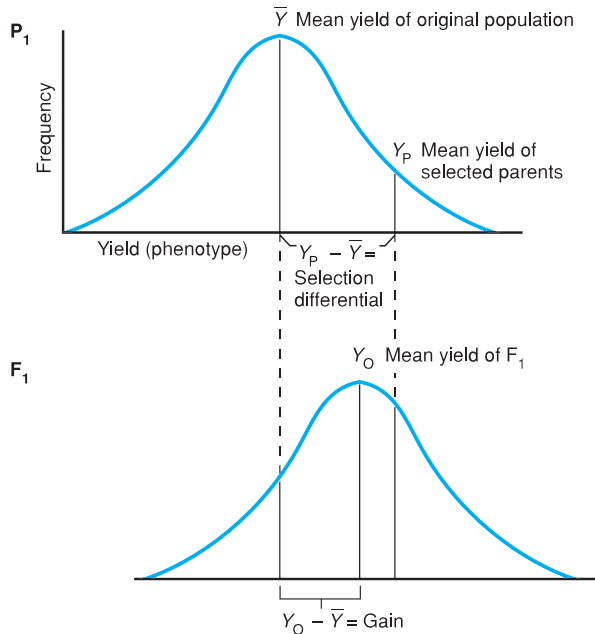


Figure 18.14 Realized heritability is the gain in yield divided by the selection differential when offspring are produced by parents with a mean yield that differs from that of the general population.

will be zero, and breeders will know that no matter how much selection they practice, they will not improve their crops and might as well not waste their time. Since this value is calculated after the breeding has been done, it is referred to as **realized heritability**. Some typical values for realized heritabilities are shown in table 18.5.

The following example may help to clarify the calculation of realized heritability. The number of bristles on the sternopleurite, a thoracic plate in *Drosophila*, is under polygenic control. In a population of flies, the mean bristle number was 6.4. Three pairs of flies served as parents; they had a mean of 7.2 bristles. Their offspring had a mean of 6.6 bristles. Hence, $Y_O = 6.6$, $\bar{Y} = 6.4$, and $Y_P = 7.2$. Dividing the gain by the selection differential—that is, substituting in equation 18.8—gives us

$$H = \frac{6.6 - 6.4}{7.2 - 6.4} = \frac{0.2}{0.8} = 0.25$$

If both a low line and a high line were begun, and if both were carried over several generations, the heritability would be measured by the final difference in means of the high and low lines (gain) divided by the cumulative selection differentials summed for both the high and low lines.

Note from figure 18.13 that the response to selection declines with time as the selected population becomes homozygous for various alleles controlling the trait. As the response declines, the calculated heritability value it-

Table 18.5 Some Realized Heritabilities

Animal	Trait	Heritability
Cattle	Birth weight	0.49
	Milk yield	0.30
Poultry	Body weight	0.31
	Egg production	0.30
	Egg weight	0.60
Swine	Birth weight	0.06
	Growth rate	0.30
	Litter size	0.15
Sheep	Wool length	0.55
	Fleece weight	0.40

self declines. After intense or prolonged selection, heritability may be zero. It does not mean that the trait is not controlled by genes, only that there is no longer a response to selection. Hence, heritability is specific for a particular population at a particular time. Intense selection exhausts the genetic variability, rendering the response to selection, and thus the heritability itself, zero.

Quantitative geneticists treat the realized heritability as an estimate of **true heritability**. True heritability is actually viewed in two different ways: as *heritability in the narrow sense* and *heritability in the broad sense*. We define these on the basis of partitioning of the variance of the quantitative character under study.

Partitioning of the Variance

Given that the variance of a distribution has genetic and environmental causes, and given that the variance is additive, we can construct the following formula:

$$V_{Ph} = V_G + V_E \quad (18.9)$$

in which

V_{Ph} = total phenotypic variance

V_G = variance due to genotype

V_E = variance due to environment

Throughout the rest of this discussion, we will stay with this model. We could construct a more complex variance model if there are interactions between variables. For example, if one genotype responded better in one soil condition than in another soil condition, this environment-genotype interaction would require a separate variance term (V_{GE}).

The variance due to the genotype (V_G) can be further broken down according to the effects of additive polygenes (V_A), dominance (V_D), and epistasis (V_I) to give us a final formula:

$$V_{Ph} = V_A + V_D + V_I + V_E \quad (18.10)$$

We can now define the two commonly used—and often confused—measures of heritability. *Heritability in the narrow sense* is

$$H_N = V_A/V_{Ph} \quad (18.11)$$

This heritability is the proportion of the total phenotypic variance caused by additive genetic effects. It is the heritability of most interest to plant and animal breeders because it predicts the magnitude of the response under selection.

Heritability in the broad sense is

$$H_B = V_G/V_{Ph} \quad (18.12)$$

This heritability is the proportion of the total phenotypic variance caused by all genetic factors, not just additive factors. It measures the extent to which individual differences in a population are caused by genetic differences. This measure is the one most often used by psychologists. We are concerned primarily with H_N , heritability in the narrow sense.

Measurement of Heritability

Three general methods are used to estimate heritability. First, as discussed earlier, we can measure heritability by the response of a population to selection. Second, we can directly estimate the components of variance by minimizing one component; the remaining variance can then be attributed to other causes. For example, by minimizing environmental causes of variance, we can estimate the genetic component directly. Or, by eliminating the genetic causes of variance, we can estimate the environmental component directly. Third, we can measure the similarity between relatives. We look now at the latter two methods.

Variance components can be minimized in several different ways. If we use genetically identical organisms, then the additive, dominance, and epistatic variances are zero, and all that is left is the environmental variance. For example, F. Robertson determined the variance components for the length of the thorax in *Drosophila*. The total variance (V_{Ph}) in a genetically heterogeneous population was 0.366 (measured directly from the distribution of the trait, as in tables 18.2 and 18.3). He then looked at the variance in flies that were genetically homogeneous. These were from isolated lines inbred in the laboratory over many generations to become virtually homozygous. Robertson studied the F_1 in several different matings of inbred lines and found the variance in thorax length to be 0.186 (V_E). By subtraction ($0.366 - 0.186$), we know that the total genetic variance (V_G) was 0.180. From this, we can calculate heritability in the broad sense as

$$H_B = V_G/V_{Ph} = 0.180/0.366 = 0.49$$

To calculate a heritability in the narrow sense, it is necessary to extract the components of the genetic variance, V_G .

Genetic variance can be measured directly by minimizing the influence of the environment. This is most easily done with plants grown in a greenhouse. Under that circumstance, environmental variables, such as soil quality, water, and sunlight, can be controlled to a very high degree. Hence, the variance among individuals grown under these circumstances is almost all genetic variance. The total phenotypic variance can be obtained from the plants grown under natural circumstances. This allows us to calculate heritability in the broad sense.

Several methods exist to sort out the additive from the dominant and epistatic portions of the genetic variance. The methods rely mostly on correlations between relatives. That is, the expected amount of genetic similarity between certain relatives can be compared with the actual similarity. The expected amount of genetic similarity is the proportion of genes shared; this is a known quantity for any form of relatedness. For example, parents and offspring have half their genes in common. The relation of observed and expected correlations between relatives is a direct measure of heritability in the narrow sense. We can thus define

$$H_N = r_{obs}/r_{exp} \quad (18.13)$$

in which r_{obs} is the observed correlation between the relatives, and r_{exp} is the expected correlation. The expected correlation is simply the proportion of the genes in common.

We must point out that the observed correlation between relatives can be artificially inflated if the environments are not random. Since we know that relatives frequently share similar (or correlated) environments, they may show a phenotypic similarity irrespective of genetic causes. It is important to keep that in mind, especially when we analyze human traits, where it may be almost impossible to rule out or quantify environmental similarity. Hence, r_{obs} may be inflated, which will inflate H_N .

In human beings, finger-ridge counts (fingerprints, fig. 18.15) have a very high heritability; there seems to be very little environmental interference in the embryonic development of the ridges (table 18.6). Monozygotic twins are from the same egg, which divides into two embryos at a very early stage. They have identical genotypes. Dizygotic twins result from the simultaneous fertilization of two eggs. They have the same genetic relationship as siblings. (However, environmental influences may be different; they may be treated differently by relatives and friends.) The data therefore suggest that human finger ridges are almost completely controlled by additive genes with a negligible input from environmental and dominance variation. Few human traits are controlled this simply (table 18.7).

This brief discussion should make it clear that the components of the total variance can be estimated. For a given quantitative trait, the total variance can be mea-

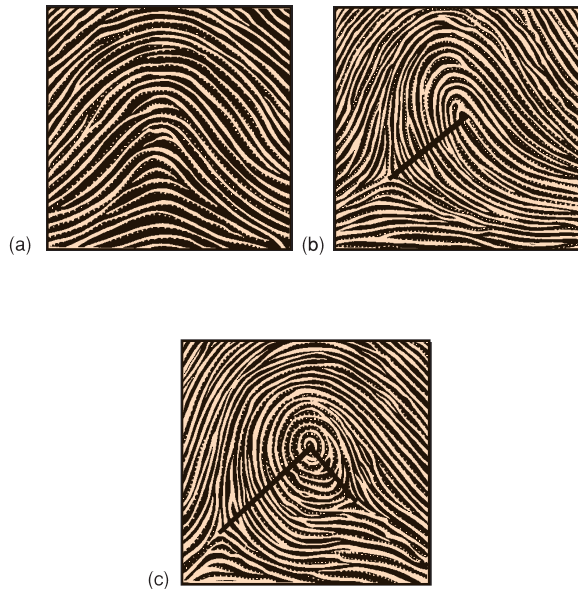


Figure 18.15 The three basic fingerprint patterns. Ridges are counted where they intersect the line connecting a triradius with a loop or whorl center. (a) An arch; there is no triradius; the ridge count is zero. (b) A loop; thirteen ridges. (c) A whorl; there are two triradii and counts of seventeen and eight (the higher one is routinely used). (From Sarah B. Holt, "Quantitative genetics of finger-print patterns," *British Medical Bulletin*, 17. Copyright © 1961 Churchill Livingstone Medical Journals, Edinburgh, Scotland. Reprinted by permission.)

Table 18.6 Correlations Between Relatives, and Heritabilities, for Finger-Ridge Counts

Relationship	r_{obs}	r_{exp}	H_N
Mother-child	0.48	0.50	0.96
Father-child	0.49	0.50	0.98
Siblings	0.50	0.50	1.00
Dizygotic twins	0.49	0.50	0.98
Monozygotic twins	0.95	1.00	0.95

Source: From Sarah B. Holt, "Quantitative genetics of finger-print patterns," *British Medical Bulletin*, 17. Copyright © 1961 Churchill Livingstone Medical Journals, Edinburgh, Scotland. Reprinted by permission.

sured directly. If identical genotypes can be used, then the environmental component of variance can be determined. By correlation of various relatives, it is possible to directly measure heritability in the narrow sense. If heritability is known, and if the total phenotypic variance is known, then all that are left, assuming no interaction, are the dominance and epistatic components. In practice, the epistatic components are usually ignored. Thus, op-

Table 18.7 Some Estimates of Heritabilities (H_N) for Human Traits and Disorders

Trait	Heritability
Schizophrenia	0.85
Diabetes Mellitus	
Early onset	0.35
Late onset	0.70
Asthma	0.80
Cleft Lip	0.76
Heart Disease, Congenital	0.35
Peptic Ulcer	0.37
Depression	0.45
Stature*	1.00+

* A heritability higher than one can be obtained when the correlation among relatives is higher than expected. This is usually the result of dominant alleles.

erationally, all that is left is the dominance variance, obtained by subtracting the additive from the total genetic variance. In addition, plant and animal breeders use sophisticated statistical techniques of covariance and variance analysis, techniques that are beyond our scope.

QUANTITATIVE INHERITANCE IN HUMAN BEINGS

As with most human studies, the measurement of heritability is limited by a lack of certain types of information. We cannot develop pure human lines, nor can we manipulate human beings into various kinds of environments or do selection experiments. However, certain kinds of information are available that allow some estimation of heritabilities.

Skin Color

Skin color is a quantitative human trait for which a simple analysis can be done on naturally occurring matings. Certain groups of people have black skin; other groups do not. Many of these groups breed true in the sense that skin colors stay the same generation after generation within a group; when groups intermarry and produce offspring, the F_1 are intermediate in skin color. In turn, when F_1 individuals intermarry and produce offspring, the skin color of the F_2 is, on the average, about the same as the F_1 , but with more variation (fig. 18.16). The data are consistent with a model of four loci, each segregating two alleles. At each locus, one allele adds a measure of color, whereas the other adds none.

IQ and Other Traits

In human beings, twin studies have been helpful in estimating the heritability of quantitative traits. One way of looking at quantitative traits is by the **concordance** among twins. Concordance means that if one twin has the trait, the other does also. Discordance means one has the trait and the other does not. Table 18.8 shows some concordance values. High concordance of monozygotic as compared with dizygotic twins is another indicator of the heritability of a trait. Concordance values for measles susceptibility and handedness, which are similar for both monozygotic and dizygotic twins, demonstrate the environmental influence on some traits.

Some monozygotic twins (MZ) have been reared apart. The same is true for dizygotic twins (DZ) and nontwin siblings. IQ (intelligence quotient) is a measure of intelligence highly correlated among relatives, indicating a strong genetic component. In three studies of monozygotic twins reared apart, the average correlation in IQ was 0.72. In thirty-four studies of monozygotic twins reared together, the average correlation in IQ was 0.86; dizygotic twins reared together have an average correlation of 0.60 in IQ. Thus, it is clear that there is a genetic influence on IQ. However, experts disagree strongly on the environmental role in shaping IQ and the exact meaning of IQ as a functional measure of intelligence (box 18.2).

At present, twin studies are emerging from the shadow of a scandal involving a knighted British psychologist, Cyril Burt (1883–1971), who did classical twin research on the inheritance of IQ. Burt was posthumously accused of fraud, an accusation that was almost universally accepted and that cast doubt on all of his data and conclusions. More recently, new infor-

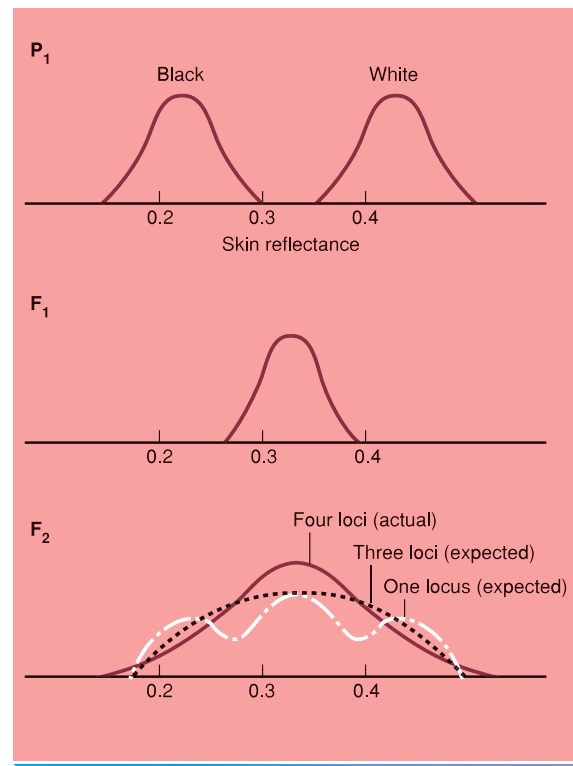


Figure 18.16 Inheritance of skin color in human beings. Four loci are probably involved.

mation seemed to cast doubt on the charges of fraud. These on-again, off-again charges have been a focus of scientific interest.

Table 18.8 Concordance of Traits Between Identical and Fraternal Twins

	Identical (MZ) Twins (%)	Fraternal (DZ) Twins (%)
Hair color	89	22
Eye color	99.6	28
Blood pressure	63	36
Handedness (left or right)	79	77
Measles	95	87
Clubfoot	23	2
Tuberculosis	53	22
Mammary cancer	6	3
Schizophrenia	80	13
Down syndrome	89	7
Spina bifida	72	33
Manic-depression	80	20

BOX 18.2

The study of human behavioral genetics was at first associated with the **eugenics** movement, founded in the late nineteenth century by Francis Galton, one of the founders of quantitative genetics. Eugenics was a movement designed to improve humanity by better breeding. This movement was tainted by bad science done by people with strong prejudices. However, although still controversial, the study of human behavioral genetics is back in vogue. The political climate has changed, and scientific methods to study human behaviors have improved. Even some of the strongest critics against these studies have changed their minds when confronted with the discovery of particular behavioral genes through the mapping of quantitative trait loci. In addition, advocates for people with many human conditions, such as homosexuality and mental illness, feel that if these traits are shown to be genetic in origin, then those who have them will be treated as people with medical conditions rather than as social outcasts.

Ethics and Genetics

*Human Behavioral
Genetics*

Even with better methods and more objective practitioners, the study of human behavioral genetics is still difficult to achieve. Some traits are very poorly defined and may be complex mixtures of phenotypes, such as schizophrenia. Other traits are just difficult to define, such as alcoholism, criminal tendency, and aggressiveness. To make things more complicated, several recent studies that seemed to isolate genes for specific behavioral traits were not verified or were later retracted. In one case, a gene for manic-depressive behavior was isolated in an Amish population. However, when the study was expanded, new cases were discovered that were not linked to the particular marker locus. The result was that a “found” genetic locus was

lost. To their credit, the workers were quick to retract their conclusions.

Currently, there are intriguing results suggesting that divorce, aggression, and dyslexia are under genetic influence. For example, in a recent study, investigators measured a heritability of 0.52 for divorce. This doesn’t mean that “divorce genes” exist, but rather that genes for certain personality traits might predispose a person to divorce. We should make it clear that genetic control does not mean that the environment does not play a role in these traits, just that there are genes that are influential also, sometimes very significantly.

In retrospect, it should not be surprising that genes influence much of our behavior. There are numerous animal studies confirming genetic control of behaviors, indicating that the same would be found in people. As long as the research is done in a competent fashion and the results are not “politicized,” human behavior genetics should not only be a reasonable area of study, but an exciting one as we learn more about ourselves.

SUMMARY

STUDY OBJECTIVE 1: To understand the patterns of inheritance of phenotypic traits controlled by many loci 531–535

Some genetically controlled phenotypes do not fall into discrete categories. This type of variation is referred to as quantitative, continuous, or metrical variation. The genetic control of this variation is referred to as polygenic control. If the number of controlling loci is small, and offspring fall into recognizable classes, it is possible to analyze the genetic control of the phenotypes with standard methods. Polygenes controlling DDT resistance are located on all chromosomes in *Drosophila*.

STUDY OBJECTIVE 2: To investigate the way that geneticists and statisticians describe and analyze normal distributions of phenotypes 535–542

When phenotypes fall into a continuous distribution, the methods of genetic analysis change. We must describe a distribution using means, variances, and standard deviations. Then we must describe the relationship between two variables using variances and correlation coefficients.

STUDY OBJECTIVE 3: To define and measure heritability, the unit of inheritance of variation in traits controlled by many loci 542–547

Equipped with statistical tools, we analyzed the genetic control of continuous traits. The heritability estimates tell us how much of the variation in the distribution of a trait can be attributed to genetic causes. Heritability in the narrow sense is the relative amount of variance due to additive

loci. Heritability in the broad sense is the relative amount of variance due to all genetic components, including dominance and epistasis. In practice, heritability can be calculated as realized heritability—gain divided by selection differential. Estimates of human heritabilities can be constructed from correlations among relatives, concordance and discordance between twins, and studies of monozygotic twins reared apart.

SOLVED PROBLEMS

PROBLEM 1: In a certain stock of wheat, grain color is controlled by four loci acting according to an additive model. How many different gametes can a tetrahybrid produce? How many different genotypes will result if tetrahybrids are self-fertilized? What will be the phenotypic distribution of these genotypes?

Answer: Assume the A , B , C , and D loci with A and a , B and b , C and c , and D and d alleles, respectively. A tetrahybrid will have the genotype $Aa Bb Cc Dd$. A gamete can get either allele at each of four independently assorting loci, so there are $2^4 = 16$ different gametes. Three genotypes are possible for each locus, two homozygotes and a heterozygote. Therefore, for four independent loci, there are $3^4 = 81$ different genotypes. Phenotypes are distributed according to the binomial distribution. Thus, there will be a pattern of $(A + a)^{2n} = (A + a)^8$; a ratio of 1:8:28:56:70:56:28:8:1 of phenotypes with decreasing red color from left to right, eight red colors plus white.

PROBLEM 2: In horses, white facial markings are inherited in an additive fashion. These markings are scored on a scale that begins at zero. In a particular population, the average score is 2.2. A group of horses with an average score of 3.4 is selected to be parents of the next generation. The offspring of this group of selected parents have a mean score of 3.1. What is the realized heritability of white facial markings in this herd of horses?

Answer: This is a simple selection experiment; the data fit our equation for realized heritability (equation 18.8). In this case:

$$Y_O = \text{offspring yield} = 3.1$$

$$\bar{Y} = \text{mean yield of the population} = 2.2$$

$$Y_P = \text{parental yield} = 3.4$$

Substituting into equation 18.8:

$$H = \frac{Y_O - \bar{Y}}{Y_P - \bar{Y}} = \frac{3.1 - 2.2}{3.4 - 2.2} = \frac{0.9}{1.2} = 0.75$$

PROBLEM 3: Corn growing in a field in Indiana had a lysine (amino acid) content of 2.0%, with a variance of 0.16. When grown in the greenhouse under controlled and uniform conditions, the mean lysine content was again 2.0%, but the variance was 0.09. What measure of heritability can you calculate?

Answer: We use equation 18.9 for the calculation of heritability by partitioning of the variance ($V_{Ph} = V_G + V_E$). In this case:

$$V_{Ph} = \text{total phenotypic variance} = 0.16$$

$$V_G = \text{variance due to genotype} = 0.09$$

$$V_E = \text{variance due to environment} = ?$$

In the greenhouse, we have minimized environmental variance, meaning the total genotypic variance = 0.09. If we subtract this from the total variance, we get the original environmental variance: $0.16 - 0.09 = 0.07$. Heritability in the broad sense is the genetic variance divided by the total phenotypic variance, or $0.09/0.16 = 0.56$.

E X E R C I S E S A N D P R O B L E M S *

TRAITS CONTROLLED BY MANY LOCI

1. A variety of squash has fruits that weigh about 5 pounds each. In a second variety, the average weight is 2 pounds. When the two varieties are crossed, the F_1 produce fruit with an average weight of 3.5 pounds. When two of these are crossed, their offspring produce a range of fruit weights, from 2 to 5 pounds. Of two hundred offspring, three produce fruits weighing about 5 pounds and three produce fruits about 2 pounds in weight. Approximately how many allelic pairs are involved in the weight difference between the varieties, and approximately how much does each effective gene contribute to the weight?
2. In rabbit variety 1, ear length averages 4 inches. In a second variety, it is 2 inches. Hybrids between the varieties average 3 inches in ear length. When these hybrids are crossed among themselves, the offspring exhibit a much greater variation in ear length, ranging from 2 to 4 inches. Of five hundred F_2 animals, two have ears about 4 inches long, and two have ears about 2 inches long. Approximately how many allelic pairs are involved in determining ear length, and how much does each effective gene seem to contribute to the length of the ear? What do the distributions of P_1 , F_1 , and F_2 probably look like?
3. Assume that height in people depends on four pairs of alleles. How can two persons of moderate height produce children who are much taller than they are? Assume that the environment is exerting a negligible effect.
4. How do polygenes differ from traditional Mendelian genes?
5. If skin color is caused by additive genes, can matings between individuals with intermediate-colored skin produce light-skinned offspring? Can such matings produce dark-skinned offspring? Can matings be-
- tween individuals with light skin produce dark-skinned offspring? (See also QUANTITATIVE INHERITANCE IN HUMAN BEINGS)
6. The tabulated data from Emerson and East ("The Inheritance of Quantitative Characters in Maize," 1913, *Univ. Nebraska Agric. Exp. Sta. Bull.*, no. 2) show the results of crosses between two varieties of corn and their F_2 offspring (see the table on ear length in corn). Provide an explanation for these data in terms of number of allelic pairs controlling ear length. Do all the genes involved affect length additively? Explain.
7. In *Drosophila*, a marker strain exists containing dominant alleles that are lethal in the homozygous condition on both chromosome 2, 3, and 4 homologues. These six lethal alleles are within inversions, so there is virtually no crossing over. The strain thus remains perpetually heterozygous for all six loci and therefore all three chromosome pairs. (Geneticists use a shorthand notation in these "balanced-lethal" systems in which only the dominant alleles on a chromosome are shown, with a slash separating the two homologous chromosomes.) The markers are: chromosome 2, Curly and Plum (Cy/Pm , shorthand for $CyPm^+/Cy^+Pm$); chromosome 3, Hairless and stubble (H/S); and chromosome 4, Cell and Minute(4) ($Ce/M[4]$). With this strain, which allows you to follow particular chromosomes by the presence or absence of phenotypic markers, construct crosses to give the strains Crow used (see fig. 18.7) to determine the location of polygenes for DDT resistance.
8. A red-flowered plant is crossed with a yellow-flowered plant to produce F_1 plants with orange flowers. The F_1 offspring are selfed, and they produce plants with flowers in a range of seven different colors. How many genes are probably involved in color production?

	Ear Length in Corn (cm)																
	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Variety P ₆₀	4	21	24	8													
Variety P ₅₄									3	11	12	15	26	15	10	7	2
F ₁					1	12	12	14	17	9	4						
F ₂ (F ₁ × F ₁)			1	10	19	26	47	73	68	68	39	25	15	9	1		

* Answers to selected exercises and problems are on page A-20.

9. A plant with a genotype of *aabb* and a height of 40 cm is crossed with a plant with a genotype of *AABB* and a height of 60 cm. If each dominant allele contributes to height additively, what is the expected height of the F_1 progeny?
10. If the F_1 generation in the cross in problem 9 is selfed, what proportion of the F_2 offspring would you expect to be 50 cm tall?
11. Two strains of wheat were compared for the time required to mature. Strain X required fourteen days, and strain Y required twenty-eight days. The strains were crossed, and the F_1 generation was selfed. One hundred F_2 progeny out of 6,200,000 matured in fourteen days or less. How many genes may be involved in maturation?

POPULATION STATISTICS

12. A geneticist wished to know if variation in the number of egg follicles produced by chickens was inherited. As a first step in his experiments, he wished to determine if the number of eggs laid could be used to predict the number of follicles. If this were true, he could then avoid killing the chickens to obtain the data he needed. He obtained the following data from fourteen chickens.

Chicken Number	Eggs Laid	Ovulated Follicles
1	39	37
2	29	34
3	46	52
4	28	26
5	31	32
6	25	25
7	49	55
8	57	65
9	51	44
10	21	25
11	42	45
12	38	26
13	34	29
14	47	30

Calculate a correlation coefficient. Graph the data, and then calculate the slope and y -intercept of the regression line. Draw the regression line on the same graph.

13. The following table (data from Ehrman and Parsons, 1976, *The Genetics of Behavior*, 121, Sunderland, Mass.: Sinauer Associates) gives heights in centimeters of eleven pairs of brothers and sisters. Calculate a correlation coefficient and a heritability. Is this realized heritability, heritability in the broad sense, or heritability in the narrow sense?

Pair	Brother	Sister	Pair	Brother	Sister
1	180	175	7	178	165
2	173	162	8	186	163
3	168	165	9	183	168
4	170	160	10	165	160
5	178	165	11	168	157
6	180	157			

How can environmental factors influence this heritability value? (See also HERITABILITY)

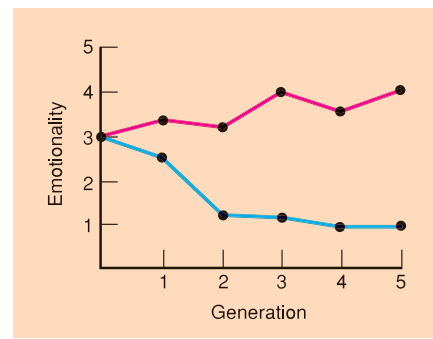
14. You determine the following variance components for leaf width in a particular species of plant:

Additive genetic variance (V_A)	4.0
Dominance genetic variance (V_D)	1.8
Epistatic variance (V_I)	0.5
Environmental variance (V_E)	2.5

Calculate the broad sense and narrow sense heritabilities. (See also HERITABILITY)

SELECTION EXPERIMENTS

15. Psychologists refer to defecation rate in rats as “emotionality.” The data shown in the accompanying figure (data modified from Broadhurst, 1960, *Experiments in Personality*, vol. 1, London: Eysenck) show mean emotionality scores during five generations in high and low selection lines. In the final generation, the parental mean was 4 for the high line and 0.9 for the low line. The cumulative selection differential is 5 for each line. Calculate realized heritability overall, and separate heritabilities for each line. Do these differ? Why? Why was the response to selection asymmetrical? (See also HERITABILITY)



16. Data were gathered during a selection experiment for six-week body weight in mice. Graph these data and calculate a realized heritability. (See also HERITABILITY)

Generation	High Line			Low Line		
	$\bar{Y}$	Y_P	Y_O	$\bar{Y}$	Y_P	Y_O
0	21			21		
1		24	22		18	20
2		24	23		18	20
3		26	23		18	20
4		26	24		16	19
5		26	23		16	18

HERITABILITY

17. Outstanding athletic ability is often found in several members of a family. Devise a study to determine to what extent athletic ability is inherited. (What is “outstanding athletic ability”?)
18. Variations in stature are almost entirely due to heredity. Yet average height has increased substantially since the Middle Ages, and the increase in the height of children of immigrants to the United States, as compared with the height of the immigrants themselves, is especially noteworthy. How can these observations be reconciled?
19. Would you expect good nutrition to increase or decrease the heritability of height?
20. Two adult plants of a particular species have extreme phenotypes for height (1 foot tall and 5 feet tall), a quantitative trait. If you had only one uniformly lighted greenhouse, how would you determine whether the variation in plant height is environmentally or genetically determined? How would you attempt to estimate the number of allelic pairs that may be involved in controlling this trait?
21. The components of variance for two characters of *D. melanogaster* are shown in the following table (data from A. Robertson, “Optimum Group Size in Progeny Testing,” *Biometrics*, 13:442–50, 1957).

Estimate the dominance and epistatic components, and calculate heritabilities in the narrow and broad sense.

Variance Components	Thorax Length	Eggs Laid in Four Days
V_{Ph}	100	100
V_A	43	18
V_E	51	38
$V_D + V_I$	?	?

22. In a mouse population, the average tail length is 10 cm. Six mice with an average tail length of 15 cm are interbred. The mean tail length in their progeny is 13.5 cm. What is the realized heritability?
23. The narrow sense heritability of egg weight in chickens in one coop is 0.5. A farmer selects for heavier eggs by breeding a few chickens with heavier eggs. He finds a difference of 9 g in the mean egg weights of selected and unselected chickens. By how much can he expect egg weight to increase in the selected chickens?
24. If, in a population of swine, the narrow sense heritability of maturation weight is 0.15, the phenotypic variance is 100 lb², the total genetic variance is 50 lb², and the epistatic variance is 0, calculate the dominance genetic variance and the environmental variance.
25. A group of four-month-old hogs has an average weight of 170 pounds. The average weight of selected breeders is 185 pounds. If the heritability of weight is 40%, what is the expected average weight of the first generation progeny?

QUANTITATIVE INHERITANCE IN HUMAN BEINGS

26. Does schizophrenia seem to have a strong genetic component (see table 18.8)? Explain.

CRITICAL THINKING QUESTIONS

1. Several cases mentioned in the text reported and then retracted the discovery of human genes controlling specific traits. Barring fraud, what might cause a scientist to retract a study of this type?
2. Monozygotic twins share identical genes. Under what conditions could they show discordance of traits?

19

POPULATION GENETICS

The Hardy-Weinberg Equilibrium and Mating Systems

STUDY OBJECTIVES

1. To understand the concept of population-level genetic processes 553
2. To learn the assumptions and nature of the Hardy-Weinberg equilibrium and its extensions 554
3. To test whether a population is in Hardy-Weinberg equilibrium 557
4. To analyze the process and consequences of nonrandom mating in diploid populations 560

STUDY OUTLINE

Hardy-Weinberg Equilibrium 553

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Assumptions of Hardy-Weinberg Equilibrium 554

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Testing for Fit to Hardy-Weinberg Equilibrium 557

Extensions of Hardy-Weinberg Equilibrium 558

Multiple Alleles 558

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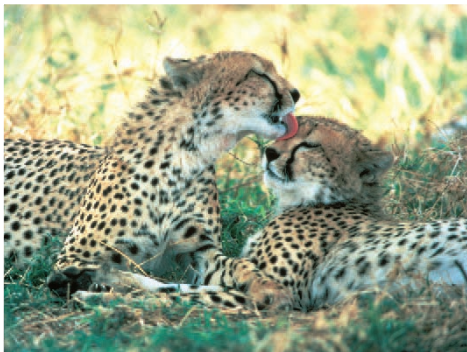
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Box 19.1 The Determination of Lethal Equivalents 562



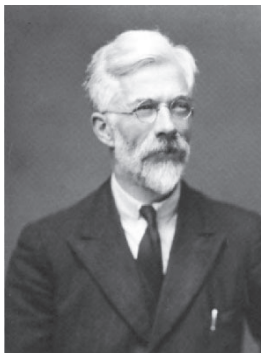
The cheetah (*Acinonyx jubatus*) is in peril of extinction;

it has very low genetic variability.

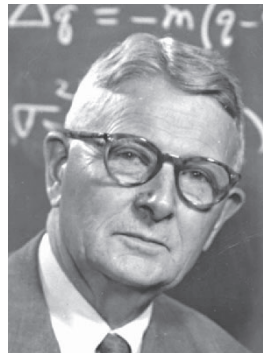
(Gregory G. Dimijian, MD/Photo Researchers, Inc.)

Evolution is a process that takes place in populations of organisms. To study evolution, we need to shift our focus to population genetics, the algebraic description of the genetic makeup of a population and the changes in allelic frequencies in populations over time. This chapter is the first of three that looks at what population genetics can tell us about the way evolution proceeds.

Almost all of the mathematical foundations of genetic changes in populations were developed in a short period of time during the 1920s and 1930s by three men: R. A. Fisher, J. B. S. Haldane, and S. Wright. Some measure of disagreement emerged among these men, but they disagreed on which evolutionary processes were more important, not on how the processes worked. Since the 1960s, excitement has arisen in the field of population genetics, primarily on three fronts. First, the high-speed computer has made it possible to do a large amount of arithmetic in a very short period of time; thus, complex simulations of real populations can be added to the repertoire of the experimental geneticist. Second, electrophoresis has provided a means of gathering the large amount of empirical data necessary to check some of the assumptions used in mathematical models. The information and interpretation of the electrophoretic data have generated some controversy about the role of “neutral” evolutionary changes in natural populations. Last, newer techniques of molecular genetics are being used to analyze the relationships among species and the rate of evolutionary processes. We consider these studies later.



Sir Ronald A. Fisher (1890–1962). (Courtesy of The National Portrait Gallery, England.)



Sewall Wright (1889–1988). (Courtesy of Dr. Sewall Wright.)

HARDY-WEINBERG EQUILIBRIUM



Let us begin with a few definitions. For the most part, we define a *species* as a group of organisms potentially capable of interbreeding. Most species are made up of **populations**, interbreeding groups of organisms that are usually subdivided into partially isolated breeding groups called **demes**. As we will see, it is these demes, or local populations, that can evolve.

In 1908, G. H. Hardy, a British mathematician, and W. Weinberg, a German physician, independently discovered a rule that relates allelic and genotypic frequencies in a population of diploid, sexually reproducing individuals if that population has random mating, large size, no mutation or migration, and no selection. The rule has three aspects:

1. The allelic frequencies at an autosomal locus in a population will not change from one generation to the next (allelic-frequency equilibrium).
2. The genotypic frequencies of the population are determined in a predictable way by the allelic frequencies (genotypic-frequency equilibrium).
3. The equilibrium is neutral. That is, if it is perturbed, it will be reestablished within one generation of random mating at the new allelic frequencies (if all the other requirements are maintained).

Calculating Allelic Frequencies



If we consider an autosomal locus in a diploid, sexually reproducing species, allelic frequencies can be measured in either of two ways. The first is simply by counting genes:

$$\text{frequency of the } a \text{ allele, } q, = \frac{\text{number of } a \text{ alleles}}{\text{total number of alleles}}$$

The expression “frequency of” can be shortened to $f()$. For example, the frequency of the a allele is written as $f(a)$. Since the homozygotes have two of a given allele and heterozygotes have only one, and since the total number of alleles is twice the number of individuals (each individual carries two alleles), we can calculate allelic frequencies in the following manner. Consider, for example, the phenotypic distribution of MN blood types (controlled by the codominant M and N alleles) among two hundred persons chosen randomly in Columbus, Ohio:

$$\text{type M (MM genotype)} = 114$$

$$\text{type MN (MN genotype)} = 76$$

$$\text{type N (NN genotype)} = \frac{10}{200}$$

Then,

$$p = f(M) = \frac{2(114) + 76}{2(200)} = \frac{304}{400} = 0.76$$

Similarly,

$$q = f(N) = \frac{2(10) + 76}{2(200)} = \frac{96}{400} = 0.24$$

Alternatively, because the frequencies of the two alleles, M and N , must add up to unity ($p + q = 1$, $q = 1 - p$, and $p = 1 - q$), if we know that $p = 0.76$, then $q = 1 - 0.76 = 0.24$.

Another way of calculating allelic frequencies is based on knowledge of the genotypic frequencies. In this example, the frequencies are

$$f(MM) = \frac{114}{200} = 0.57$$

$$f(MN) = \frac{76}{200} = 0.38$$

$$f(NN) = \frac{10}{200} = 0.05$$

We derive an expression for calculating p and q based on genotypic frequencies as follows:

$$\begin{aligned} p = f(M) &= \frac{2 \times \text{number of } MM + \text{number of } MN}{2 \times \text{total number}} \\ &= \frac{2 \times \text{number of } MM}{2 \times \text{total number}} + \frac{\text{number of } MN}{2 \times \text{total number}} \\ &= f(MM) + (1/2)f(MN) \end{aligned}$$

and,

$$\begin{aligned} q = f(N) &= \frac{2 \times \text{number of } NN + \text{number of } MN}{2 \times \text{total number}} \\ &= \frac{2 \times \text{number of } NN}{2 \times \text{total number}} + \frac{\text{number of } MN}{2 \times \text{total number}} \\ &= f(NN) + (1/2)f(MN) \end{aligned}$$

Thus, allelic frequencies can be calculated as the frequency of homozygotes, plus half the frequency of heterozygotes, as follows:

$$\begin{aligned} p = f(M) &= f(MM) + (1/2)f(MN) \\ &= 0.57 + (1/2)0.38 = 0.76 \\ q = f(N) &= f(NN) + (1/2)f(MN) \\ &= 0.05 + (1/2)0.38 = 0.24 \end{aligned}$$

or

$$q = 1 - p = 1 - 0.76 = 0.24$$

Note that these two methods (counting alleles and using genotypic frequencies) are algebraically identical and thus give identical results.

Assumptions of Hardy-Weinberg Equilibrium



We will consider a population of diploid, sexually reproducing organisms with a single autosomal locus segregating two alleles (i.e., every individual is one of three genotypes— MM , MN , or NN). Later on, we generalize the discussion to include multiple alleles and multiple loci. For the moment, the focus is on a genetic system such as the MN locus in human beings. The following major assumptions are necessary for the Hardy-Weinberg equilibrium to hold.

Random Mating

The first assumption is **random mating**, which means that the probability that two genotypes will mate is the product of the frequencies (or probabilities) of the genotypes in the population. If the MM genotype makes up 90% of a population, then any individual has a 90% chance (probability = 0.9) of mating with a person with an MM genotype. The probability of an MM by MM mating is $(0.9)(0.9)$, or 0.81.

Deviations from random mating come about for two reasons: choice or circumstance. If members of a population choose individuals of a particular phenotype as mates more or less often than at random, the population is engaged in **assortative mating**. If individuals with similar phenotypes are mating more often than at random, *positive assortative mating* is in force; if matings occur between individuals with dissimilar phenotypes more often than at random, *negative assortative mating*, or **disassortative mating**, is at work.

Deviations from random mating also arise when mating individuals are either more closely related genetically or more distantly related than individuals chosen at random from the population. **Inbreeding** is the mating of related individuals, and **outbreeding** is the mating of genetically unrelated individuals. Inbreeding is a consequence of pedigree relatedness (e.g., cousins) and small population size.

One of the first counterintuitive observations of population genetics is that deviations from random mating alter genotypic frequencies but not allelic frequencies. Envision a population in which every individual is the parent of two children. On the average, each individual will pass on one copy of each of his or her alleles. Assortative mating and inbreeding will change the zygotic (genotypic) combinations from one generation to the next, but will not change which alleles are passed into

the next generation. Thus genotypic, but not allelic, frequencies change under nonrandom mating.

Large Population Size

Even when an extremely large number of gametes is produced in each generation, each successive generation is the result of a sampling of a relatively small portion of the gametes of the previous generation. A sample may not be an accurate representation of a population, especially if the sample is small. Thus, the second assumption of the Hardy-Weinberg equilibrium is that the population is infinitely large. A large population produces a large sample of successful gametes. The larger the sample, the greater the probability that the allelic frequencies of the offspring will accurately represent the allelic frequencies in the parental population. When populations are small or when alleles are rare, changes in allelic frequencies take place due to chance alone. These changes are referred to as **random genetic drift**, or just *genetic drift*.

No Mutation or Migration

Allelic and genotypic frequencies may change through the loss or addition of alleles through mutation or migration (immigration or emigration) of individuals from or into a population. The third and fourth assumptions of the Hardy-Weinberg equilibrium are that neither mutation nor migration causes such allelic loss or addition in the population.

No Natural Selection

The final assumption necessary to the Hardy-Weinberg equilibrium is that no individual will have a reproductive advantage over another individual because of its genotype. In other words, no natural selection is occurring. (Artificial selection, as practiced by animal and plant breeders, will also perturb the Hardy-Weinberg equilibrium of captive populations.)

In summary, the Hardy-Weinberg equilibrium holds (is exactly true) for an infinitely large, randomly mating population in which mutation, migration, and natural selection do not occur. In view of these assumptions, it seems that such an equilibrium would never be characteristic of natural populations. However, this is not the case. Hardy-Weinberg equilibrium is approximated in natural populations for two major reasons. First, the consequences of violating some of the assumptions, such as no mutation or infinitely large population size, are small. Mutation rates, for example, are on the order of one change per locus per generation per 10^6 gametes. Thus, there is virtually no measurable effect of mutation in a single generation. In addition, populations do not have to be infinitely large to act as if they were. As we will see, a

relatively small population can still closely approximate Hardy-Weinberg equilibrium. In other words, minor deviations from the other assumptions can still result in a good fit to the equilibrium; only major deviations can be detected statistically. Second, the Hardy-Weinberg equilibrium is extremely resilient to change because, regardless of the perturbation, the equilibrium is usually reestablished after only one generation of random mating. The new equilibrium will be, however, at the new allelic frequencies—the Hardy-Weinberg equilibrium does not “return” to previous allelic values.

Proof of Hardy-Weinberg Equilibrium

The three properties of the Hardy-Weinberg equilibrium are that (1) allelic frequencies do not change from generation to generation, (2) allelic frequencies determine genotypic frequencies, and (3) the equilibrium is achieved in one generation of random mating. We will concentrate for a moment on the second property. In a population of individuals segregating the *A* and *a* alleles at the *A* locus, each individual will be one of three genotypes: *AA*, *Aa*, or *aa*. If $p = f(A)$ and $q = f(a)$, then we can predict the genotypic frequencies in the next generation. If all the assumptions of the Hardy-Weinberg equilibrium are met, the three genotypes should occur in the population in the same frequencies at which gametes would be randomly drawn in pairs from a **gene pool**. A gene pool is defined as all of the alleles available among the reproductive members of a population from which gametes can be drawn. Thus,

$$f(AA) = (p \times p) = p^2$$

$$f(Aa) = (p \times q) + (q \times p) = 2pq$$

$$f(aa) = (q \times q) = q^2$$

demonstrates the second property of the Hardy-Weinberg equilibrium (fig. 19.1).

		♂	
		<i>A</i> $f(A) = p$	<i>a</i> $f(a) = q$
♀	<i>A</i> $f(A) = p$	<i>AA</i> $f(AA) = p^2$	<i>Aa</i> $f(Aa) = pq$
	<i>a</i> $f(a) = q$	<i>Aa</i> $f(Aa) = pq$	<i>aa</i> $f(aa) = q^2$

Figure 19.1 Gene pool concept of zygote formation. Males and females have the same frequencies of the two alleles: $f(A) = p$ and $f(a) = q$. After one generation of random mating, the three genotypes, *AA*, *Aa*, and *aa*, have the frequencies of p^2 , $2pq$, and q^2 , respectively.

Another way of demonstrating the properties of the Hardy-Weinberg equilibrium for the one-locus, two-allele case in sexually reproducing diploids is by simply observing the offspring of a randomly mating, infinitely large population. Let the initial frequencies of the three genotypes be any values that sum to one; for example, let X , Y , and Z be the proportions of the AA , Aa , and aa genotypes, respectively. The proportions of offspring after one generation of random mating are as shown in table 19.1. For example, the probability that an AA individual will mate with an AA individual is $X \times X$, or X^2 . Since all the offspring of this mating are AA , they are counted only under the AA column of offspring in table 19.1. When all possible matings are counted, the offspring with each genotype are summed. The proportion of AA offspring is $X^2 + XY + (1/4)Y^2$, which factors to $(X + [1/2]Y)^2$. Recall that the frequency of an allele is the frequency of its homozygote plus half the frequency of the heterozygote. Hence, $X + (1/2)Y$ is the frequency of A , since $X = f(AA)$ and $Y = f(Aa)$. If $p = f(A)$, then $(X + [1/2]Y)^2$ is p^2 . Thus, after one generation of random mating, the proportion of AA homozygotes is p^2 . Similarly, the frequency of aa homozygotes after one generation of random mating is $Z^2 + YZ + (1/4)Y^2$, which factors to $(Z + [1/2]Y)^2$, or q^2 . The frequency of heterozygotes when summed and factored (table 19.1) is $2(X + [1/2]Y)(Z + [1/2]Y)$, or $2pq$. Therefore, after one generation of random mating, the three genotypes (AA , Aa , and aa) occur as p^2 , $2pq$, and q^2 .

Looking at the first property of the Hardy-Weinberg equilibrium, that allelic frequencies do not change generation after generation, we can ask, Have the allelic frequencies changed from one generation to the next (from

the parents to the offspring)? Before random mating, the frequency of the A allele is, by definition, p :

$$f(A) = p = f(AA) + (1/2)f(Aa) = X + (1/2)Y$$

After random mating, the frequency of the A homozygote is p^2 , and the frequency of the heterozygote is $2pq$. Thus, the frequency of the A allele, the frequency of its homozygote plus half the frequency of the heterozygotes, is

$$\begin{aligned} f(A) &= f(AA) + (1/2)f(Aa) \\ &= p^2 + (1/2)(2pq) \\ &= p^2 + pq = p(p + q) \\ &= p \text{ (remember, } p + q = 1) \end{aligned}$$

Thus, in a randomly mating population of sexually reproducing diploid individuals, the allelic frequency, p , does not change from generation to generation. Here, by observing the offspring of a randomly mating population, we have proven all three properties of the Hardy-Weinberg equilibrium.

Generation Time

Although generation interval is commonly thought of as the average age of the parents when their offspring are born, the statistical concept of a generation is more complex. Demographers use formulas that relate generation time to the age of reproducing females, the reproductive level of each age group, and the probability of survival in each age group. Here, to avoid these complexities, we will use **discrete generations**, unless otherwise noted. That is, we will assume that all the individuals drawn in a sam-

Table 19.1 Proportions of Offspring in a Randomly Mating Population Segregating the A and a Alleles at the A locus: $X = f(AA)$, $Y = f(Aa)$, and $Z = f(aa)$

Mating	Proportion	Offspring		
		AA	Aa	aa
$AA \times AA$	X^2	X^2		
$AA \times Aa$	XY	$(1/2)XY$	$(1/2)XY$	
$AA \times aa$	XZ		XZ	
$Aa \times AA$	XY	$(1/2)XY$	$(1/2)XY$	
$Aa \times Aa$	Y^2	$(1/4)Y^2$	$(1/2)Y^2$	$(1/4)Y^2$
$Aa \times aa$	YZ		$(1/2)YZ$	$(1/2)YZ$
$aa \times AA$	XZ		XZ	
$aa \times Aa$	YZ		$(1/2)YZ$	$(1/2)YZ$
$aa \times aa$	Z^2			Z^2
Sum	$(X + Y + Z)^2$	$(X + [1/2]Y)^2$	$2(X + [1/2]Y)(Z + [1/2]Y)$	$(Z + [1/2]Y)^2$

ple, for purposes of determining allelic and genotypic frequencies, are drawn from the same generation, and that, in resampling the population, the second sample represents the offspring of the first generation. The discrete-generation model holds for organisms such as annual plants and fruit flies maintained under laboratory conditions, with no breeding among individuals of different generations. Generations that overlap, as in populations of human beings and many other organisms, usually are better described by somewhat more complex mathematical models.

Testing for Fit to Hardy-Weinberg Equilibrium

There are several ways to determine whether a given population conforms to the Hardy-Weinberg equilibrium at a particular locus. However, the question usually arises when there is just a single sample from a population, representing only one generation. Can the existence of the Hardy-Weinberg equilibrium be determined with just one sample? The answer is that we can determine whether the three genotypes (AA , Aa , and aa) occur with the frequencies p^2 , $2pq$, and q^2 . If they do, then the population is considered to be in Hardy-Weinberg proportions; if not, then the population is not considered to be in Hardy-Weinberg proportions.

MN Blood Types

To determine whether observed and expected allelic frequencies are the same, we can use the chi-square statistical test. In a chi-square test, we compare an observed number with an expected number. In this case, the observed values are the actual numbers of the three genotypes in the sample, and the expected values come from the prediction that the genotypes will occur in the p^2 , $2pq$, and q^2 proportions. An analysis for the Ohio MN blood-type data is presented in table 19.2. The agreement between observed and expected numbers is very good, obvious even before the calculation of the chi-square value. Since the critical chi-square for one degree of freedom at the 0.05 level is 3.841 (see table 4.4), we find that

the Ohio population does not deviate from Hardy-Weinberg proportions at the MN locus.

Earlier (chapter 4), we used the chi-square statistic to test how well real data fit an expected data set based on a ratio predicted before the test. For example, we tested the data against a 3:1 ratio in table 4.2. In that case, the number of degrees of freedom was simply the number of independent categories: the total number of categories minus one. Here, however, our expected ratio is derived from the data set itself. The values p^2 , $2pq$, and q^2 came from p and q , which were estimated from the data. In this case, we lose one additional degree of freedom for every independent value we estimate from the data. If we calculate p from a sample, we lose one degree of freedom. However, we do not lose a degree of freedom for estimating q , since q is no longer an independent variable: $q = 1 - p$. So in the previous case, we lose two degrees of freedom—one for estimating p and one for independent categories. The general rule of thumb in using chi-square analysis to test for data fit to Hardy-Weinberg proportions is that the number of degrees of freedom must equal the number of phenotypes minus the number of alleles (in this case, $3 - 2 = 1$).

The chi-square analysis in table 19.2 may seem paradoxical. Because the observed allelic frequencies calculated from the original genotypic data are used to calculate the expected genotypic frequencies, it may appear to some individuals that the analysis must, by its very nature, show that the population is in Hardy-Weinberg proportions. To demonstrate that this is not necessarily the case, a counterexample appears in table 19.3. We use data similar to the Ohio sample, except that the original number of heterozygotes has been distributed equally among the two homozygote classes. The same allelic frequencies are maintained, yet the genotypic distribution differs. The chi-square value of 200.00 for these data demonstrates that the population represented in table 19.3 is not in Hardy-Weinberg proportions. Thus, a chi-square analysis of fit to the Hardy-Weinberg proportions by no means represents circular reasoning.

Table 19.2 Chi-Square Test of Goodness-of-Fit to the Hardy-Weinberg Proportions of a Sample of 200 Persons for MN Blood Types for Which $p = 0.76$ and $q = 0.24$

	MM	MN	NN	Total
Observed Numbers	114	76	10	200
Expected Proportions	p^2 (0.5776)	$2pq$ (0.3648)	q^2 (0.0576)	1.0 1.0
Expected Numbers	115.52	72.96	11.52	200.0
$\chi^2 = (O - E)^2/E$	0.020	0.127	0.201	0.348

Table 19.3 Chi-Square Test of Goodness-of-Fit to the Hardy-Weinberg Proportions of a Second Sample of 200 Persons for MN Blood Types for Which $p = 0.76$ and $q = 0.24$ and Heterozygotes Are Absent

	MM	MN	NN	Total
Observed Numbers	152	0	48	200
Expected Proportions	p^2 (0.5776)	$2pq$ (0.3648)	q^2 (0.0576)	1.0 1.0
Expected Numbers	115.52	72.96	11.52	200.0
$\chi^2 = (O - E)^2/E$	11.52	72.96	115.52	200.00

PKU

Circumstances sometimes do not allow us to test for Hardy-Weinberg proportions. In the case of a dominant trait, for example, allelic frequencies cannot be calculated from the genotypic classes because the homozygous dominant individuals cannot be distinguished from the heterozygotes. However, we can estimate allelic frequencies by assuming that the Hardy-Weinberg equilibrium exists and, thereby, assuming that the frequency of the recessive homozygote is q^2 , from which q and then p can be estimated.

If, for example, Hardy-Weinberg equilibrium is assumed for a disease such as phenylketonuria (PKU), which is expressed only in the homozygous recessive state, it is possible to calculate the proportion of the population that is heterozygous (carriers of the PKU allele). But is it fair to assume Hardy-Weinberg equilibrium here? Until recent medical advances allowed intervention, there was a good deal of selection against individuals with PKU, who were usually mentally retarded. Thus the assumption of no selection, required for equilibrium, is violated. However, only one child in ten thousand live births has PKU. When a genotype is as rare as one in ten thousand, selection has a negligible effect on allelic frequencies. Therefore, because of the rarity of the trait, we can assume Hardy-Weinberg equilibrium and calculate

$$\text{frequency of recessive homozygote} = q^2 =$$

$$1/10,000 = 0.0001$$

so,

$$q = \sqrt{0.0001} = 0.01$$

and

$$p = 1 - q = 0.99$$

Therefore,

$$\text{frequency of normal homozygote} = p^2 = (0.99)^2$$

$$\cong 0.98 \text{ or } 98 \text{ in } 100$$

$$\text{frequency of heterozygote} = 2pq$$

$$= 2(0.01)(0.99) \cong 0.02 \text{ or } 2 \text{ in } 100.$$

By assuming the Hardy-Weinberg equilibrium, we have discovered something not intuitively obvious: A recessive gene causing a trait as rare as one in ten thousand is carried in the heterozygous state by one individual in fifty. Obviously, the chi-square test cannot be used to verify the Hardy-Weinberg proportions since we derived the allelic frequencies by assuming Hardy-Weinberg proportions to begin with. In statistical terms, the number of phenotypes minus the number of alleles = $2 - 2 = 0$ degrees of freedom, which precludes doing a chi-square test.

EXTENSIONS OF HARDY-WEINBERG EQUILIBRIUM

The Hardy-Weinberg equilibrium can be extended to include, among other cases, multiple alleles and multiple loci.

Multiple Alleles

Multinomial Expansion

The expected genotypic array under Hardy-Weinberg equilibrium is p^2 , $2pq$, and q^2 , which form the terms of the binomial expansion $(p + q)^2$. If males and females each have the same two alleles in the proportions of p and q , then genotypes will be distributed as a binomial expansion in the frequencies p^2 , $2pq$, and q^2 (see fig. 19.1). To generalize to more than two alleles, one need only add terms to the binomial expansion and thus create a multinomial expansion. For example, with alleles a , b , and c with frequencies p , q , and r , the genotypic distribution should be $(p + q + r)^2$, or

$$p^2 + q^2 + r^2 + 2pq + 2pr + 2qr$$

Homozygotes will occur with frequencies p^2 , q^2 , and r^2 , and heterozygotes will occur with frequencies $2pq$, $2pr$, and $2qr$. The ABO blood-type locus in human beings is an interesting example because it has multiple alleles and dominance.

ABO Blood Groups

The ABO locus has three alleles: I^A , I^B , and i , with the I^A and I^B alleles codominant, and both dominant to the i allele. These alleles control the production of a surface antigen on red blood cells (see fig. 2.13). Table 19.4 contains blood-type data from a sample of five hundred persons from Massachusetts. Is the population in Hardy-Weinberg proportions? The answer is not apparent from the data in table 19.4 alone, since there are two possible genotypes for both the A and the B phenotypes. No estimate of the allelic frequencies is possible without making assumptions about the number of each genotype within these two phenotypic classes. Is it possible to estimate the allelic frequencies? The answer is yes, if we assume that Hardy-Weinberg equilibrium exists.

One procedure follows. Let us assume that $p = f(I^A)$, $q = f(I^B)$, and $r = f(i)$. Blood type O has the ii genotype; if the population is in Hardy-Weinberg proportions, this genotype should occur at a frequency of r^2 . Thus

$$f(ii) = 231/500 = 0.462 = r^2$$

and

$$r = f(i) = \sqrt{0.462} = 0.680$$

From table 19.4, we see that blood type A plus blood type O include only the genotypes $I^A I^A$, $I^A i$, and ii . If the population is in Hardy-Weinberg proportions, these together should be $(p + r)^2$, in which $p^2 = f(I^A I^A)$, $2pr = f(I^A i)$, and $r^2 = f(ii)$:

$$(p + r)^2 = (199 + 231)/500 = 0.860$$

Then, taking the square root of each side

$$p + r = \sqrt{0.860} = 0.927$$

and

$$p = 0.927 - r = 0.927 - 0.680 = 0.247$$

The frequency of allele I^B , q , can be obtained by similar logic with blood types B and O, or simply by subtraction:

$$q = 1 - (p + r) = 1 - 0.927 = 0.073$$

Thus, the Hardy-Weinberg equilibrium can be extended to include multiple alleles and can be used to make estimates of the allelic frequencies in the ABO blood groups. With ABO, it is statistically feasible to do a chi-square test because there is one degree of freedom (number of phenotypes – number of alleles = 4 – 3 = 1). We are really testing only the AB and B categories; if we did our calculations as shown, the observed and expected values of phenotypes A and O must be equal.

Multiple Loci

The Hardy-Weinberg equilibrium can also be extended to consider several loci at the same time in the same population. This situation deserves mention because the whole genome is likely involved in evolutionary processes and we must, eventually, consider simultaneous allelic changes in all loci segregating alleles in an organism. (Even with a high-speed computer, simultaneous consideration of many loci is a bit far off in the future.) When two loci, A and B , on the same chromosome are in equilibrium with each other, the combinations of alleles on a chromosome in a gamete follow the product rule of probability. Consider the A locus with alleles A and a and the B locus with alleles B and b , respectively, with allelic frequencies p_A and q_A for A and a , respectively, and p_B and q_B for B and b , respectively. Given completely random circumstances, the chromosome with the A and B alleles should occur at the frequency $p_A p_B$. This is referred to as **linkage equilibrium**. When alleles of different loci are not in equilibrium (i.e., not randomly distributed in gametes), the condition is referred to as **linkage disequilibrium**. The approach to linkage equilibrium is gradual and is a function of the recombination distance between the two loci.

For example, let's start with a population out of equilibrium so that all chromosomes are AB (70%) or ab (30%). Then $p_A = 0.7$, $q_A = 0.3$, $p_B = 0.7$, and $q_B = 0.3$. We expect the Ab chromosome to occur $0.7 \times 0.3 = 0.21$, or 21% of the time. The frequency of the Ab chromosome is zero. Assume the map distance between the two loci is 0.1; in other words, 10% of chromatids in gametes are recombinant. Initially, we consider that each locus is in Hardy-Weinberg proportions, or the frequency of AB/AB individuals = 0.49 (0.7×0.7); the frequency of ab/ab individuals is 0.09 (0.3×0.3); and the frequency of AB/ab individuals is 0.42 ($2 \times 0.7 \times 0.3$).

Table 19.4 ABO Blood-Type Distribution in 500 Persons from Massachusetts

Blood Type	Genotype	Number
A	$I^A I^A$ or $I^A i$	199
B	$I^B I^B$ or $I^B i$	53
AB	$I^A I^B$	17
O	ii	231
Total		500

After one generation of random mating, gametes will be as follows:

from *AB/AB* individuals (49%): only *AB* gametes, 49% of total

from *ab/ab* individuals (9%): only *ab* gametes, 9% of total

from *AB/ab* individuals (42%):

AB gametes, 18.9% of total (0.45×0.42)

ab gametes, 18.9% of total (0.45×0.42)

Ab gametes, 2.1% of total (0.05×0.42)

aB gametes, 2.1% of total (0.05×0.42)

(The values of 18.9% and 2.1% for the dihybrids result from the fact that since map distance is 0.1, 10% of gametes will be recombinant, split equally between the two recombinant classes—5% and 5%. Ninety percent will be parental, split equally between the two parental classes—45% and 45%. Each of these numbers must be multiplied by 0.42 because the dihybrid makes up 42% of the total number of individuals.)

Although we expect 21% of the chromosomes to be of the *Ab* type, only 2.1%, 10% of the expected, appear in the gene pool after one generation of random mating. You can see that linkage equilibrium is achieved at a rate dependent on the map distance between loci. Unlinked genes, appearing 50 map units apart, also gradually approach linkage equilibrium.

Although we will not derive these extensions here, we note two others. If the frequencies of alleles at an autosomal locus differ in the two sexes, it takes two generations of random mating to achieve equilibrium. In the first generation, the allelic frequencies in the two sexes are averaged so that each sex now has the same allelic frequencies. Genotypic frequencies then come into Hardy–Weinberg proportions in the second generation. However, if the allelic frequencies differ in the two sexes for a sex-linked locus, Hardy–Weinberg proportions are established only gradually. The reasoning is straightforward. Females, with an X chromosome from each parent, average the allelic frequencies from the previous generation. However, males, who get their X chromosomes from their mothers, have the allelic frequencies of the females in the previous generation. Hence, the allelic frequencies are not the same in the two sexes after one generation of random mating, and equilibrium is achieved slowly.

NONRANDOM MATING

The Hardy–Weinberg equilibrium is based on the assumption of random mating. Deviations from random mating come about when phenotypic resemblance or re-

latedness influences mate choice. When phenotypic resemblance influences mate choice, either *assortative* or *disassortative* mating occurs, depending on whether individuals choose mates on the basis of similarity or dissimilarity, respectively. For example, in human beings, assortative mating occurs for height—short men tend to marry short women, and tall men tend to marry tall women. When relatedness influences mate choice, either *inbreeding* or *outbreeding* occurs, depending on whether mates are more or less related than two randomly chosen individuals from the population. An example of inbreeding in human beings is marriage between first cousins. Both types of nonrandom mating (assortative-disassortative mating and inbreeding-outbreeding) have the same qualitative effects on the Hardy–Weinberg equilibrium: assortative mating and inbreeding increase homozygosity without changing allelic frequencies, whereas disassortative mating and outbreeding increase heterozygosity without changing allelic frequencies.

Two differences are apparent, however, between the effects of phenotypic resemblance and relatedness on mate choice. First, assortative or disassortative mating disturbs the Hardy–Weinberg equilibrium only when the phenotype and genotype are closely related. That is, if assortative mating occurs for a nongenetic trait, then the Hardy–Weinberg equilibrium will not be distorted. Inbreeding and outbreeding affect the genome directly. A second difference between the two types of mating is that the effects of inbreeding or outbreeding are felt across the whole genome, whereas the disturbances to the equilibrium caused by assortative and disassortative mating occur only for the particular trait being considered (and for closely linked loci). Given the similarities in the consequences of the two types of matings, we will concentrate our discussion on inbreeding.

Inbreeding

Inbreeding comes about in two ways: (1) the systematic choice of relatives as mates and (2) the subdivision of a population into small subunits, leaving individuals little choice but to mate with relatives. We will concentrate on inbreeding as the systematic choice of relatives as mates. The consequences of both are similar.

Common Ancestry

An inbred individual is one whose parents are related—that is, there is **common ancestry** in the family tree. The extent of inbreeding thus depends on the degree of common ancestry that the parents of an inbred individual share. When mates share ancestral genes, each may pass on copies of the same ancestral allele to their offspring. An inbred individual can then carry identical copies of a single ancestral allele. In other words, an in-

What are the effects of inbreeding on the Hardy-Weinberg equilibrium? Let us for a moment return to the gene pool concept to produce zygotes. Assume that an allele drawn from this gene pool is of the A type, drawn with a probability of p . On the second draw, the probability of autozygosity, that is, of drawing a copy of the same allele A , is F , the inbreeding coefficient. Thus the probability of an autozygous AA individual is pF . On the second draw, however, with probability $(1 - F)$, either the A or a allele can be drawn, with probabilities of $p^2(1 - F)$ and $pq(1 - F)$, respectively. Note that a second A allele produces a homozygote that is not inbred

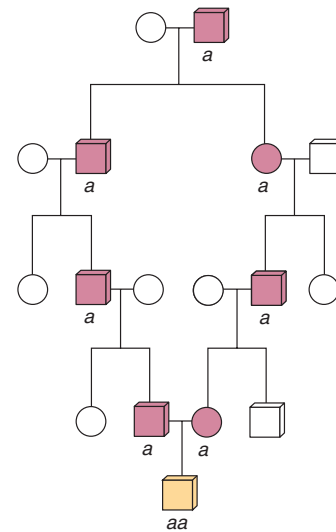


Figure 19.2 Homozygosity by descent of copies of the same ancestral allele, a . The individual at the bottom of the pedigree is inbred with the aa genotype.

(allozygous). If the first allele drawn was an a allele, with probability q , then the probability of drawing the same allele (copy of the same ancestral allele) is F , and thus the probability of autozygosity is qF . However, the probability of drawing an a or A allele that does not contribute to inbreeding is $(1 - F)$ and, therefore, the probability of an aa or Aa genotype is $q^2(1 - F)$ and $pq(1 - F)$, respectively. These calculations are summarized in table 19.5, a summary of the genotypic proportions in a population with inbreeding.

Several points emerge from table 19.5. First, when the inbreeding coefficient is zero (completely random mating), the table reduces to Hardy-Weinberg proportions. Second, compared with Hardy-Weinberg proportions, inbreeding increases the proportion of homozygotes in the population (identity by descent implies homozygosity). With complete inbreeding ($F = 1$), only homozygotes will occur in the population.

How does inbreeding affect allelic frequencies? Recall that an allelic frequency is calculated as the frequency of homozygotes for one allele plus half the frequency of the heterozygotes. Here we let p_{n+1} be the frequency of the A allele after one generation of inbreeding:

$$\begin{aligned} p_{n+1} &= p^2(1 - F) + pF + (1/2)(2pq)(1 - F) \\ &= p^2(1 - F) + pF + pq(1 - F) \\ &= p^2 + pq + F(p - p^2 - pq) \\ &= p(p + q) + pF(1 - p - q) \\ &= p(1) + pF(0) \\ &= p \end{aligned}$$

BOX 19.1

The average person carries about four lethal-equivalent alleles that are hidden because they are recessive. Four lethal equivalents means four alleles that are lethal when homozygous, or eight alleles conferring a 50% chance of mortality when homozygous, or any similar combination of lethal and semilethal alleles. The exact arrangement cannot be determined with current analytical methods. We arrive at the estimate of hidden defective and lethal alleles by using inbreeding data.

J. Crow and M. Kimura, in 1970, analyzed data showing that in Swedish families in which marriages occurred between first cousins, between 16 and 28% of the offspring had genetic diseases. For unrelated parents, the comparable figure is between 4 and 6%. Therefore, it is estimated that the offspring of first cousins have an added risk of 12 to 22% of having a genetic defect. The children of first cousins have an inbreeding coefficient of one-

Experimental Methods

The Determination of Lethal Equivalents

sixteenth. Hence, a theoretical individual who is completely inbred has the risk of genetic defect increased sixteenfold over an individual whose parents are first cousins. If 100% risk is considered 1 lethal equivalent, then a completely inbred individual would carry 2 to 3.5 lethal equivalents ($16 \times 12\% - 16 \times 22\%$). However, a completely inbred individual is, in essence, a doubled gamete. Since our interest is in the number of deleterious alleles a normal person carries, it is necessary to further multiply the risk by a factor of two to determine the number of lethal-equivalent alleles carried by a normal individual. The conclusion is that the average person carries the equivalent

of four to seven alleles that would, in the homozygous state, cause a genetic defect.

A similar calculation can be made using viability data rather than genetic defects to determine the occurrence of lethal equivalents. A study from rural France, also analyzed by Crow and Kimura, showed that the mortality rate of offspring of first cousins was 25%, whereas the analogous figure for the offspring of unrelated parents was about 12%, an increased risk of 13% for the offspring of cousins. Multiplying this risk figure of 0.13 by 32 (16×2) presents a figure of four lethal equivalents per average person in the population. In 1971, L. Cavalli-Sforza and W. Bodmer, using data primarily from Japanese populations, reported an estimate of about two lethal equivalents per average person. Despite some interpopulation differences in these estimates, they are about the same order of magnitude—two to seven lethal equivalents per person.

Thus, inbreeding does not change allelic frequencies. We can also see intuitively that inbreeding affects zygotic combinations (genotypes), but not allelic frequencies: Although inbreeding may determine the genotypes of offspring, inbreeding does not change the numbers of each allele that an individual transmits into the next generation.

In summary, inbreeding causes an increase in homozygosity, affects all loci in a population equally, and, in itself, has no effect on allelic frequencies, although it can expose deleterious alleles to selection. The results of inbreeding are evident in the appearance of recessive traits that are often deleterious. Inbreeding increases the rate of fetal deaths and congenital malformations in human beings and in other species that normally outbreed. In outbred agricultural crops and farm animals, decreases in size, fertility, vigor, and yield often result from inbreeding. Once deleterious traits appear due to inbreeding, natural selection can cause their removal from the population. However, in species adapted to inbreeding, including many crop plants and farm animals,

inbreeding does not expose deleterious alleles because those alleles have generally been eliminated already.

*Pedigree Analysis**Path Diagram Construction*

The inbreeding coefficient, F , of an individual (the probability of autozygosity) can be determined by pedigree analysis. This is done by converting a pedigree to a **path diagram** by eliminating all extraneous individuals, those who cannot contribute to the inbreeding coefficient of the individual in question. A path diagram shows the direct line of descent from common ancestors. An example of the conversion of a pedigree to a path diagram is shown in figure 19.3, in which individuals C and F are omitted from the path of descent because they are not related to anyone on the other side of the family tree and, therefore, do not contribute to the “common ancestry” of individual I. The pedigree in figure 19.3 shows an offspring who is the daughter of first cousins. Since first

Table 19.5 Genotypic Proportions in a Population with Inbreeding

Genotype	Due to Random Mating (1 - F)		Due to Inbreeding (F)		Observed Proportions
AA	$p^2(1 - F)$	+	pF	=	$p^2 + Fpq$
Aa	$2pq(1 - F)$			=	$2pq(1 - F)$
aa	$q^2(1 - F)$	+	qF	=	$q^2 + Fpq$
Total	$(p^2 + 2pq + q^2)(1 - F)$	+	$(p + q)F$	=	
	$(1 - F)$	+	F	=	1

cousins are the offspring of siblings, they share a set of common grandparents. Thus, individual I can be autozygous for alleles from either ancestor A or B, her great-grandparents. The path diagram shows the only routes by which autozygosity can occur.

The inbreeding coefficient of the offspring of first cousins can be calculated as follows. The path diagram of figure 19.3 is shown again in figure 19.4, with lowercase letters designating gametes. Two paths of autozygosity appear in this diagram, one path for each grandparent as a common ancestor: A to D and E, then to G and H, and finally to I; or B to D and E, then to G and H, and finally to I.

In the path with A as the common ancestor, A contributes a gamete to D and a gamete to E. The probability is one-half that D and E each carry a copy of the same allele. That is, there are four possible allelic combinations for the two gametes, a_1 and a_2 : $A-A$; $A-a$; $a-A$; and $a-a$. Of these combinations, the first and last ($A-A$ and $a-a$) give a copy of the same allele to the two offspring, D and E, and can thus contribute to autozygosity. The probability that gametes a_1 and d carry copies of the same allele is one-half, and the probability that d and g carry copies of the same allele is also one-half. Similarly, on the other side of the pedigree, the probability is one-half that a_2 and e carry copies of the same allele and one-half that e and h carry copies of the same allele. Thus, the overall probability that the alleles that g and h carry are identical by descent (autozygous) is $(1/2)^5$. In general, it would be $(1/2)^n$ for each path, where n is the number of ancestors in the path.

You may have spotted an additional factor here. Of the possible combinations of allelic copies passed on to D and E, one-half ($A-A$ and $a-a$) are autozygous combinations. However, the other half of the combinations, $A-a$ and $a-A$, can lead to autozygosity if A is itself inbred. If we let F_A be the inbreeding coefficient of A (the probability that any two alleles at a locus in A are identical by descent), then F_A is the probability that the $A-a$ and $a-A$ combinations are also autozygous. Thus, the probability that a common ancestor, A, passes on copies of an

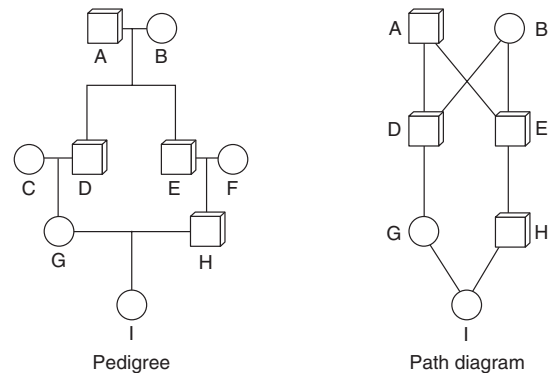


Figure 19.3 Conversion of a pedigree to a path diagram. This pedigree depicts the mating of first cousins. In the path diagram, all extraneous individuals are removed, leaving only those who could contribute to the inbreeding of individual I. Individuals in the line of descent are connected directly with straight lines, indicating the paths along which gametes are passed.

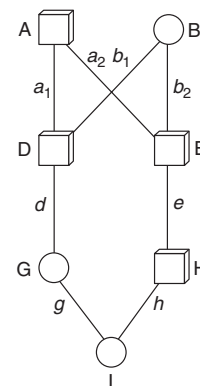


Figure 19.4 The path diagram of the mating of first cousins with gametes labeled in lowercase letters.

identical ancestral allele is $1/2 + (1/2)F_A$, or $(1/2)(1 + F_A)$. In other words, there is a one-half probability that the alleles transmitted from A to D and E are copies of the same allele. In the other half of the cases, these alleles can be identical if A is inbred. The probability of identity of A's two alleles is F_A . The expression for the inbreeding coefficient of I, F_I , can now be changed from $(1/2)^n$ by substituting $(1/2)(1 + F_A)$ for one of the $(1/2)$ s to

$$F_I = (1/2)^n(1 + F_A)$$

This equation accounts only for the inbreeding of I by the path involving the common ancestor, A, and does not account for the symmetrical path with B as the common ancestor. To obtain the total probability of inbreeding, the values from each path must be added (because these are mutually exclusive events; see chapter 4). Thus the complete formula for the inbreeding coefficient of the offspring of first cousins is

$$F_I = \Sigma[(1/2)^n(1 + F_j)] \quad (19.1)$$

in which F_I is the probability that the two alleles in I are identical by descent, n is the number of ancestors in a given path, F_j is the inbreeding coefficient of the common ancestor of that path, and all paths are summed.

In the example of the mating of first cousins (fig. 19.4)

$$F_I = (1/2)^5(1 + F_A) + (1/2)^5(1 + F_B)$$

If we assume that F_A and F_B are zero (which we must assume when the pedigrees of A and B are unknown), then

$$F_I = 2(1/2)^5 = (1/2)^4 = 0.0625$$

This can be interpreted to mean that about 6.25% of individual I's loci are autozygous, or that there is a 6.25% chance of autozygosity at any one of I's loci.

The inbreeding coefficient of the offspring of siblings (fig. 19.5) can also be calculated, assuming that A and B are not themselves inbred (F_A and F_B are zero), as

$$F_I = 2(1/2)^3 = 0.25$$

Thus, about 25% of the loci in an offspring of siblings are autozygous.

Path Diagram Rules

The following points should be kept in mind when calculating an inbreeding coefficient:

1. All possible paths must be counted. A path is possible if gametes can actually pass in that direction. Paths that violate the rules of inheritance cannot be used. For example, in figure 19.4, the following path is unacceptable: I G E A D H I.
2. In any path, an individual can be counted only once.
3. Every path must have one and only one common ancestor. The inbreeding coefficient of any other individual in the path is immaterial.

In figure 19.6, we present a complex pedigree produced from repeated sib mating, a pattern found in livestock and laboratory animals. This pedigree has two interesting points. First, common ancestors occur in several different generations. Second, some of the paths are complex. Thus, we must be sure to count all paths (paths 5 and 6 might not be immediately obvious). Although not shown in figure 19.6, one of the common ancestors, A, is also inbred ($F_A = 0.05$)—a fact that we must take into consideration in paths 3 and 5. Thus, F_I is as follows:

$$\text{From path 1: } (1/2)^3 = 0.1250$$

$$\text{From path 2: } (1/2)^3 = 0.1250$$

$$\text{From path 3: } (1/2)^5(1 + 0.05) = 0.0328$$

$$\text{From path 4: } (1/2)^5 = 0.0313$$

$$\text{From path 5: } (1/2)^5(1 + 0.05) = 0.0328$$

$$\text{From path 6: } (1/2)^5 = 0.0313$$

$$F_I = 0.3782$$

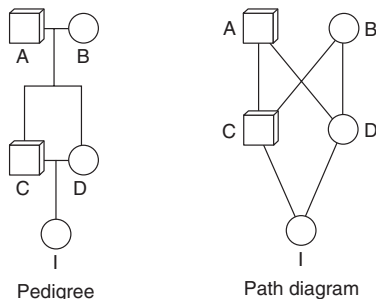


Figure 19.5 Conversion of a sib-mating pedigree to a path diagram. Individual I is inbred.

Population Analysis

It is also possible to define the inbreeding coefficient, F , of a population as the relative reduction in heterozygosity in the population due to inbreeding. In an individual, F is the probability of autozygosity; it represents an increase in homozygosity, which is therefore a decrease in heterozygosity. In a population, it also represents the reduction in heterozygosity. From the definition, we can calculate the population F as follows:

$$F = \frac{(2pq - H)}{2pq}$$

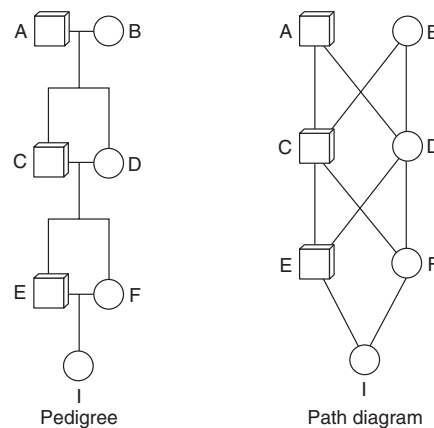
where H is the actual proportion of heterozygotes in a population, and $2pq$ is the expected proportion of het-

erozygotes based on Hardy-Weinberg proportions. This equation reduces to

$$F = 1 - \frac{H}{2pq} \quad (19.2)$$

This equation shows that when $H = 2pq$, F is zero, meaning that there is no decrease in heterozygotes and therefore, apparently, no inbreeding. When there are no heterozygotes, $F = 1$. This could be the case in a completely inbred population—for example, a self-fertilizing plant species.

As an example of an intermediate case, take the sample of one hundred individuals segregating the A_1 and A_2 alleles at the A locus: A_1A_1 , fifty-four; A_1A_2 , thirty-two; and A_2A_1 , fourteen. In this example, $p = 0.7$, $q = 0.3$, and $H = 0.32$. Since $2pq = 0.42$, $H/2pq = 0.32/0.42 = 0.76$, and $F = 1 - 0.76$, or 0.24. Thus, the inbreeding coefficient of this population is 0.24; there is a 24% reduction in heterozygotes, due presumably to inbreeding.



Paths

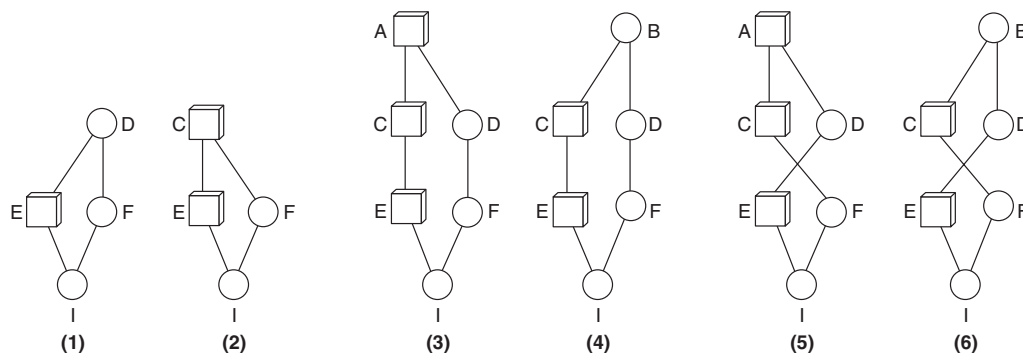


Figure 19.6 Pedigree and path diagram of two generations of sib matings. The six paths involving the potential for autozygosity are shown. $F_A = 0.05$. The paths involve common ancestors in two generations.

S U M M A R Y

STUDY OBJECTIVE 1: To understand the concept of population-level genetic processes 553–554

In a large, randomly mating population of sexually reproducing diploid organisms, not subject to the influences of mutation, migration, or selection, an equilibrium will be achieved for an autosomal locus with two alleles.

STUDY OBJECTIVE 2: To learn the assumptions and nature of the Hardy-Weinberg equilibrium and its extensions 554–557

The Hardy-Weinberg equilibrium predicts that (1) allelic frequencies (p , q) will not change from generation to generation; (2) genotypes will occur according to the binomial distribution $p^2 = f(AA)$, $2pq = f(Aa)$, and $q^2 = f(aa)$; and (3) if perturbed, equilibrium will reestablish itself in just one generation of random mating.

STUDY OBJECTIVE 3: To test whether a population is in Hardy-Weinberg equilibrium 557–560

To determine whether a population is in Hardy-Weinberg proportions, the observed and expected distribution of genotypes can be compared by the chi-square statistical test. In some circumstances, when it is reasonable to assume equilibrium, we can estimate allelic and genotypic frequencies even when dominance occurs. The Hardy-Weinberg equilibrium is easily extended to the prediction of the frequencies of multiple alleles, multiple loci, and different frequencies of alleles in the two sexes, for both sex-linked and autosomal loci.

STUDY OBJECTIVE 4: To analyze the process and consequences of nonrandom mating in diploid populations 560–565

Random mating is required for the Hardy-Weinberg equilibrium to hold. Deviations from random mating fall into two categories, depending on whether phenotypic resemblance or relatedness is involved in mate choice. Phenotypic resemblance is the basis for assortative and disassortative mating, in which individuals choose similar or dissimilar mates, respectively. Assortative mating causes increased homozygosity only among loci controlling the traits that influence mate choice. There are no changes in allelic frequencies. Similarly, disassortative mating causes increased heterozygosity without changing allelic frequencies.

Mating among relatives, or inbreeding, is represented by F , the inbreeding coefficient, which measures the probability of autozygosity (homozygosity by descent). It can be calculated from pedigrees by using the formula

$$F = \sum [(1/2)]^n (1 + F_i)$$

where n is the number of ancestors in a given path and F_i is the inbreeding coefficient of the common ancestor of that path. Inbreeding exposes recessive deleterious traits already present in the population and causes homozygosity throughout the genome. It does not, by itself, change allelic frequencies. F can also be calculated from the reduction in heterozygosity in a population.

S O L V E D P R O B L E M S

PROBLEM 1: One hundred fruit flies (*Drosophila melanogaster*) from California were tested for their genotype at the alcohol dehydrogenase locus using starch-gel electrophoresis. Two alleles were present, S and F , for slow and fast migration, respectively. The following results were noted: SS , sixty-six; SF , twenty; FF , fourteen. What are the allelic and genotypic frequencies in this population?

Answer: Since the sample size is one hundred, the proportions of the three genotypes, SS , SF , and FF , are 0.66, 0.20, and 0.14, respectively. We can calculate allelic frequencies directly from these genotypes, remembering that the frequency of an allele is the frequency of its homozygote plus half the frequency of the heterozygote, or

$$\begin{aligned} p &= f(S) = f(SS) + (1/2)f(SF) \\ &= 0.66 + (1/2)(0.20) = 0.76 \end{aligned}$$

$$\begin{aligned} q &= f(F) = f(FF) + (1/2)f(SF) \\ &= 0.14 + (1/2)(0.20) = 0.24 \end{aligned}$$

Alternatively, we could get allelic frequencies by counting alleles. Thus,

$$p = \frac{2 \times \text{number of } SS + \text{number of } SF}{2 \times \text{total number}} \\ = \frac{2(66) + 20}{2(100)} = \frac{152}{200} = 0.76$$

$$p = \frac{2 \times \text{number of } FF + \text{number of } SF}{2 \times \text{total number}} \\ = \frac{2(14) + 20}{2(100)} = \frac{48}{200} = 0.24$$

PROBLEM 2: Is the population described in problem 1 in Hardy-Weinberg equilibrium?

Answer: We can determine whether the numbers of the three genotypes (*SS*, *SF*, and *FF*) are in Hardy-Weinberg proportions through the chi-square statistical test. The observed numbers of the three genotypes are sixty-six, twenty, and fourteen, respectively. Using allelic frequencies of $p = f(S) = 0.76$ and $q = f(F) = 0.24$, we expect p^2 , $2pq$, and q^2 , respectively, of the three genotypes. That is,

$$p^2 = (0.76)^2 = 0.5776, \text{ or } 57.76 \text{ in } 100$$

$$2pq = 2(0.76)(0.24) = 0.3648, \text{ or } 36.48 \text{ in } 100$$

$$q^2 = (0.24)^2 = 0.0576, \text{ or } 5.76 \text{ in } 100$$

We can now set up a chi-square table as follows:

	<i>SS</i>	<i>SF</i>	<i>FF</i>	Total
Observed Numbers	66	20	14	100
Expected Proportions	p^2	$2pq$	q^2	1.0
	(0.5776)	(0.3648)	(0.0576)	1.0
Expected Numbers	57.76	36.48	5.76	100
$\chi^2 = (O - E)^2/E$	1.176	7.445	11.788	20.408

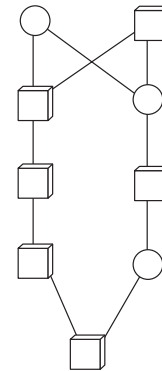
The critical chi-square value (0.05 at one degree of freedom) is 3.841, so we reject the hypothesis that this population is in Hardy-Weinberg proportions. From inspection of the table, it appears that there are too few heterozygotes and too many homozygotes, indicating that inbreeding could be the cause of the discrepancy.

PROBLEM 3: Convert the pedigree in figure 19.2 into a path diagram, and determine the inbreeding coefficient of the inbred individual, assuming that the common ancestors are not themselves inbred.

Answer: There are two paths (see the figure), each with seven ancestors. Thus, the inbreeding coefficient is

$$F = \sum [(1/2)^n (1 + F_i)] = 2(1/2)^7 = 0.016$$

Hence, the inbreeding coefficient is 0.016; about 1.6% of the loci of the inbred individual are autozygous.



EXERCISES AND PROBLEMS *

HARDY-WEINBERG EQUILIBRIUM

- One hundred persons from a small town in Pennsylvania were tested for their MN blood types. Is the population they represent in Hardy-Weinberg proportions? The genotypic data are: *MM*, forty-one; *MN*, thirty-eight; and *NN*, twenty-one.
- From the following two sets of data, calculate allelic and genotypic frequencies, and determine whether

the populations are in Hardy-Weinberg proportions. Do a statistical test if one is appropriate.

- Allele *A* is dominant to *a*; *A*, 91; *aa*, 9.
 - Electrophoretic alleles *F* and *S* are codominant at the malate dehydrogenase locus in *Drosophila*; *FF*, 137; *FS*, 196; *SS*, 87.
- The dominant ability to taste PTC comes from the allele *T*. Among a sample of 215 individuals from a population in Vancouver, 150 could detect the taste of PTC, and 65 could not. Calculate the allelic frequencies of *T* and *t*. Is the population in Hardy-Weinberg proportions?

* Answers to selected exercises and problems are on page A-21.

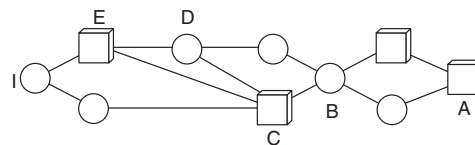
4. The frequency of children homozygous for the recessive allele for cystic fibrosis is about one in twenty-five hundred. What is the percentage of heterozygotes in the population?
5. PTC tasting is dominant in human beings.
 - a. Should most human populations be heading toward a 3:1 ratio of tasters to nontasters? Explain.
 - b. Confronted with a population sample of human beings of unknown origin, would you expect more or less than half the sample to be tasters?
6. Graph the relationship of the proportions of genotype (AA , Aa , aa) as allelic frequencies change.
7. A particular recessive disorder is present in one in ten thousand individuals. If the population is in Hardy-Weinberg equilibrium, what are the frequencies of the two alleles?
8. What allelic frequency will generate twice as many recessive homozygotes as heterozygotes?
9. Assume brown eye color is the result of a dominant allele at one locus. Attack or defend mathematically the following statement: With time, the frequency of brown-eyed individuals will increase, until about three out of four individuals are brown-eyed.
10. A particular human population has five hundred MM individuals, three hundred MN , and seven hundred NN . Calculate the allelic frequencies, and determine whether the population is in Hardy-Weinberg equilibrium.
11. Assume random mating occurs among the individuals of the population described in problem 10. What will be the frequency of each type of individual in the next generation?
12. On a small island, 235 mating individuals are all true-breeding for brown eyes. An epidemic eliminates all the population except ten young women, two young men, and four older (postmenopausal) women. A boatload of foreigners arrives; the foreign population consists of six heterozygous brown-eyed females, four homozygous brown-eyed males, and ten blue-eyed males. Assuming that one locus controls eye color, that mating is random with respect to eye color, and that each male and female capable of breeding does so, calculate the genotypic frequencies of their offspring.
13. In a given population, only the I^A and I^B alleles are present in the ABO system; there are no individuals with type O blood or with i alleles. If two hundred people have type A blood, seventy-five have type AB blood, and twenty-five have type B blood, what are the allelic frequencies in this population?

EXTENSIONS OF HARDY-WEINBERG EQUILIBRIUM

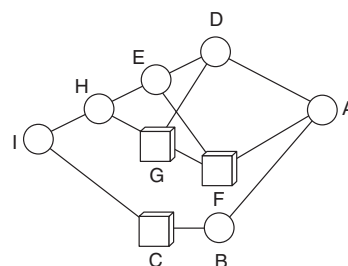
14. The following data are ABO phenotypes from a population sample of one hundred persons. Determine the frequencies of the three alleles: type A, seven; type B, seventy-two; type AB, twelve; type O, nine. What do you have to assume? Is the population in Hardy-Weinberg proportions?
15. How quickly and in what manner is Hardy-Weinberg equilibrium achieved under the following initial conditions (assuming a diploid, sexually reproducing population)?
 - a. One locus, five alleles
 - b. Two unlinked loci, two alleles each
16. A sample of fruit flies was testcrossed to determine the allelic arrangements of two linked loci in the gametes of that generation. With the following data, can you determine whether linkage equilibrium holds? Gametic arrangements are AB , fifty-eight; ab , eight; Ab , twelve; and aB , twenty-two.
17. In a large, randomly mating human population, the frequencies of the I^A , I^B , and i alleles are 0.7, 0.2, and 0.1, respectively. Calculate the expected frequencies for each blood type.
18. In a human population of one hundred people, seventeen have type A blood, seventeen have type B, two have type AB, and sixty-four have type O. If this population is in equilibrium, what are the allelic frequencies?

NONRANDOM MATING

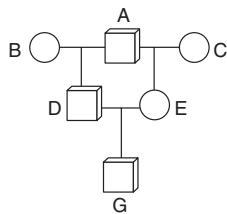
19. Under what circumstances is inbreeding deleterious?
20. What is the inbreeding coefficient of I in the following pedigree? Assume that the inbreeding coefficients of other members of the pedigree are zero unless other information tells you differently.



21. What is the inbreeding coefficient of individual I in this pedigree? $F_A = 0.01$; $F_B = 0.02$; $F_C = 0.02$.



22. The following is the pedigree of an offspring produced by the mating of half siblings. Individuals A and C have inbreeding coefficients of 0.2; all others are zero. Convert the pedigree to a path diagram and determine the inbreeding coefficient of individual G.



23. Given the population in Exercises and Problems problem 1, what is its inbreeding coefficient?
24. In a sample of one hundred people, are fourteen *MM*, thirty-two *MN*, and fifty-four *NN* individuals. Calculate the inbreeding coefficient.
25. If, in a population with two alleles at an autosomal locus, $p = 0.8$, $q = 0.2$, and the frequency of heterozygotes is 0.20, what is the inbreeding coefficient?

CRITICAL THINKING QUESTIONS

1. Prove that two generations are needed for the establishment of Hardy-Weinberg proportions when an autosomal locus with two alleles in a sexually reproducing species has frequencies of the two alleles that differ in the two sexes.
2. What might the ramifications to conservation efforts be of zoos maintaining captive breeding programs for rare and endangered species?

Suggested Readings for chapter 19 are on page B-19.

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POPULATION GENETICS

Processes That Change Allelic Frequencies

STUDY OBJECTIVES

1. To develop ways to analyze population genetics problems 571
2. To analyze the effects of mutation, migration, and population size on the Hardy-Weinberg equilibrium 571
3. To study the ways in which natural selection results in organisms adapted to their environments 577

STUDY OUTLINE

Models for Population Genetics 571

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Mutational Equilibrium 571

Stability of Mutational Equilibrium 571

Migration 573

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Natural Selection 577

How Natural Selection Acts 577

Selection Against the Recessive Homozygote 578

Selection-Mutation Equilibrium 581

Types of Selection Models 581

Summary 585

Solved Problems 585

Exercises and Problems 586

Critical Thinking Questions 587

Box 20.1 A General Computer Program to Simulate the Approach to Allelic Equilibrium Under Heterozygous Advantage 583



Natural selection works on the variation found in nature, shown here by different banding patterns in tree snails (*Liguus fasciatus*), found mainly in southern Florida. (© J. H. Robinson/Photo Researchers, Inc.)